

# **MATH 3010: Vector Integral Calculus**

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June 27, 2026



# Preface

These are the first edition of these lecture notes for MATH 3010 (Vector Integral Calculus). Consequently, there may be several typographical errors, missing exposition on necessary background, and more advanced topics for which there will not be time in class to cover. Future iterations of these notes will hopefully be fairly self-contained provided one has the necessary background. If you come across any typos, errors, omissions, or unclear expositions, please feel free to contact me so that I may continually improve these notes.



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# Chapter 1

## Differential Forms

In single-variable calculus, differentiation and integration are tied together by the Fundamental Theorem of Calculus:

$$\int_a^b f'(t) \, dt = f(b) - f(a)$$

This theorem says that integrating the derivative of a function over an interval depends only on the values of the function at the boundary points. One of the central goals of multivariable calculus is to understand how this principle extends to curves, surfaces, and higher-dimensional regions.

In earlier courses, we encountered several results that appear to be different versions of the same idea. The Fundamental Theorem for Line Integrals relates the integral of a gradient field along a curve to the values of a potential function at the endpoints. Green's Theorem relates a line integral around a closed planar curve to a double integral over the enclosed region. Stokes's Theorem relates circulation around the boundary of a surface to the curl across that surface, while the Divergence Theorem relates flux across a closed surface to a triple integral over the enclosed volume. Differential forms provide a single language in which all of these theorems become instances of one general principle.

The objects that we integrate in this setting are called differential forms. A differential 0-form is simply a smooth function. A differential 1-form may be integrated along an oriented curve and naturally represents quantities such as work or circulation. A differential 2-form may be integrated over an oriented surface and naturally represents quantities such as flux. More generally, a differential  $k$ -form is an object designed to be integrated over an oriented  $k$ -dimensional manifold.

Differential forms are built using multilinear algebra and determinants. Determinants measure signed volume, and differential forms generalize this idea by measuring signed projected lengths, areas, volumes, and higher-dimensional analogues. The wedge product allows us to combine lower-degree forms into higher-degree forms, while the exterior derivative provides a natural notion of differentiation for forms. The exterior derivative increases degree by one and satisfies the fundamental identity

$$d^2 = 0$$

In the familiar language of vector calculus, this identity contains the facts that the curl of a gradient field is zero and that the divergence of a curl field is zero.

A second essential idea is that differential forms behave naturally under smooth maps. The pullback of a form allows us to transfer an integral from a curve, surface, or manifold in one space back to a parametrizing domain in another space. In this way, pullbacks provide a conceptual foundation for substitution and change of variables in multiple integrals.

## 1.1 The Multilinear Setup

In multivariable calculus, determinants measure signed  $n$ -dimensional volume. For example, if  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \in \mathbb{R}^n$ , then

$$\det \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

is the signed volume of the parallelepiped spanned by these vectors.

Our eventual goal is to integrate over objects of many possible dimensions: curves, surfaces, and higher-dimensional manifolds. Therefore, we need objects that can measure not only signed  $n$ -dimensional volume in  $\mathbb{R}^n$ , but also signed  $k$ -dimensional projected volume for  $k \leq n$ . The appropriate algebraic objects are alternating multilinear functions, which will form the algebraic foundation for differential forms.

Throughout this section, all vector spaces are real vector spaces.

**Definition 1.1.1.** Let  $V$  and  $W$  be vector spaces. Function  $T^k : V^k \rightarrow W$  is called a  $k$ -linear map, or *multilinear map*, if it is linear in each argument separately. That is, for every  $1 \leq i \leq k$ , the vectors  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in V$ , vectors  $\vec{u}, \vec{w} \in V$ , and scalars  $a, b \in \mathbb{R}$ ,

$$T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_{i-1}, a\vec{u} + b\vec{w}, \vec{v}_{j+1}, \dots, \vec{v}_k) = aT(\vec{v}_1, \dots, \vec{v}_{i-1}, \vec{u}, \vec{v}_{j+1}, \dots, \vec{v}_k)$$



$$+ bT(\vec{v}_1, \dots, \vec{v}_{j-1}, \vec{w}, \vec{v}_{j+1}, \dots, \vec{v}_k)$$

When  $k = 1$ , this is simply the usual definition of a linear map.

**Example 1.1.2.** The dot product

$$\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$$

is bilinear. Indeed,

$$\langle a\vec{v} + b\vec{w}, \vec{u} \rangle = a\langle \vec{v}, \vec{u} \rangle + b\langle \vec{w}, \vec{u} \rangle$$

and

$$\langle \vec{v}, a\vec{w} + b\vec{u} \rangle = a\langle \vec{v}, \vec{w} \rangle + b\langle \vec{v}, \vec{u} \rangle$$

However, the dot product is not alternating, since in general,

$$\langle \vec{v}, \vec{v} \rangle = \|\vec{v}\|^2 \geq 0$$

**Example 1.1.3.** The determinant defines an  $n$ -linear map.

$$\det : (\mathbb{R}^n)^n \rightarrow \mathbb{R}$$

given by

$$\det(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n) = \det \begin{bmatrix} \uparrow & \uparrow & \cdots & \uparrow \\ \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_n \\ \downarrow & \downarrow & \cdots & \downarrow \end{bmatrix}$$

It is linear in every column separately. Moreover, it changes sign whenever two columns are interchanged and is equal to 0 whenever two columns are equal. The determinant is the fundamental example of an alternating multilinear map.

**Definition 1.1.4.** A  $k$ -linear map  $T : V^k \rightarrow \mathbb{R}$  is called *alternating* if

$$T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k) = 0$$

whenever two of the vectors  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k$  are equal. Equivalently,  $T$  is alternating if interchanging any two arguments changes its sign.

**Lemma 1.1.5.** Let  $T : V^k \rightarrow \mathbb{R}$  be an alternating multilinear map. If two adjacent arguments are interchanged, then

$$T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_i, \vec{v}_{i+1}, \dots, \vec{v}_k) = -T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_{i+1}, \vec{v}_i, \dots, \vec{v}_k)$$

*Proof.* Since  $T$  is alternating,

$$0 = T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_i + \vec{v}_{i+1}, \vec{v}_i, \vec{v}_{i+1}, \dots, \vec{v}_k)$$

Using multilinearity in the two displayed positions gives

$$\begin{aligned} 0 &= T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_i, \vec{v}_i, \dots, \vec{v}_k) + T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_i, \vec{v}_{i+1}, \dots, \vec{v}_k) \\ &\quad + T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_{i+1}, \vec{v}_i, \dots, \vec{v}_k) + T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_{i+1}, \vec{v}_{i+1}, \dots, \vec{v}_k) \end{aligned}$$

The first and last terms are 0 because  $T$  is alternating. Hence,

$$T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_i, \vec{v}_{i+1}, \dots, \vec{v}_k) = -T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_{i+1}, \vec{v}_i, \dots, \vec{v}_k)$$

which proves the result.  $\square$

**Corollary 1.1.6.** *Let  $T : V^k \rightarrow \mathbb{R}$  be alternating and multilinear. If  $\sigma$  is a permutation of  $(1, 2, \dots, k)$ , then*

$$T(\vec{v}_{\sigma(1)}, \vec{v}_{\sigma(2)}, \dots, \vec{v}_{\sigma(k)}) = \text{sgn}(\sigma)T(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k)$$

*Proof.* Every permutation can be obtained by a sequence of adjacent interchanges. Each interchange contributes a factor of  $-1$ . Therefore, the overall sign is determined by whether the permutation requires an even or odd number of interchanges.  $\square$

**Remark 1.1.7.** For a 1-linear map, the alternating condition is automatic. There are no two distinct argument positions that could contain repeated vectors. Thus, every linear functional is an alternating 1-linear map.

For a 2-linear map, alternating means

$$T(\vec{v}, \vec{v}) = 0$$

for every  $\vec{v}$ , or equivalently,

$$T(\vec{v}, \vec{w}) = -T(\vec{w}, \vec{v})$$

For a 3-linear map, alternating means that the value changes sign whenever any two arguments are swapped.

We next define the basic linear functionals from which all of our alternating multilinear maps will be built.

**Definition 1.1.8.** For each  $1 \leq i \leq n$ , define  $dx_i : \mathbb{R}^n \rightarrow \mathbb{R}$  by

$$dx_i(\vec{v}) = v_i$$

where  $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ . Thus  $dx_i$  extracts the  $i$ th coordinate of a vector.

For example, in  $\mathbb{R}^3$ ,

$$dx_1(x, y, z) = x, \quad dx_2(x, y, z) = y, \quad dx_3(x, y, z) = z$$

As usual, when working with  $\mathbb{R}^3$ , we often write

$$dx = dx_1, \quad dy = dx_2, \quad dz = dx_3$$

**Example 1.1.9.** Let  $\vec{v} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} \in \mathbb{R}^3$ . Then

$$dx(\vec{v}) = 3, \quad dy(\vec{v}) = -2, \quad dz(\vec{v}) = 5$$

Thus,  $dx$ ,  $dy$ , and  $dz$  do not measure length or direction. They simply select coordinates.

**Definition 1.1.10.** The vector space of all linear maps from  $\mathbb{R}^n$  to  $\mathbb{R}$  is called the *dual space of  $\mathbb{R}^n$* , denoted by  $(\mathbb{R}^n)^*$ . An element of  $(\mathbb{R}^n)^*$  is called a *covector*, *linear functional*, or in this context, a *constant coefficient 1-form*.

**Proposition 1.1.11.** *The set  $\{dx_1, dx_2, \dots, dx_n\}$  is a basis for  $(\mathbb{R}^n)^*$ .*

*Proof.* Let  $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$  be linear, and let  $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n$  be the standard basis vectors of  $\mathbb{R}^n$ . For each  $1 \leq i \leq n$ , define

$$a_i = \varphi(\vec{e}_i)$$

Every vector  $\vec{v} \in \mathbb{R}^n$  written as

$$\vec{v} = v_1\vec{e}_1 + v_2\vec{e}_2 + \dots + v_n\vec{e}_n$$

satisfies

$$\varphi(\vec{v}) = \varphi(v_1\vec{e}_1 + v_2\vec{e}_2 + \dots + v_n\vec{e}_n)$$

$$\begin{aligned}
&= v_1\varphi(\vec{e}_1) + v_2\varphi(\vec{e}_2) + \cdots + v_n\varphi(\vec{e}_n) \\
&= a_1v_1 + a_2v_2 + \cdots + a_nv_n \\
&= a_1dx_1(\vec{v}) + a_2dx_2(\vec{v}) + \cdots + a_ndx_n(\vec{v})
\end{aligned}$$

Therefore,

$$\varphi = a_1dx_1 + a_2dx_2 + \cdots + a_ndx_n$$

Hence,  $dx_1, dx_2, \dots, dx_n$  span  $(\mathbb{R}^n)^*$ .

To prove linear independence, suppose

$$a_1dx_1 + a_2dx_2 + \cdots + a_ndx_n = 0$$

Evaluating both sides at  $\vec{e}_i$  gives  $a_i = 0$  for every  $1 \leq i \leq n$ . Thus, the coordinate covectors are linearly independent. Therefore,  $\{dx_1, dx_2, \dots, dx_n\}$  is a basis.  $\square$

**Remark 1.1.12.** The notation  $dx_i$  is motivated by differentiation. If  $x_i : \mathbb{R}^n \rightarrow \mathbb{R}$  denotes the  $i$ th coordinate function, then the derivative of  $x_i$  is precisely the linear functional  $dx_i$

$$D(x_i)(\vec{v}) = dx_i(\vec{v})$$

For the moment, however, we should view  $dx_i$  as the  $i$ th coordinate covector.

We now use determinants to build alternating multilinear maps involving several vector arguments.

**Definition 1.1.13.** An ordered  $k$ -tuple  $I = (i_1, i_2, \dots, i_k)$  where each  $1 \leq i_j \leq n$ , is called a *multi-index of length  $k$* . The entries of  $I$  are allowed to repeat, unless otherwise stated.

**Definition 1.1.14.** Let  $I = (i_1, i_2, \dots, i_k)$  be an ordered  $k$ -tuple. Define  $dx_I : (\mathbb{R}^n)^k \rightarrow \mathbb{R}$  by

$$\det(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k) = \det \begin{bmatrix} dx_{i_1}(\vec{v}_1) & dx_{i_1}(\vec{v}_2) & \cdots & dx_{i_1}(\vec{v}_k) \\ dx_{i_2}(\vec{v}_1) & dx_{i_2}(\vec{v}_2) & \cdots & dx_{i_2}(\vec{v}_k) \\ dx_{i_3}(\vec{v}_1) & dx_{i_3}(\vec{v}_2) & \cdots & dx_{i_3}(\vec{v}_k) \\ \vdots & \vdots & & \vdots \\ dx_{i_k}(\vec{v}_1) & dx_{i_k}(\vec{v}_2) & \cdots & dx_{i_k}(\vec{v}_k) \end{bmatrix}$$

The map  $dx_I$  is obtained by keeping the coordinates numbered  $i_1, i_2, \dots, i_k$ , and then taking the determinant of the resulting  $k \times k$  matrix.

**Proposition 1.1.15.** *For every ordered  $k$ -tuple  $I = (i_1, i_2, \dots, i_k)$ , the map  $d\vec{x}_I : (\mathbb{R}^n)^k \rightarrow \mathbb{R}$  is an alternating multilinear map.*

*Proof.* Each vector  $\vec{v}_j$  appears as the  $j$ th column of the determinant defining  $d\vec{x}_I$ . Since the determinant is linear in each column,  $d\vec{x}_I$  is linear in each vector argument.

If two vector arguments are equal, then the corresponding columns of the determinant are equal. Therefore, the determinant is 0. Hence,  $d\vec{x}_I$  is alternating.  $\square$

**Example 1.1.16.** Let  $\vec{v} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$  and  $\vec{w} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$ . Then

$$d\vec{x}_{(1,3)}(\vec{v}, \vec{w}) = \det \begin{bmatrix} dx_1(\vec{v}) & dx_1(\vec{w}) \\ dx_3(\vec{v}) & dx_3(\vec{w}) \end{bmatrix} = \det \begin{bmatrix} 2 & 1 \\ 4 & -2 \end{bmatrix} = -8$$

**Example 1.1.17.** Using the vectors from the previous example,

$$d\vec{x}_{(1,3)}(\vec{w}, \vec{v}) = \det \begin{bmatrix} 1 & 2 \\ -2 & 4 \end{bmatrix} = 8$$

Hence,

$$d\vec{x}_{(1,3)}(\vec{v}, \vec{w}) = -d\vec{x}_{(1,3)}(\vec{w}, \vec{v})$$

Also,

$$d\vec{x}_{(1,3)}(\vec{v}, \vec{v}) = \det \begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} = 0$$

These calculations illustrate both sign reversal and vanishing on repeated arguments.

**Example 1.1.18.** Let  $\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ ,  $\vec{v} = \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix}$ , and  $\vec{w} = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}$ . Then

$$d\vec{x}_{(1,2,3)}(\vec{u}, \vec{v}, \vec{w}) = \det \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 1 \\ 0 & 4 & 5 \end{bmatrix} = 37$$

Thus, the parallelepiped spanned by  $\vec{u}$ ,  $\vec{v}$ , and  $\vec{w}$  has signed volume 37.

**Definition 1.1.19.** A multi-index  $I = (i_1, i_2, \dots, i_k)$  is called *increasing* if  $i_1 < i_2 < \dots < i_k$ . For example,  $(1, 3, 5)$  is increasing, while  $(3, 1, 5)$  and  $(1, 1, 4)$  are not.

**Definition 1.1.20.** Suppose  $I = (i_1, i_2, \dots, i_k)$  has no repeated entries. The increasing multi-index obtained by rearranging the entries of  $I$  into increasing order is defined by  $I^<$ . For example,

$$(2, 4, 5, 1)^< = (1, 2, 4, 5)$$

**Proposition 1.1.21.** Let  $I = (i_1, i_2, \dots, i_k)$  be a multi-index.

1. If  $I$  contains a repeated index, then  $d\vec{x}_I = 0$ .
2. If  $I$  has no repeated indices and  $I^<$  is the increasing rearrangement of  $I$ , then

$$d\vec{x}_I = (-1)^s dx_{I^<}$$

where  $s$  is the number of changes required to rearrange  $I$  into increasing order.

*Proof.* If  $I$  contains a repeated index, then the determinant defining  $d\vec{x}_I$  has two identical rows. Hence,  $d\vec{x}_I = 0$ .

If two indices in  $I$  are interchanged, then two rows of the determinant are interchanged. Therefore, the determinant changes sign. Reordering  $I$  into increasing order requires  $s$  such row interchanges, so

$$d\vec{x}_I = (-1)^s d\vec{x}_{I^<}$$

□

**Example 1.1.22.** Consider  $I = (2, 4, 5, 1)$ . To rearrange  $I$  into increasing order,

$$(2, 4, 5, 1) \rightarrow (2, 4, 1, 5) \rightarrow (2, 1, 4, 5) \rightarrow (1, 2, 4, 5)$$

This requires three interchanges. Therefore,

$$d\vec{x}_{(2,4,5,1)} = (-1)^3 d\vec{x}_{(1,2,4,5)} = -d\vec{x}_{(1,2,4,5)}$$

**Example 1.1.23.** Since the multi-index  $(1, 3, 1)$  contains a repeated index,

$$d\vec{x}_{(1,3,1)} = 0$$

Indeed, the determinant defining this map always has its first and third rows equal.

**Definition 1.1.24.** The vector space of alternating  $k$ -linear maps from  $(\mathbb{R}^n)^k$  to  $\mathbb{R}$  is denoted by  $\mathcal{A}^k(\mathbb{R}^n)^*$ . An element of  $\mathcal{A}^k(\mathbb{R}^n)^*$  is called an *alternating  $k$ -covector* or a constant coefficient  $k$ -form.

**Theorem 1.1.25.** *The set  $\{d\vec{x}_I : I \text{ is an increasing } k\text{-tuple}\}$  is a basis for  $\mathcal{A}^k(\mathbb{R}^n)^*$ . Consequently,*

$$\dim(\mathcal{A}^k(\mathbb{R}^n)^*) = \binom{n}{k}$$

*Proof.* Let  $T \in \mathcal{A}^k(\mathbb{R}^n)^*$ . For every increasing multi-index  $I = (i_1, i_2, \dots, i_k)$ , define

$$a_I = T(\vec{e}_{i_1}, \vec{e}_{i_2}, \dots, \vec{e}_{i_k})$$

We claim that

$$T = \sum_{I \text{ increasing}} a_I d\vec{x}_I$$

Let

$$\tilde{T} = \sum_{I \text{ increasing}} a_I d\vec{x}_I$$

Since  $T$  and  $\tilde{T}$  are multilinear, it is enough to verify that they agree on all ordered  $k$ -tuples of standard basis vectors.

First suppose that  $\{\vec{e}_{j_1}, \vec{e}_{j_2}, \dots, \vec{e}_{j_k}\}$  contains a repeated basis vector. Since  $T$  and  $\tilde{T}$  are alternating, then

$$T(\vec{e}_{j_1}, \vec{e}_{j_2}, \dots, \vec{e}_{j_k}) = \tilde{T}(\vec{e}_{j_1}, \vec{e}_{j_2}, \dots, \vec{e}_{j_k}) = 0$$

Now suppose  $j_1, j_2, \dots, j_k$  are distinct. Rearrange the tuple into increasing order

$$(j_1, j_2, \dots, j_k) \rightarrow (r_1, r_2, \dots, r_k)$$

where  $r_1 < r_2 < \dots < r_k$ . By alternation,

$$T(\vec{e}_{j_1}, \vec{e}_{j_2}, \dots, \vec{e}_{j_k}) = \pm T(\vec{e}_{r_1}, \vec{e}_{r_2}, \dots, \vec{e}_{r_k})$$

By the definition of  $a_I$ ,

$$T(\vec{e}_{r_1}, \vec{e}_{r_2}, \dots, \vec{e}_{r_k}) = a_{(r_1, r_2, \dots, r_k)}$$

On the other hand, the only basis element  $d\vec{x}_I$  that is nonzero on  $(\vec{e}_{j_1}, \vec{e}_{j_2}, \dots, \vec{e}_{j_k})$  is the one corresponding to the increasing rearrangement  $I = (r_1, r_2, \dots, r_k)$ . Its value is the same sign determined by the reordering. Hence,

$$\tilde{T}(\vec{e}_{j_1}, \vec{e}_{j_2}, \dots, \vec{e}_{j_k}) = T(\vec{e}_{j_1}, \vec{e}_{j_2}, \dots, \vec{e}_{j_k})$$

Therefore,  $T = \tilde{T}$ , so the increasing  $d\vec{x}_I$  span  $\mathcal{A}^k(\mathbb{R}^n)^*$ .

To prove linear independence, suppose

$$\sum_{I \text{ increasing}} a_I d\vec{x}_I = 0$$

Fix an increasing multi-index  $I = (i_1, i_2, \dots, i_k)$ . Evaluating at  $(\vec{e}_{i_1}, \vec{e}_{i_2}, \dots, \vec{e}_{i_k})$  gives  $a_I = 0$  because

$$d\vec{x}_I(\vec{e}_{i_1}, \vec{e}_{i_2}, \dots, \vec{e}_{i_k}) = 1$$

and all other basis elements vanish on this tuple. Thus, every coefficient  $a_I$  is zero. Therefore, the increasing  $d\vec{x}_I$  form a basis.

Finally, an increasing  $k$ -tuple corresponds exactly to a choice of  $k$  distinct indices from  $(1, 2, \dots, n)$ . There are  $\binom{n}{k}$  such choices. Hence,

$$\dim(\mathcal{A}^k(\mathbb{R}^n)^*) = \binom{n}{k}$$

□

**Corollary 1.1.26.** *Every  $T \in \mathcal{A}^k(\mathbb{R}^n)^*$  can be written uniquely in the form*

$$T = \sum_{I \text{ increasing}} a_I d\vec{x}_I$$

where  $a_I = T(\vec{e}_{i_1}, \vec{e}_{i_2}, \dots, \vec{e}_{i_k})$  for  $I = (i_1, i_2, \dots, i_k)$ .

**Remark 1.1.27.** The dimensions of the spaces of alternating  $k$ -forms in  $\mathbb{R}^3$  are

$$\begin{aligned} \dim(\mathcal{A}^0(\mathbb{R}^3)^*) &= 1, \\ \dim(\mathcal{A}^1(\mathbb{R}^3)^*) &= 3, \\ \dim(\mathcal{A}^2(\mathbb{R}^3)^*) &= 3, \\ \dim(\mathcal{A}^3(\mathbb{R}^3)^*) &= 1, \end{aligned}$$

Thus,

$$\begin{aligned} \mathcal{A}^1(\mathbb{R}^3)^* &= \text{span}\{dx, dy, dz\}, \\ \mathcal{A}^2(\mathbb{R}^3)^* &= \text{span}\{dx \wedge dy, dy \wedge dz, dz \wedge dx\}, \\ \mathcal{A}^3(\mathbb{R}^3)^* &= \text{span}\{dx \wedge dy \wedge dz\}, \end{aligned}$$

The wedge notation will be defined formally below.



**Remark 1.1.28.** Let  $I = (i_1, i_2, \dots, i_k)$  be increasing. For vectors  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in \mathbb{R}^n$ , the quantity  $d\vec{x}_I(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k)$  is the signed  $k$ -dimensional volume of the projection of the parallelepiped spanned by  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k$  onto the coordinate  $k$ -plane  $\text{span}\{\vec{e}_{i_1}, \vec{e}_{i_2}, \dots, \vec{e}_{i_k}\}$ . The sign is determined by the orientation of the ordered coordinate basis  $(\vec{e}_{i_1}, \vec{e}_{i_2}, \dots, \vec{e}_{i_k})$ . Thus,  $d\vec{x}_I$  measures oriented projected volume rather than necessarily the actual  $k$ -dimensional volume of the original parallelepiped.

**Example 1.1.29.** Let  $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$  and  $\vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$  be vectors in  $\mathbb{R}^3$ . Then

$$(dx \wedge dz)(\vec{v}, \vec{w}) = d\vec{x}_{(1,3)}(\vec{v}, \vec{w}) = \det \begin{bmatrix} v_1 & w_1 \\ v_3 & w_3 \end{bmatrix}$$

This is the signed area of the projection of the parallelogram spanned by  $\vec{v}$  and  $\vec{w}$  onto the  $xz$ -plane.

Similarly,

$$(dx \wedge dy)(\vec{v}, \vec{w}) = \det \begin{bmatrix} v_1 & w_1 \\ v_2 & w_2 \end{bmatrix}$$

is the signed area of the projection onto the  $xy$ -plane.

**Remark 1.1.30.** In  $\mathbb{R}^3$ , the 2-form  $dy \wedge dz$  measures the signed area projected onto the  $yz$ -plane, while  $dz \wedge dx$  measures signed area projected onto the  $zx$ -plane, and  $dx \wedge dy$  measures the signed area projected onto the  $xy$ -plane. These forms will later be connected with the three coordinate components of the cross product.

We now introduce the operation that combines alternating multilinear maps.

**Definition 1.1.31.** Let  $I = (i_1, i_2, \dots, i_k)$  and  $J = (j_1, j_2, \dots, j_l)$  be ordered multi-indices. Define

$$d\vec{x}_I \wedge d\vec{x}_J = d\vec{x}_{(I,J)}$$

where

$$(I, J) = (i_1, i_2, \dots, i_k, j_1, j_2, \dots, j_l)$$

is the concatenation of  $I$  and  $J$ .

Thus, the *wedge product* combines the indices of the two coordinate forms.

For example,

$$d\vec{x}_{(1,2)} \wedge d\vec{x}_{(3)} = d\vec{x}_{(1,2,3)}$$

**Definition 1.1.32.** Suppose

$$\omega = \sum_I a_I d\vec{x}_I$$

is an alternating  $k$ -form, and

$$\eta = \sum_J b_J d\vec{x}_J$$

is an alternating  $l$ -form. Define

$$\omega \wedge \eta = \sum_I \sum_J a_I b_J d\vec{x}_{(I,J)}$$

The form  $\omega \wedge \eta$  is an alternating  $(k+l)$ -form.

**Example 1.1.33.** We have

$$dx_1 \wedge dx_2 \wedge dx_3 = d\vec{x}_{(1,2,3)}$$

Also,

$$d\vec{x}_{(1,5)} \wedge d\vec{x}_{(4,2)} = d\vec{x}_{(1,5,4,2)}$$

To put the indices into increasing order,

$$(1, 5, 4, 2) \rightarrow (1, 4, 5, 2) \rightarrow (1, 4, 2, 5) \rightarrow (1, 2, 4, 5)$$

This requires three interchanges, so

$$d\vec{x}_{(1,5)} \wedge d\vec{x}_{(4,2)} = -d\vec{x}_{(1,2,4,5)}$$

**Example 1.1.34.** Consider

$$d\vec{x}_{(1,3,2)} \wedge d\vec{x}_{(3,4)}$$

Then

$$d\vec{x}_{(1,3,2)} \wedge d\vec{x}_{(3,4)} = d\vec{x}_{(1,3,2,3,4)}$$

Since the index 3 appears twice,

$$d\vec{x}_{(1,3,2,3,4)} = 0$$

Therefore,

$$d\vec{x}_{(1,3,2)} \wedge d\vec{x}_{(3,4)} = 0$$

**Proposition 1.1.35.** *Let  $\omega \in \mathcal{A}^k(\mathbb{R}^n)^*$ ,  $\eta \in \mathcal{A}^l(\mathbb{R}^n)^*$ , and  $\theta \in \mathcal{A}^m(\mathbb{R}^n)^*$ . Then the wedge product has the following properties.*

1. *Bilinearity:*  $(c\omega + \eta) \wedge \theta = c(\omega \wedge \theta) + \eta \wedge \theta$ .
2. *Skew-Commutativity:*  $\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega$ .
3. *Associativity:*  $(\omega \wedge \eta) \wedge \theta = \omega \wedge (\eta \wedge \theta)$ .

*Proof.* The bilinearity and associativity follow directly from the definition and the distributive law for finite sums.

To prove skew-commutativity, it is enough to consider basis forms  $d\vec{x}_I$  and  $d\vec{x}_J$ , where  $I$  has length  $k$  and  $J$  has length  $l$ . We have

$$d\vec{x}_I \wedge d\vec{x}_J = d\vec{x}_{(I,J)}$$

To reorder  $(I, J)$  into  $(J, I)$ , each of the  $l$  indices in  $J$  must pass across all  $k$  indices in  $I$ . Thus, the total number of interchanges is  $kl$ . Therefore,

$$d\vec{x}_{(I,J)} = (-1)^{kl} d\vec{x}_{(J,I)}$$

Therefore,

$$d\vec{x}_I \wedge d\vec{x}_J = (-1)^{kl} d\vec{x}_J \wedge d\vec{x}_I$$

By bilinearity, the identity holds for all  $\omega$  and  $\eta$ . □

**Corollary 1.1.36.** *If  $\alpha$  and  $\beta$  are 1-forms, then*

$$\alpha \wedge \beta = -\beta \wedge \alpha$$

*In particular,*

$$\alpha \wedge \alpha = 0$$

*Proof.* Since  $\alpha$  and  $\beta$  have degree 1,

$$\alpha \wedge \beta = (-1)^1 \beta \wedge \alpha = -\beta \wedge \alpha$$

Taking  $\beta = \alpha$  gives

$$\alpha \wedge \alpha = -\alpha \wedge \alpha$$

Therefore,

$$2\alpha \wedge \alpha = 0$$

and since we work over  $\mathbb{R}$ ,

$$\alpha \wedge \alpha = 0$$

□

**Corollary 1.1.37.** *If  $\omega \in \mathcal{A}^k(\mathbb{R}^n)^*$  and  $k$  is odd, then*

$$\omega \wedge \omega = 0$$

*Proof.* Since  $k$  is odd,

$$(-1)^{k^2} = -1$$

Thus,

$$\omega \wedge \omega = (-1)^{k^2} \omega \wedge \omega = -\omega \wedge \omega$$

Hence,  $\omega \wedge \omega = 0$ . □

**Remark 1.1.38.** The conclusion of the previous corollary need not hold when  $k$  is even. For example,

$$(dx \wedge dy) \wedge (dx \wedge dy) = 0$$

because of repeated factors, but in  $\mathbb{R}^4$ ,

$$(dx_1 \wedge dx_2 + dx_3 \wedge dx_4) \wedge (dx_1 \wedge dx_2 + dx_3 \wedge dx_4)$$

is nonzero. In fact,

$$(dx_1 \wedge dx_2 + dx_3 \wedge dx_4) \wedge (dx_1 \wedge dx_2 + dx_3 \wedge dx_4) = 2dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4$$

The sign rule depends on the degrees of the form.

**Corollary 1.1.39.** *If  $\omega \in \mathcal{A}^k(\mathbb{R}^n)^*$  with  $k > n$ , then  $\omega = 0$ .*

*Proof.* Any  $k$ -tuple of indices chosen from  $(1, 2, \dots, n)$  must contain a repeated index when  $k > n$ . Therefore, every coordinate form  $d\vec{x}_I$  of degree  $k$  is 0. Hence, every alternating  $k$ -form is 0. □

**Example 1.1.40.** Let  $\omega = a_1 dx + a_2 dy$  and  $\eta = b_1 dx + b_2 dy$ . Then

$$\begin{aligned} \omega \wedge \eta &= (a_1 dx + a_2 dy) \wedge (b_1 dx + b_2 dy) \\ &= a_1 b_1 dx \wedge dx + a_1 b_2 dx \wedge dy + a_2 b_1 dy \wedge dx + a_2 b_2 dy \wedge dy \\ &= a_1 b_2 dx \wedge dy + a_2 b_1 dy \wedge dx \\ &= a_1 b_2 dx \wedge dy - a_2 b_1 dx \wedge dy \\ &= (a_1 b_2 - a_2 b_1) dx \wedge dy \end{aligned}$$

The coefficient is the determinant

$$a_1 b_2 - a_2 b_1 = \det \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix}$$

Thus,

$$\omega \wedge \eta = \det \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix} dx \wedge dy$$

**Example 1.1.41.** Let  $\alpha = 2dx - dy + 3dz$  and  $\beta = dx + 4dy - dz$ . Then

$$\begin{aligned}\alpha \wedge \beta &= (2dx - dy + 3dz) \wedge (dx + 4dy - dz) \\ &= 2dx \wedge dx + 8dx \wedge dy - 2dx \wedge dz - dy \wedge dx - 4dy \wedge dy + dy \wedge dz \\ &\quad + 3dz \wedge dx + 12dz \wedge dy - 3dz \wedge dz \\ &= 9dx \wedge dy - 5dx \wedge dz - 11dy \wedge dz\end{aligned}$$

**Example 1.1.42.** Let  $\alpha = 3dx - 2dy + 5dz$ . Then  $\alpha \wedge \alpha = 0$ . This follows immediately from Corollary 1.1.36. It can also be verified directly: every term appears with an equal and opposite partner, while the terms involving repeated factors vanish.

**Proposition 1.1.43.** Suppose  $\omega = a_1dx + a_2dy$  and  $\eta = b_1dx + b_2dy$ . Then

$$\omega \wedge \eta = \det \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix} dx \wedge dy$$

*Proof.* This is exactly the computation of Example 1.1.40. The wedge product is built from the same alternating behaviour that defines determinants.  $\square$

**Remark 1.1.44.** The appearance of determinants is not accidental. Differential forms are designed so that the wedge product records oriented areas, volumes, and their higher-dimensional analogues.

For example, if  $\omega = a_1dx + a_2dy$  and  $\eta = b_1dx + b_2dy$ , then the coefficient  $a_1b_2 - a_2b_1$  is the signed area of the parallelogram spanned by  $(a_1, a_2)$  and  $(b_1, b_2)$ .

This determinant-based structure is one of the reason differential forms are the correct objects to integrate over oriented geometric objects.

## 1.2 Differential Forms on $\mathbb{R}^n$ and the Exterior Derivative

In Section 1.1, we constructed alternating multilinear maps from the coordinate covectors  $dx_1, dx_2, \dots, dx_n$ . These constant-coefficient forms measure signed projected lengths, areas, volumes, and higher-dimensional analogues.

We now allow the coefficients of these forms to vary smoothly from point to point. This produces the differential forms that will be integrated over curves, surfaces, and manifolds. Most importantly, differential forms can be differentiated by a natural operator called the exterior derivative:

$$d : \mathcal{A}^k(\mathbb{R}^n) \rightarrow \mathcal{A}^{k+1}(\mathbb{R}^n)$$

The exterior derivative will eventually unify the gradient, curl, and divergence operators of vector calculus.

**Definition 1.2.1.** Let  $U \subseteq \mathbb{R}^n$  be an open set. A *differential 0-form* on  $U$  is a smooth function  $f : U \rightarrow \mathbb{R}$ . Thus,  $\mathcal{F}^0(U) = \mathcal{C}^\infty(U)$  the vector space of smooth real-valued functions on  $U$ .

**Remark 1.2.2.** The terminology may initially seem artificial: why we call a smooth function a 0-form?

The answer is that a smooth function behaves naturally as the degree 0 member of a sequence of objects: 0-forms, 1-forms, 2-forms, ...,  $n$ -forms. The exterior derivative will send a 0-form to a 1-form, a 1-form to a 2-form, and so forth.

**Definition 1.2.3.** Let  $U \subseteq \mathbb{R}^n$  be an open set, and let  $0 \leq k \leq n$ . A *differential  $k$ -form* on  $U$  is an expression of the form

$$\omega = \sum_{I \text{ increasing}} f_I(\vec{x}) d\vec{x}_I$$

where  $I = (i_1, i_2, \dots, i_k)$  ranges over all increasing  $k$ -tuples, and each coefficient  $f_I : U \rightarrow \mathbb{R}$  is smooth, and

$$d\vec{x}_I = dx_{i_1} \wedge d\vec{x}_{i_2} \wedge \cdots \wedge d\vec{x}_{i_k}$$

The vector space of smooth differential  $k$ -forms on  $U$  is denoted by  $\mathcal{F}^k(U)$ . When  $U = \mathbb{R}^n$ , we simply write  $\mathcal{F}^k(\mathbb{R}^n)$ .

**Remark 1.2.4.** A  $k$ -form is not merely an alternating  $k$ -linear map. Rather, a differential  $k$ -form assigns an alternating  $k$ -linear map to every point of  $U$ . More precisely, if

$$\omega = \sum_{I \text{ increasing}} f_I(\vec{x}) d\vec{x}_I$$

then at each point  $a \in U$ ,

$$\omega(\vec{a}) = \sum_{I \text{ increasing}} f_I(\vec{a}) d\vec{x}_I$$

is an alternating  $k$ -linear map on  $\mathbb{R}^n$ . Thus, the coefficients of a differential form vary smoothly with position.

**Definition 1.2.5.** A *differential 1-form* on  $U \subseteq \mathbb{R}^n$  has the form

$$\omega = f_1(\vec{x})dx_1 + f_2(\vec{x})dx_2 + \cdots + f_n(\vec{x})dx_n$$

where each  $f_i$  is smooth on  $U$ . In  $\mathbb{R}^3$ , we usually write

$$\omega = Pdx + Qdy + Rdz$$

**Definition 1.2.6.** A *differential 2-form* on an open set  $U \subseteq \mathbb{R}^3$  has the form

$$\omega = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$$

where  $P$ ,  $Q$ , and  $R$  are smooth functions on  $U$ .

This particular ordering is useful because these three coordinate 2-forms later correspond naturally to the three coordinate directions in  $\mathbb{R}^3$ .

**Definition 1.2.7.** An  *$n$ -form* on  $U \subseteq \mathbb{R}^n$  has the form

$$\omega = f(\vec{x})dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n$$

for some smooth function  $f$ . The standard  $n$ -form

$$dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n$$

is called the *standard volume form* on  $\mathbb{R}^n$ .

**Remark 1.2.8.** Since there are no nonzero alternating  $k$ -forms on  $\mathbb{R}^n$  when  $k > n$ , we write  $\mathcal{F}^k(U) = \{\vec{0}\}$  whenever  $k > n$ . For example, every 4-form on  $\mathbb{R}^3$  is  $\vec{0}$ .

The algebraic operations from Section 1.1 extend naturally to differential forms whose coefficients are smooth functions.

**Definition 1.2.9.** Let  $\omega, \eta \in \mathcal{F}^k(U)$  and let  $f \in \mathcal{F}^0(U)$  be a smooth function. If  $\omega = \sum_I a_I d\vec{x}_I$  and  $\eta = \sum_I b_I d\vec{x}_I$ , then define *addition* by

$$\omega + \eta = \sum_I (a_I + b_I) d\vec{x}_I$$

and *scalar multiplication* by

$$f\omega = \sum_I (fa_I) d\vec{x}_I$$

Thus,  $\mathcal{F}^k(U)$  is naturally a vector space over  $\mathbb{R}$  and more strongly, a module over the ring of smooth functions on  $U$ .

**Definition 1.2.10.** Let  $\omega \in \mathcal{F}^k(U)$  and  $\eta \in \mathcal{F}^l(U)$ . If  $\omega = \sum_I a_I d\vec{x}_I$  and  $\eta = \sum_J b_J d\vec{x}_J$ , define

$$\omega \wedge \eta = \sum_I \sum_J a_I b_J d\vec{x}_I \wedge d\vec{x}_J$$

Then  $\omega \wedge \eta \in \mathcal{F}^{k+l}(U)$

**Proposition 1.2.11.** Let  $\omega \in \mathcal{F}^k(U)$ ,  $\eta \in \mathcal{F}^l(U)$ , and  $\theta \in \mathcal{F}^m(U)$ . Then the following hold.

1. If  $k = l$ , then  $\omega + \eta = \eta + \omega$  and  $(\omega + \eta) + \theta = \omega + (\eta + \theta)$
2. The wedge product is graded skew-commutative:  $\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega$
3. The wedge product is associative:  $(\omega \wedge \eta) \wedge \theta = \omega \wedge (\eta \wedge \theta)$ .
4. The wedge product distributes over addition:  $(\omega \wedge \eta) \wedge \theta = \omega \wedge \theta + \eta \wedge \theta$ .
5. For every smooth function  $f$ ,  $(f\omega) \wedge \eta = f(\omega \wedge \eta) = \omega \wedge (f\eta)$

*Proof.* All of these properties follow from the corresponding properties of addition and scalar multiplication of smooth functions, together with the bilinearity, associativity, and graded skew-commutativity of the wedge product of constant-coefficient forms.  $\square$

**Example 1.2.12.** Let  $\omega = (x + y)dx + (x - y)dy$  and  $\eta = (2x)dx + (3y)dy$ . Then

$$\begin{aligned} \omega \wedge \eta &= ((x + y)dx + (x - y)dy) \wedge ((2x)dx + (3y)dy) \\ &= 3y(x + y)dx \wedge dx + 2x(x - y)dy \wedge dx \end{aligned}$$

Since  $dy \wedge dx = -dx \wedge dy$ , we obtain

$$\omega \wedge \eta = (3xy + 3y^2 - 2x^2 + 2xy)dx \wedge dy = (-2x^2 + 5xy + 3y^2)dx \wedge dy$$

**Example 1.2.13.** Let  $\omega = Pdx + Qdy + Rdz$  be a 1-form on  $\mathbb{R}^3$ . Then  $\omega \wedge \omega = 0$ . indeed, every 1-form has odd degree, so

$$\omega \wedge \omega = (-1)^1 \omega \wedge \omega = -\omega \wedge \omega$$

Therefore,  $\omega \wedge \omega = 0$ .

**Remark 1.2.14.** A common mistake is to think that  $\omega \wedge \eta = -\eta \wedge \omega$  for all differential forms. This is true only when both forms have odd degree. In general,

$$\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega$$



For example, if  $\omega$  is a 2-form and  $\eta$  is a 1-form, then

$$\omega \wedge \eta = (-1)^2 \eta \wedge \omega = \eta \wedge \omega$$

Thus, a 2-form commutes with a 1-form under the wedge product.

The exterior derivative is the fundamental differentiation operator for differential forms.

**Definition 1.2.15.** Let  $f \in \mathcal{F}^0(U)$  be a smooth function. The *exterior derivative* of  $f$  is the 1-form

$$df = \sum_{j=1}^n \frac{\partial f}{\partial x_j} dx_j = \sum_{j=1}^n f_{x_j} dx_j$$

Thus,

$$df = f_{x_1} dx_1 + f_{x_2} dx_2 + \cdots + f_{x_n} dx_n$$

In  $\mathbb{R}^3$ ,

$$df = f_x dx + f_y dy + f_z dz$$

**Remark 1.2.16.** At each point  $\vec{a} \in U$ , the 1-form  $df(\vec{a})$  is exactly the derivative of  $f$  at  $\vec{a}$

$$df(\vec{a})(\vec{v}) = Df(\vec{a})(\vec{v})$$

Thus, if  $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ , then

$$df(\vec{a})(\vec{v}) = \sum_{j=1}^n f_{x_j}(\vec{a}) v_j$$

**Example 1.2.17.** Let  $f(x, y, z) = x^2 y + e^{yz} + \sin(z)$ . Then

$$f_x = 2xy, \quad f_y = x^2 + ze^{yz}, \quad f_z = ye^{yz} + \cos(z)$$

Therefore,

$$df = 2xydx + (x^2 + ze^{yz})dy + (ye^{yz} + \cos(z))dz$$

**Definition 1.2.18.** Let

$$\omega = \sum_{I \text{ increasing}} f_I(\vec{x}) d\vec{x}_I$$

be a differential  $k$ -form on  $U$ . The *exterior derivative* of  $\omega$  is the  $(k+1)$ -form

$$d\omega = \sum_{I \text{ increasing}} df_I \wedge d\vec{x}_I = \sum_{I \text{ increasing}} (f_I)_{x_j} dx_j \wedge d\vec{x}_I$$

Terms containing repeated differential factors automatically vanish.

**Remark 1.2.19.** The exterior derivative increases degree by one

$$d : \mathcal{F}^k(U) \rightarrow \mathcal{F}^{k+1}(U)$$

In particular,

$$\begin{aligned} d : \mathcal{F}^0(U) &\rightarrow \mathcal{F}^1(U), \\ d : \mathcal{F}^1(U) &\rightarrow \mathcal{F}^2(U), \\ d : \mathcal{F}^{n-1}(U) &\rightarrow \mathcal{F}^n(U), \end{aligned}$$

Since every  $(n+1)$ -form on  $\mathbb{R}^n$  is zero,  $d\omega = 0$  for every  $n$ -form  $\omega$ .

**Example 1.2.20.** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be smooth. Then  $df = f'(x)dx$ . Thus, in one-dimension, the exterior derivative of a function is the usual derivative multiplied by  $dx$ .

**Example 1.2.21.** Let  $\omega = ydx + xdy$ . Then

$$d\omega = d(y) \wedge dx + d(x) \wedge dy = dy \wedge dx + dx \wedge dy = 0$$

In fact,  $\omega = d(xy)$ .

**Example 1.2.22.** Let  $\omega = -ydx + xdy$ . Then

$$d\omega = d(-y) \wedge dx + d(x) \wedge dy = -dy \wedge dx + dx \wedge dy = 2dx \wedge dy$$

**Example 1.2.23.** Consider the 1-form

$$\omega = -\frac{y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$$

on  $\mathbb{R}^2 \setminus \{\vec{0}\}$ . Write  $\omega = Pdx + Qdy$ , where

$$P = -\frac{y}{x^2 + y^2}, \quad Q = \frac{x}{x^2 + y^2}$$

Then

$$d\omega = (Q_x - P_y)dx \wedge dy$$

We compute

$$Q_x = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad P_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

Therefore,

$$d\omega = 0$$

Later, we will see that although  $\omega$  is closed, it is not the exterior derivative of a globally defined function on the punctured plane. This will be one of our first examples in which topology affects calculus.

**Example 1.2.24.** Let  $\omega = x_1 dx_2 + x_3 dx_4 + x_5 dx_6$ . Then

$$d\omega = dx_1 \wedge dx_2 + dx_3 \wedge dx_4 + dx_5 \wedge dx_6$$

**Example 1.2.25.** Let

$$\omega = (x^2 + e^{yz})dy \wedge dz + (y^2 + \sin(x^3 z))dz \wedge dx + (z^2 + \tan^{-1}(x^2 + y^2))dx \wedge dy$$

Denoting  $P = x^2 + e^{yz}$ ,  $Q = y^2 + \sin(x^3 z)$ , and  $R = z^2 + \tan^{-1}(x^2 + y^2)$ , we compute each term separately. First

$$d(Pdy \wedge dz) = d(x^2 + e^{yz}) \wedge dy \wedge dz$$

Only the  $dx$  term survives, since terms involving  $dy$  or  $dz$  produce repeated wedge factors. Hence,

$$d(Pdy \wedge dz) = 2x dx \wedge dy \wedge dz$$

Similarly,

$$d(Qdz \wedge dx) = 2y dy \wedge dz \wedge dx = 2y dx \wedge dy \wedge dz$$

Finally,

$$d(Rdx \wedge dy) = 2z dz \wedge dx \wedge dy = 2z dx \wedge dy \wedge dz$$

Therefore,

$$d\omega = 2(x + y + z)dx \wedge dy \wedge dz$$

**Proposition 1.2.26.** *Let  $\omega = P(x, y)dx + Q(x, y)dy$ . Then*

$$d\omega = (Q_x - P_y)dx \wedge dy$$

*Proof.* Using the definition of the exterior derivative

$$d\omega = dP \wedge dx + dQ \wedge dy$$

Now

$$dP = P_x dx + P_y dy, \quad dQ = Q_x dx + Q_y dy$$

and so

$$\begin{aligned} dP \wedge dx &= P_x dx \wedge dx + P_y dy \wedge dx = -P_y dx \wedge dy, \\ dQ \wedge dy &= Q_x dx \wedge dy + Q_y dy \wedge dy = Q_x dx \wedge dy \end{aligned}$$

Therefore,

$$d\omega = (Q_x - P_y)dx \wedge dy$$

□

**Remark 1.2.27.** The coefficient  $Q_x - P_y$  is the scalar curl of the planar vector field  $\vec{F} = \begin{bmatrix} P \\ Q \end{bmatrix}$ . Thus, the exterior derivative of a 1-form in  $\mathbb{R}^2$  measures the infinitesimal circulation of the corresponding vector field.

**Proposition 1.2.28.** *Let  $\omega = Pdx + Qdy + Rdz$ . Then*

$$d\omega = (R_y - Q_z)dy \wedge dz + (P_z - R_x)dz \wedge dx + (Q_x - P_y)dx \wedge dy$$

*Proof.* We have

$$d\omega = dP \wedge dx + dQ \wedge dy + dR \wedge dz$$

expanding each, we have

$$\begin{aligned} dP \wedge dx &= P_y dy \wedge dx + P_z dz \wedge dx, \\ dQ \wedge dy &= Q_x dx \wedge dy + Q_z dz \wedge dy, \\ dR \wedge dz &= R_x dx \wedge dz + R_y dy \wedge dz \end{aligned}$$

Now reorder each wedge product into the basis  $dy \wedge dz$ ,  $dz \wedge dx$ , and  $dx \wedge dy$ . Thus, using skew-commutativity, we obtain

$$d\omega = (R_y - Q_z)dy \wedge dz + (P_z - R_x)dz \wedge dx + (Q_x - P_y)dx \wedge dy$$

□

**Remark 1.2.29.** If the vector field  $\vec{F} = \begin{bmatrix} P \\ Q \\ R \end{bmatrix}$  corresponds to the 1-form  $\omega = Pdx + Qdy + Rdz$ , then the components appearing in  $d\omega$  are precisely the components of  $\nabla \times \vec{F}$ . Thus, the exterior derivative of a 1-form is the differential-form version of curl.

**Proposition 1.2.30.** Let  $\omega = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$ . Then

$$d\omega = (P_x + Q_y + R_z)dx \wedge dy \wedge dz$$

*Proof.* Using the definition,

$$d\omega = dP \wedge dy \wedge dz + dQ \wedge dz \wedge dx + dR \wedge dx \wedge dy$$

Only one coordinate derivative survives in each term

$$\begin{aligned} dP \wedge dy \wedge dz &= P_x dx \wedge dy \wedge dz, \\ dQ \wedge dz \wedge dx &= Q_y dy \wedge dz \wedge dx, \\ dR \wedge dx \wedge dy &= R_z dz \wedge dx \wedge dy \end{aligned}$$

Since

$$dx \wedge dy \wedge dz = dy \wedge dz \wedge dx = dz \wedge dx \wedge dy$$

we conclude that

$$d\omega = (P_x + Q_y + R_z)dx \wedge dy \wedge dz$$

□

**Remark 1.2.31.** The coefficient  $P_x + Q_y + R_z$  is the divergence of the vector field  $\vec{F} = \begin{bmatrix} P \\ Q \\ R \end{bmatrix}$ . Thus, exterior differentiation of a 2-form in  $\mathbb{R}^3$  corresponds to divergence. This connection will be developed systematically in Chapter 5.

**Proposition 1.2.32.** The exterior derivative  $d : \mathcal{F}^k(U) \rightarrow \mathcal{F}^{k+1}(U)$  is linear. That is, if  $\omega, \eta \in \mathcal{F}^k(U)$  and  $c \in \mathbb{R}$ , then

$$d(c\omega + \eta) = cd\omega + d\eta$$

*Proof.* Write

$$\omega = \sum_I f_I d\vec{x}_I, \quad \eta = \sum_I g_I d\vec{x}_I$$

Then

$$c\omega + \eta = \sum_I (cf_I + g_I) d\vec{x}_I$$

Therefore,

$$\begin{aligned} d(c\omega + \eta) &= \sum_I d(cf_I + g_I) \wedge d\vec{x}_I \\ &= \sum_I (cdf_I + dg_I) \wedge d\vec{x}_I \\ &= cd\omega + d\eta \end{aligned}$$

□

**Proposition 1.2.33.** *Let  $f \in \mathcal{F}^0(U)$  be smooth and  $\omega \in \mathcal{F}^k(U)$ . Then*

$$d(f\omega) = df \wedge \omega + f d\omega$$

*Proof.* Write

$$\omega = \sum_I g_I d\vec{x}_I$$

Then

$$f\omega = \sum_I (fg_I) d\vec{x}_I$$

Therefore,

$$\begin{aligned} d(f\omega) &= \sum_I d(fg_I) \wedge d\vec{x}_I \\ &= \sum_I (g_I df + f dg_I) \wedge d\vec{x}_I \\ &= df \wedge \sum_I g_I d\vec{x}_I + f \sum_I dg_I \wedge d\vec{x}_I \\ &= df \wedge \omega + f d\omega \end{aligned}$$

□

**Proposition 1.2.34.** *Let  $\omega \in \mathcal{F}^k(U)$  and  $\eta \in \mathcal{F}^l(U)$ . Then*

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$$

*Proof.* It suffices to prove the result when  $\omega = f d\vec{x}_I$  and  $\eta = g d\vec{x}_J$ , where  $d\vec{x}_I$  has degree  $k$ . Then

$$\omega \wedge \eta = f g d\vec{x}_I \wedge d\vec{x}_J$$

Therefore,

$$\begin{aligned} d(\omega \wedge \eta) &= d(fg) \wedge d\vec{x}_I \wedge d\vec{x}_J \\ &= (gdf + fdg) \wedge d\vec{x}_I \wedge d\vec{x}_J \end{aligned}$$

The first term is

$$gdf \wedge d\vec{x}_I \wedge d\vec{x}_J = d\omega \wedge \eta$$

For the second term,  $dg$  is a 1-form and must pass through the  $k$  factors in  $d\vec{x}_I$ . Thus,

$$fdg \wedge d\vec{x}_I \wedge d\vec{x}_J = (-1)^k f d\vec{x}_I \wedge dg \wedge d\vec{x}_J$$

Hence,

$$fdg \wedge d\vec{x}_I \wedge d\vec{x}_J = (-1)^k \omega \wedge d\eta$$

Therefore,

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$$

□

**Example 1.2.35.** Let  $f(x, y) = x^2 + y$  and let  $\omega = xdy$ . Then

$$f\omega = x(x^2 + y)dy$$

We can differentiate directly

$$\begin{aligned} d(f\omega) &= d(x^3 + xy) \wedge dy \\ &= (3x^2 + y)dx \wedge dy + xdy \wedge dy \\ &= (3x^2 + y)dx \wedge dy \end{aligned}$$

Now use the product rule. We have

$$df = 2xdx + dy, \quad d\omega = d(x) \wedge dy = dx \wedge dy$$

Therefore,

$$\begin{aligned} df \wedge \omega + f d\omega &= (2xdx + dy) \wedge xdy + (x^2 + y)dx \wedge dy \\ &= 2x^2 dx \wedge dy + (x^2 + y)dx \wedge dy \\ &= (3x^2 + y)dx \wedge dy \end{aligned}$$

This agrees with the direct computation.

**Theorem 1.2.36.** For every differential form  $\omega \in \mathcal{F}^k(U)$ , we have  $d(d\omega) = 0$ . Equivalently,  $d^2 = 0$ .

*Proof.* It suffices to consider a form of the form  $\omega = f d\vec{x}_I$ , where  $I = (i_1, i_2, \dots, i_k)$ . Then

$$d\omega = df \wedge d\vec{x}_I = \sum_{j=1}^n f_{x_j} dx_j \wedge d\vec{x}_I$$

Applying  $d$  gives again

$$d(d\omega) = \sum_{r=1}^n \sum_{j=1}^n f_{x_j x_r} dx_r \wedge dx_j \wedge d\vec{x}_I$$

The terms with  $r = j$  vanish and for each pair  $r \neq j$ , the term involving  $f_{x_j x_r} dx_r \wedge dx_j \wedge d\vec{x}_I$  is paired with the term involving  $f_{x_r x_j} dx_j \wedge dx_r \wedge d\vec{x}_I$ . Since mixed partial derivatives agree,

$$f_{x_j x_r} = f_{x_r x_j}$$

while  $dx_j \wedge dx_r = -dx_r \wedge dx_j$  the two terms cancel. Therefore,  $d(d\omega) = 0$ .  $\square$

**Definition 1.2.37.** A differential  $k$ -form  $\omega$  is called *exact* if there exists a differential  $(k-1)$ -form  $\eta$  such that  $\omega = d\eta$ .

**Definition 1.2.38.** A differential  $k$ -form  $\omega$  is called *closed* if  $d\omega = 0$ .

**Corollary 1.2.39.** Every exact form is closed.

*Proof.* Suppose  $\omega = d\eta$ . Then

$$d\omega = d(d\eta) = 0$$

Therefore,  $\omega$  is closed.  $\square$

**Remark 1.2.40.** The converse need not be true. That is, a closed form need not be exact. We already saw an example

$$\omega = -\frac{y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$$

is closed on  $\mathbb{R}^2 \setminus \{\vec{0}\}$ , but it is not exact on that domain.

The failure of a closed form to be exact is often caused by the topology of the region on which the form is defined. This idea will become increasingly important in Chapters 2 and 6.



**Corollary 1.2.41.** *Let  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$  be a  $\mathcal{C}^2$  function. Then*

$$\nabla \times (\nabla f) = \vec{0}$$

*Proof.* The 1-form associated with  $\nabla f$  is  $df$ . By Theorem 1.2.36,  $d(df) = 0$ . By Proposition 1.2.28, the exterior derivative of a 1-form in  $\mathbb{R}^3$  corresponds to the curl. Therefore,

$$\nabla \times (\nabla f) = \vec{0}$$

□

**Corollary 1.2.42.** *Let  $\vec{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be a  $\mathcal{C}^2$  vector field. Then*

$$\nabla \cdot (\nabla \times \vec{F}) = 0$$

*Proof.* Let  $\omega = Pdx + Qdy + Rdz$  be the 1-form corresponding to  $\vec{F} = \begin{bmatrix} P \\ Q \\ R \end{bmatrix}$ .

The 2-form  $d\omega$  corresponds to  $\nabla \times \vec{F}$ . By Theorem 1.2.36,  $d(d\omega) = 0$ . By Proposition 1.2.30, exterior differentiation of a 2-form corresponds to divergence. Hence,

$$\nabla \cdot (\nabla \times \vec{F}) = 0$$

□

### 1.3 Pullback of Differential Forms

In one-variable calculus, substitution is a familiar procedure. If  $x = g(u)$ , then  $dx = g'(u)du$ , and an integral involving  $x$  may be rewritten in terms of the parameter  $u$ . For example,

$$\int_a^b f(g(u))g'(u)du = \int_{g(a)}^{g(b)} f(x)dx$$

The pullback is the differential-form version of substitution. Given a smooth map  $\vec{g} : U \rightarrow V$ , the pullback transfers differential forms on the target space  $V$  back to differential forms on the domain  $U$ . It tells us how a form changes when expressed in the coordinates or parameters used to describe  $\vec{g}$ .

The pullback will be the essential tool for integrating forms over parametrized curves, surfaces, and higher-dimensional surfaces.

**Definition 1.3.1.** Let  $U \subseteq \mathbb{R}^m$  be open. Recall that  $\mathcal{A}^k(\mathbb{R}^m)^*$  denotes the finite-dimensional vector space of alternating  $k$ -covectors on  $\mathbb{R}^m$ .

We denote the vector space of smooth differential  $k$ -forms on  $U$  by  $\mathcal{F}^k(U)$ . In particular, an element  $\omega \in \mathcal{F}^k(U)$  assigns to every point  $\vec{a} \in U$  an alternating  $k$ -covector  $\omega(\vec{a}) \in \mathcal{A}^k(\mathbb{R}^m)^*$ .

**Remark 1.3.2.** Suppose  $U \subseteq \mathbb{R}^m$  and  $V \subseteq \mathbb{R}^n$  are open sets. Then the forms in  $\mathcal{F}^k(U)$  take values in  $\mathcal{A}^k(\mathbb{R}^m)^*$ , whereas the forms in  $\mathcal{F}^k(V)$  take values in  $\mathcal{A}^k(\mathbb{R}^n)^*$ . This distinction is important because a smooth map  $\vec{g} : U \rightarrow V$  may map between spaces of different dimensions.

**Definition 1.3.3.** Let  $U \subseteq \mathbb{R}^m$  have coordinates  $(u_1, u_2, \dots, u_m)$  and let  $V \subseteq \mathbb{R}^n$  have coordinates  $(x_1, x_2, \dots, x_n)$ . Suppose  $\vec{g} : U \rightarrow V$  is smooth. We write

$$\vec{g}(u_1, u_2, \dots, u_m) = \begin{bmatrix} g_1(u_1, u_2, \dots, u_m) \\ g_2(u_1, u_2, \dots, u_m) \\ \vdots \\ g_n(u_1, u_2, \dots, u_m) \end{bmatrix}$$

Thus,  $g_i = x_i \circ \vec{g}$  is the  $i$ th coordinate function of  $\vec{g}$ .

**Definition 1.3.4.** Let  $f \in \mathcal{F}^0(V)$ . The *pullback* of  $f$  by  $\vec{g}$  is the function  $\vec{g}^*f \in \mathcal{F}^0(U)$  defined by

$$\vec{g}^*f = f \circ \vec{g}$$

Thus,

$$(\vec{g}^*f)(\vec{u}) = f(\vec{g}(\vec{u}))$$

**Example 1.3.5.** Let  $\vec{g} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be given by

$$\vec{g}(u, v) = (u^2 - v, uv)$$

Let  $f(x, y) = x^2 + y$ . Then

$$\vec{g}^*f = f \circ \vec{g} = f(u^2 - v, uv) = (u^2 - v)^2 + uv$$

**Definition 1.3.6.** Let  $U \subseteq \mathbb{R}^m$  and  $V \subseteq \mathbb{R}^n$  be open sets, and let  $\vec{g} : U \rightarrow V$  be smooth. The *pullback* of the coordinate 1-form  $dx_i$  is defined by

$$\vec{g}^*(dx_i) = dg_i$$

Since  $g_i : U \rightarrow \mathbb{R}$  is a smooth function, its exterior derivative is

$$dg_i = \sum_{j=1}^m (g_i)_j du_j$$

Therefore,

$$\bar{g}^*(dx_i) = \sum_{j=1}^m (g_i)_j du_j$$

**Remark 1.3.7.** The equation  $\bar{g}^*(dx_i) = dg_i$  is the rigorous differential-form interpretation of the informal substitution rule

$$dx_i = \sum_{j=1}^m (x_i)_j du_j$$

In particular, when  $x = g(u)$  is a one-variable substitution, we recover the familiar formula  $dx = g'(u)du$ .

**Example 1.3.8.** Let  $g : \mathbb{R} \rightarrow \mathbb{R}$  be given by  $g(u) = u^3 - 2u$ . Then  $g_1(u) = u^3 - 2u$  so

$$g^*(dx) = dg_1 = (3u^2 - 2)du$$

**Example 1.3.9.** Let  $g : \mathbb{R} \rightarrow \mathbb{R}^2$  be given by  $g(t) = (t^2, \sin(t))$ . Then  $g_1(t) = t^2$  and  $g_2(t) = \sin(t)$ . Hence,

$$g^*(dx) = d(t^2) = 2t dt$$

and

$$g^*(dy) = d(\sin(t)) = \cos(t) dt$$

**Definition 1.3.10.** Let  $I = (i_1, i_2, \dots, i_k)$  be an ordered  $k$ -tuple. Define

$$\bar{g}^*(dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_k}) = dg_{i_1} \wedge dg_{i_2} \wedge \dots \wedge dg_{i_k}$$

Equivalently,

$$\bar{g}^*(d\vec{x}_I) = d\vec{g}_I$$

**Definition 1.3.11.** Let  $\omega \in \mathcal{F}^k(V)$  have the coordinate expression

$$\omega = \sum_{I \text{ increasing}} f_I(\vec{x}) d\vec{x}_I$$

The *pullback* of  $\omega$  by  $\bar{g}$  is the  $k$ -form  $\bar{g}^*\omega \in \mathcal{F}^k(U)$  defined by

$$\bar{g}^*\omega = \sum_{I \text{ increasing}} (f_I \circ \bar{g}) d\vec{g}_I$$

Equivalently,

$$\bar{g}^* \left( \sum_{I \text{ increasing}} f_I d\vec{x}_I \right) = \sum_{I \text{ increasing}} (f_I \circ g) dg_{i_1} \wedge dg_{i_2} \wedge \dots \wedge dg_{i_k}$$

**Remark 1.3.12.** The pullback preserves degree:  $\vec{g}^* : \mathcal{F}^k(V) \rightarrow \mathcal{F}^k(U)$ . A  $k$ -form on the target remains a  $k$ -form on the domain. However, its coefficient functions and coordinate covectors are transformed according to the map  $\vec{g}$ .

**Example 1.3.13.** Let  $g : \mathbb{R} \rightarrow \mathbb{R}$  be smooth, and let  $\omega = f(x)dx$ . Then

$$g^*\omega = (f \circ g)g^*(dx) = (f \circ g)g'(u)du$$

Therefore,

$$g^*(f(x)dx) = f(g(u))g'(u)du$$

This is precisely the usual substitution expression from one-variable calculus.

**Example 1.3.14.** Let  $\vec{g} : (0, 2\pi) \rightarrow \mathbb{R}^2$  be the parametrization

$$\vec{g}(t) = (\cos(t), \sin(t))$$

Consider the 1-form

$$\omega = -ydx + xdy$$

We have

$$\vec{g}^*(dx) = d(\cos(t)) = -\sin(t)dt, \quad \vec{g}^*(dy) = d(\sin(t)) = \cos(t)dt$$

Moreover,

$$x \circ \vec{g} = \cos(t), \quad y \circ \vec{g} = \sin(t)$$

Therefore,

$$\begin{aligned} \vec{g}^*\omega &= -(y \circ \vec{g})\vec{g}^*(dx) + (x \circ \vec{g})\vec{g}^*(dy) \\ &= -(\sin(t))(-\sin(t)dt) + (\cos(t))(\cos(t)dt) \\ &= (\sin^2(t) + \cos^2(t))dt \\ &= dt \end{aligned}$$

This calculation will later imply that

$$\int_C -ydx + xdy = 2\pi$$

where  $C$  is the unit circle traversed counterclockwise.

**Example 1.3.15.** Let  $\vec{g} : (0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}^2$  be given by

$$\vec{g}(u, v) = (u \cos(v), u \sin(v))$$

Here,  $x = u \cos(v)$  and  $y = u \sin(v)$ . We compute

$$\begin{aligned}\vec{g}^*(dx) &= d(u \cos(v)) = \cos(v)du - u \sin(v)dv, \\ \vec{g}^*(dy) &= d(u \sin(v)) = \sin(v)du + u \cos(v)dv\end{aligned}$$

Now consider the standard area 2-form  $dx \wedge dy$ . Then

$$\begin{aligned}\vec{G}^*(dx \wedge dy) &= \vec{g}^*(dx) \wedge \vec{g}^*(dy) \\ &= (\cos(v)du - u \sin(v)dv) \wedge (\sin(v)du + u \cos(v)dv)\end{aligned}$$

Expanding

$$\begin{aligned}\vec{g}^*(dx \wedge dy) &= u \cos^2(v)du \wedge dv - u \sin^2(v)dv \wedge du \\ &= u \cos^2(v)du \wedge dv + u \sin^2(v)du \wedge dv \\ &= u du \wedge dv\end{aligned}$$

This is the differential-form origin of the polar-coordinate Jacobian factor  $u$ .

**Example 1.3.16.** Using the polar-coordinate map from Example 1.3.15, let

$$\omega = (x^2 + y^2)dx \wedge dy$$

Then  $x^2 + y^2 = u^2$ , so

$$\vec{g}^*\omega = ((x^2 + y^2) \circ \vec{g})\vec{g}^*(dx \wedge dy) = u^2(udu \wedge dv) = u^3du \wedge dv$$

**Remark 1.3.17.** The coefficient  $u$  in  $\vec{g}^*(dx \wedge dy) = u du \wedge dv$  arises because polar coordinates distort areas. A small rectangle of dimensions  $du$  and  $dv$  in the  $uv$ -plane corresponds approximately to a thin sector in the  $xy$ -plane whose area is  $u du dv$ . The pullback records this scaling factor automatically.

**Example 1.3.18.** Let  $\vec{g}: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  be given by

$$\vec{g}(u, v) = (u \cos(v), u \sin(v), v)$$

Thus,  $x = u \cos(v)$ ,  $y = u \sin(v)$ , and  $z = v$ . We have

$$\begin{aligned}\vec{g}^*(dx) &= \cos(v)du - u \sin(v)dv, \\ \vec{g}^*(dy) &= \sin(v)du + u \cos(v)dv, \\ \vec{g}^*(dz) &= dv\end{aligned}$$

We now compute the pullbacks of the standard 2-forms. First,

$$\vec{g}^*(dx \wedge dy) = u du \wedge dv$$

Next,

$$\begin{aligned}\bar{g}^*(dx \wedge dz) &= \bar{g}^*(dy) \wedge \bar{g}^*(dz) \\ &= (\cos(v)du - u \sin(v)dv) \wedge dv \\ &= \cos(v)du \wedge dv\end{aligned}$$

Similarly,

$$\begin{aligned}\bar{g}^*(dy \wedge dz) &= \bar{g}^*(dy) \wedge \bar{g}^*(dz) \\ &= (\sin(v)du + u \cos(v)dv) \wedge dv \\ &= \sin(v)du \wedge dv\end{aligned}$$

**Example 1.3.19.** Continue with the parametrization from Example 1.3.18,

$$\vec{g}(u, v) = (u \cos(v), u \sin(v), v)$$

Let

$$\omega = (x^2 + y^2)dx \wedge dy + xdx \wedge dz + ydy \wedge dz$$

Since  $x^2 + y^2 = u^2$ ,  $x = u \cos(v)$  and  $y = u \sin(v)$ , we have

$$\begin{aligned}\bar{g}^*\omega &= u^2(udu \wedge dv) + u \cos(v)(\cos(v)du \wedge dv) + u \sin(v)(\sin(v)du \wedge dv) \\ &= (u^3 + u \cos^2(v) + u \sin^2(v))du \wedge dv \\ &= (u^3 + u)du \wedge dv\end{aligned}$$

The pullback of a coordinate  $k$ -form can be expressed using determinants of submatrices of the derivative matrix.

**Theorem 1.3.20.** Let  $U \subseteq \mathbb{R}^m$  and  $V \subseteq \mathbb{R}^n$  be open sets, and let  $\vec{g}: U \rightarrow V$  be smooth. Let  $I = (i_1, i_2, \dots, i_k)$  be an increasing  $k$ -tuple of indices from  $(1, 2, \dots, n)$ . Then

$$\bar{g}^*(dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_k}) = \sum_{J \text{ increasing}} \det[(g_{i_p})_{j_q}]_{p,q=1}^k du_{j_1} \wedge du_{j_2} \wedge \dots \wedge du_{j_k}$$

where the sum is over all increasing  $k$ -tuples  $J = (j_1, j_2, \dots, j_k)$  from  $(1, 2, \dots, m)$ .

*Proof.* By definition,

$$\bar{g}^*(dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_k}) = dg_{i_1} \wedge dg_{i_2} \wedge \dots \wedge dg_{i_k}$$

For each  $1 \leq p \leq k$ ,

$$dg_{i_p} = \sum_{j=1}^m (g_{i_p})_j du_j$$

Therefore,  $dg_{i_1} \wedge dg_{i_2} \wedge \cdots \wedge dg_{i_k}$  is obtained by expanding the wedge product of these sums.

Any term containing a repeated  $du_j$  vanishes. The surviving terms correspond precisely to choosing  $k$  distinct indices  $j_1, j_2, \dots, j_k$ . For a fixed increasing multi-index  $J = (j_1, j_2, \dots, j_k)$ , the coefficient of  $du_{j_1} \wedge du_{j_2} \wedge \cdots \wedge du_{j_k}$  is the determinant  $\det [(g_{i_p})_{j_q}]_{p,q=1}^k$ .

This follows from the alternating signs produced when the  $du_{j_q}$  factors are reordered into increasing order. Therefore,

$$\bar{g}^*(d\vec{x}_I) = \sum_{J \text{ increasing}} \det [(g_{i_p})_{j_q}]_{p,q=1}^k d\vec{u}_J$$

□

**Corollary 1.3.21.** *Let  $U \subseteq \mathbb{R}^m$  and  $V \subseteq \mathbb{R}^n$  be open sets, and let  $\vec{g}: U \rightarrow V$  be smooth. Then*

$$\bar{g}^*(dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n) = \det(D\vec{g}) du_1 \wedge du_2 \wedge \cdots \wedge du_n$$

*Proof.* Apply Theorem 1.3.20 with  $I = (1, 2, \dots, n)$ . Since there is only one increasing  $n$ -tuple chosen from  $\{1, 2, \dots, n\}$  namely  $J = (1, 2, \dots, n)$  we obtain

$$\bar{g}^*(dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n) = \det [(g_i)_j]_{i,j=1}^n du_1 \wedge du_2 \wedge \cdots \wedge du_n$$

The matrix in the determinant is the derivative matrix  $D\vec{g}$ . □

**Remark 1.3.22.** The Jacobian determinant in the Change of Variables Theorem is not an additional rule that must be memorized separately. It appears naturally when pulling back the standard volume form  $dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n$ . The determinant measures both local volume scaling and orientation change.

- If  $\det(D\vec{g}) > 0$ , then  $\vec{g}$  preserves orientation locally.
- If  $\det(D\vec{g}) < 0$ , then  $\vec{g}$  reverses orientation locally.

**Example 1.3.23.** Let  $\vec{g}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be given by

$$\vec{g}(u, v) = (3u + v, 2u - v)$$

Then

$$\det(D\vec{g}) = \det \begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix} = -5$$

Therefore,

$$\bar{g}^*(dx \wedge dy) = -5du \wedge dv$$

The negative sign indicates that  $\vec{g}$  reverses orientation, while the factor 5 indicates that it scales signed area by a factor of 5.

The coordinate formulas above have a simple geometric interpretation.

**Proposition 1.3.24.** *Let  $U \subseteq \mathbb{R}^m$  and  $V \subseteq \mathbb{R}^n$  be open sets, let  $\vec{g} : U \rightarrow V$  be smooth, and let  $\omega \in \mathcal{F}^k(V)$ . For every point  $\vec{a} \in U$  and  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in \mathbb{R}^m$ ,*

$$(\vec{g}^* \omega)(\vec{a})(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k) = \omega(\vec{g}(\vec{a}))(D\vec{g}(\vec{a})\vec{v}_1, D\vec{g}(\vec{a})\vec{v}_2, \dots, D\vec{g}(\vec{a})\vec{v}_k)$$

*Proof.* It suffices to verify the formula when

$$\omega = d\vec{x}_I = dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_k}$$

By definition,

$$\vec{g}^*(d\vec{x}_I) = dg_{i_1} \wedge dg_{i_2} \wedge \dots \wedge dg_{i_k}$$

Therefore,

$$(\vec{g}^* d\vec{x}_I)(\vec{a})(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k) = \det \begin{bmatrix} dg_{i_1}(\vec{a})(\vec{v}_1) & dg_{i_1}(\vec{a})(\vec{v}_2) & \dots & dg_{i_1}(\vec{a})(\vec{v}_k) \\ dg_{i_2}(\vec{a})(\vec{v}_1) & dg_{i_2}(\vec{a})(\vec{v}_2) & \dots & dg_{i_2}(\vec{a})(\vec{v}_k) \\ \vdots & \vdots & \ddots & \vdots \\ dg_{i_k}(\vec{a})(\vec{v}_1) & dg_{i_k}(\vec{a})(\vec{v}_2) & \dots & dg_{i_k}(\vec{a})(\vec{v}_k) \end{bmatrix}$$

Since  $dg_i(\vec{a})(\vec{v}) = Dg_i(\vec{a})(\vec{v})$ , the entries of this matrix are the  $i_p$ th coordinate components of  $D\vec{g}(\vec{a})\vec{v}_q$ . Hence,

$$(\vec{g}^* d\vec{x}_I)(\vec{a})(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k) = d\vec{x}_I(D\vec{g}(\vec{a})\vec{v}_1, D\vec{g}(\vec{a})\vec{v}_2, \dots, D\vec{g}(\vec{a})\vec{v}_k)$$

By linearity, the formula holds for all  $\omega \in \mathcal{F}^k(V)$ . □

**Remark 1.3.25.** The pullback works by applying the derivative  $D\vec{g}(\vec{a})$  to vectors in the domain before evaluating the form in the target. Thus, forms move in the direction opposite to vectors: vectors are pushed forward by  $D\vec{g}$ , while forms are pulled back by  $\vec{g}^*$ . For this reason, the pullback is called a contravariant operation.

**Corollary 1.3.26.** *Let  $U \subseteq \mathbb{R}^m$  and  $V \subseteq \mathbb{R}^n$  be open sets, and let  $\vec{g} : U \rightarrow V$  be smooth. If  $\omega \in \mathcal{F}^k(V)$  and  $k > m$ , then*

$$\vec{g}^* \omega = \vec{0}$$

*Proof.* Every  $k$ -form on the  $m$ -dimensional domain  $U$  is zero when  $k > m$ . Equivalently, there are no nonzero alternating  $k$ -covectors on  $\mathbb{R}^m$  when  $k > m$ . □



**Corollary 1.3.27.** *Let  $U \subseteq \mathbb{R}^m$  and  $V \subseteq \mathbb{R}^n$  be open sets, let  $\vec{g} : U \rightarrow V$  be smooth, and assume  $\text{rank}(D\vec{g}(\vec{a})) < k$  at a point  $\vec{a} \in U$ . Then for every  $\omega \in \mathcal{F}^k(V)$ ,*

$$(\vec{g}^*\omega)(\vec{a}) = \vec{0}$$

*Proof.* The image of  $D\vec{g}(\vec{a}) : \mathbb{R}^m \rightarrow \mathbb{R}^n$  has dimension less than  $k$ . Therefore, for any vectors  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in \mathbb{R}^m$ , the vectors

$$D\vec{g}(\vec{a})\vec{v}_1, D\vec{g}(\vec{a})\vec{v}_2, \dots, D\vec{g}(\vec{a})\vec{v}_k$$

are linearly independent. Since  $\omega(\vec{g}(\vec{a}))$  is alternating, it vanishes on linearly dependent vectors. Thus,

$$(\vec{g}^*\omega)(\vec{a})(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k) = \vec{0}$$

for all  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k$ . Hence,

$$(\vec{g}^*\omega)(\vec{a}) = \vec{0}$$

□

**Corollary 1.3.28.** *Let  $\vec{g} : U \rightarrow V$  be a constant map. Then for every positive-degree form  $\omega \in \mathcal{F}^k(V)$  with  $k \geq 1$ , we have*

$$\vec{g}^*\omega = \vec{0}$$

*Proof.* If  $\vec{g}$  is constant, then  $D\vec{g}(\vec{a}) = \vec{0}$  for every  $\vec{a} \in U$ . Thus,  $\text{rank}(D\vec{g}(\vec{a})) = 0 < k$ . The result follows from Corollary 1.3.27. □

**Proposition 1.3.29.** *Let  $\omega, \eta \in \mathcal{F}^k(V)$  and let  $c \in \mathbb{R}$ . Then*

$$\vec{g}^*(c\omega + \eta) = c\vec{g}^*\omega + \vec{g}^*\eta$$

*Proof.* The result follows directly from the coordinate definition of pullback and the linearity of composition, exterior differentiation, and the wedge product. □

**Proposition 1.3.30.** *Let  $f \in \mathcal{F}^0(V)$  and let  $\omega \in \mathcal{F}^k(V)$ . Then*

$$\vec{g}^*(f\omega) = (\vec{g}^*f)(\vec{g}^*\omega)$$

*Proof.* Write

$$\omega = \sum_I a_I d\vec{x}_I$$

Then

$$f\omega = \sum_I (fa_I)d\vec{x}_I$$

Therefore,

$$\begin{aligned}\vec{g}^*(f\omega) &= \sum_I ((fa_I) \circ \vec{g})d\vec{g}_I \\ &= \sum_I (f \circ \vec{g})(a_I \circ \vec{g})d\vec{g}_I \\ &= (f \circ \vec{g}) \sum_I (a_I \circ \vec{g})d\vec{g}_I \\ &= (\vec{g}^*f)(\vec{g}^*\omega)\end{aligned}$$

□

**Proposition 1.3.31.** *Let  $\omega \in \mathcal{F}^k(V)$  and  $\eta \in \mathcal{F}^l(V)$ . Then*

$$\vec{g}^*(\omega \wedge \eta) = \vec{g}^*\omega \wedge \vec{g}^*\eta$$

*Proof.* It is enough to consider forms of the form  $\omega = fd\vec{x}_I$  and  $\eta = hd\vec{x}_J$ . Then

$$\omega \wedge \eta = fhd\vec{x}_I \wedge d\vec{x}_J$$

Therefore,

$$\begin{aligned}\vec{g}^*(\omega \wedge \eta) &= ((fh) \circ \vec{g})d\vec{g}_I \wedge d\vec{g}_J \\ &= (f \circ \vec{g})(h \circ \vec{g})d\vec{g}_I \wedge d\vec{g}_J \\ &= ((f \circ \vec{g})d\vec{g}_I) \wedge ((h \circ \vec{g})d\vec{g}_J) \\ &= \vec{g}^*\omega \wedge \vec{g}^*\eta\end{aligned}$$

By linearity, the result holds for all forms. □

**Example 1.3.32.** Let

$$\vec{g}(u, v) = (u \cos(v), u \sin(v))$$

We know that

$$\begin{aligned}\vec{g}^*(dx) &= \cos(v)du - u \sin(v)dv, \\ \vec{g}^*(dy) &= \sin(v)du + u \cos(v)dv\end{aligned}$$

Therefore,

$$\begin{aligned}\vec{g}^*(dx \wedge dy) &= \vec{g}^*(dx) \wedge \vec{g}^*(dy) \\ &= udu \wedge dv\end{aligned}$$

This illustrates that the pullback of a wedge product is the wedge product of the products.

**Theorem 1.3.33.** *Let  $W \subseteq \mathbb{R}^l$ ,  $U \subseteq \mathbb{R}^m$ , and  $V \subseteq \mathbb{R}^n$  be open sets. Let  $\vec{h}: W \rightarrow U$  and  $\vec{g}: U \rightarrow V$  be smooth maps. Then for every  $\omega \in \mathcal{F}^k(V)$ , we have*

$$(\vec{g} \circ \vec{h})^* \omega = \vec{h}^*(\vec{g}^* \omega)$$

*Proof.* For a function  $f \in \mathcal{F}^0(V)$ , we have

$$(\vec{g} \circ \vec{h})^* f = f \circ (\vec{g} \circ \vec{h}) = (f \circ \vec{g}) \circ \vec{h} = \vec{h}^*(\vec{g}^* f)$$

Now consider a coordinate 1-form  $dx_i$ . We have

$$(\vec{g} \circ \vec{h})^*(dx_i) = d(g_i \circ \vec{h})$$

By the chain rule,

$$d(g_i \circ \vec{h}) = \vec{h}^*(dg_i) = \vec{h}^*(\vec{g}^*(dx_i))$$

Since pullbacks are linear and respect wedge products, the identity holds for all differential forms.  $\square$

**Remark 1.3.34.** The order in the composition formula is important:

$$(\vec{g} \circ \vec{h})^* = \vec{h}^* \circ \vec{g}^*$$

A form begins on the target  $V$ . It is first pulled back from  $V$  to  $U$  by  $\vec{g}^*$  and then from  $U$  to  $W$  by  $\vec{h}^*$ . Thus, pullback reverses the direction of composition.

The next theorem is one of the most important properties of pullbacks. It says that differentiating a form before pulling it back gives the same result as pulling it back first and then differentiating.

**Theorem 1.3.35.** *Let  $U \subseteq \mathbb{R}^m$  and  $V \subseteq \mathbb{R}^n$  be open sets, and let  $\vec{g}: U \rightarrow V$  be smooth. Then for every  $\omega \in \mathcal{F}^k(V)$ , we have*

$$\vec{g}^*(d\omega) = d(\vec{g}^* \omega)$$

*Proof.* We first consider a smooth function  $f \in \mathcal{F}^0(V)$ . By definition,  $\vec{g}^* f = f \circ \vec{g}$ . Therefore,

$$d(\vec{g}^* f) = d(f \circ \vec{g})$$

By the multivariable chain rule,

$$d(f \circ \vec{g}) = \sum_{j=1}^m (f \circ \vec{g})_j du_j$$

On the other hand,

$$df = \sum_{i=1}^n f_{x_i} dx_i$$

so

$$\vec{g}^*(df) = \sum_{i=1}^n (f_{x_i} \circ \vec{g}) \vec{g}^*(dx_i) = \sum_{i=1}^n (f_{x_i} \circ \vec{g}) dg_i$$

This is exactly the chain-rule expression for  $d(f \circ \vec{g})$ . Hence,

$$\vec{g}^*(df) = d(\vec{g}^* f)$$

Now let  $\omega = f d\vec{x}_I$ . Then

$$d\omega = df \wedge d\vec{x}_I$$

Using the fact that pullback respects wedge products,

$$\vec{g}^*(d\omega) = \vec{g}^*(df \wedge d\vec{x}_I) = \vec{g}^*(df) \wedge \vec{g}^*(d\vec{x}_I)$$

Since  $\vec{g}^*(df) = d(\vec{g}^* f)$ , we obtain

$$\vec{g}^*(d\omega) = d(\vec{g}^* f) \wedge d\vec{g}_I$$

But  $\vec{g}\omega = (\vec{g}^* f) d\vec{g}_I$ . Since each  $dg_i = d(dg_i)$  is closed, because  $d(dg_i) = 0$ , the graded product rule gives

$$d((\vec{g}^* f) d\vec{g}_I) = d(\vec{g}^* f) \wedge d\vec{g}_I$$

Therefore,

$$\vec{g}^*(d\omega) = d(\vec{g}^* \omega)$$

By linearity, the identity holds for every differential form  $\omega \in \mathcal{F}^k(V)$ .  $\square$

**Remark 1.3.36.** The identity  $\vec{g}^*(d\omega) = d(\vec{g}^*\omega)$  is the differential-form version of the chain rule. It is one of the reasons differential forms are so useful: the exterior derivative behaves naturally under smooth coordinate changes and parametrizations.

**Example 1.3.37.** Let  $\vec{g}: \mathbb{R} \rightarrow \mathbb{R}^2$  be given by

$$\vec{g}(t) = (t, t^2)$$

Let  $\omega = xdy$ . First compute

$$d\omega = d(x) \wedge dy = dx \wedge dy$$

Since  $dx \wedge dy$  is a 2-form on  $\mathbb{R}^2$ , while the domain of  $\vec{g}$  is one-dimensional,  $\vec{g}^*(d\omega) = 0$ . Now compute the pullback first

$$\vec{g}^*\omega = (t)d(t^2) = 2t^2dt$$

Since this is a 1-form on  $\mathbb{R}$ ,

$$d(\vec{g}^*\omega) = d(2t^2) \wedge dt = 4tdt \wedge dt = 0$$

Therefore,

$$\vec{g}^*(d\omega) = d(\vec{g}^*\omega) = 0$$



## Chapter 2

# Line Integration and Green's Theorem

In Chapter 1, we introduced differential forms, the exterior derivative, and pullbacks. These ideas now allow us to define integration over parametrized curves in a natural way. Instead of integrating ordinary functions along curves, we will integrate differential 1-forms. This viewpoint gives a unified and coordinate-friendly way to describe work, circulation, and path-dependent quantities.

The first major result of this chapter is the Fundamental Theorem of Calculus for Line Integrals. It says that if a 1-form is the exterior derivative of a function, then its integral along a curve depends only on the endpoints of the curve. In the language of vector calculus, this is the theory of conservative vector fields and potential functions.

We will then study how to find potential functions when they exist, and we will examine the important distinction between closed forms and exact forms. The identity  $d^2 = 0$  tells us that every exact form is closed, but the converse may fail when the domain has holes. This is one of the first places where topology begins to influence calculus.

Finally, we will prove Green's Theorem, which relates a line integral around the boundary of a planar region to a double integral over the region itself:

$$\int_{\partial S} \omega = \int_S d\omega.$$

This theorem is the two-dimensional prototype of Stokes's Theorem. It shows how the exterior derivative measures the infinitesimal circulation of a 1-form and how this local information accumulates over a region to determine the

boundary integral.

## 2.1 The Fundamental Theorem of Calculus for Line Integrals

In Chapter 1, we defined differential forms and learned how to pull them back under smooth maps. We now use pullbacks to define line integrals of 1-forms along parametrized curves.

The main idea is simple: if  $U \subseteq \mathbb{R}^n$  is open and if a curve is parametrized by

$$\vec{g} : [a, b] \rightarrow U$$

and if  $\omega$  is a 1-form on  $U$ , then the pullback  $\vec{g}^*\omega$  is a 1-form on the interval  $[a, b]$ . Since every 1-form on an interval has the form  $h(t)dt$ , we can integrate it using ordinary single-variable integration.

This is the first place where the pullback becomes essential.

**Definition 2.1.1.** Let  $U \subseteq \mathbb{R}^n$  be open. A *parametrized curve* in  $U$  is a smooth map  $\vec{g} : [a, b] \rightarrow U$ . The image of  $\vec{g}$  is denoted by  $C = \vec{g}([a, b])$ . The point  $\vec{g}(a)$  is called the *initial point* of the parametrized curve, and  $\vec{g}(b)$  is called the *terminal point*.

**Definition 2.1.2.** A *piecewise smooth curve* in  $U$  is a curve  $C$  that can be written as a finite union

$$C = C_1 \cup C_2 \cup \cdots \cup C_s$$

where each  $C_j$  is the image of a smooth parametrized curve

$$\vec{g}_j : [a_j, b_j] \rightarrow U$$

and the terminal point of each piece agrees with the initial point of the next piece

$$\vec{g}_j(b_j) = \vec{g}_{j+1}(a_{j+1})$$

A path is called *closed* if its initial point and terminal point agree.

**Remark 2.1.3.** In these notes, the word “curve” often refers not only to the set of points traced out in space, but also to the direction in which the curve is traversed. This direction is called *orientation*.

For example, the unit circle traversed counterclockwise is considered differently from the unit circle traversed clockwise, even though the underlying set of points is the same.



**Definition 2.1.4.** Let  $\omega \in \mathcal{F}^1(U)$  be a differential 1-form on an open set  $U \subseteq \mathbb{R}^n$ . Suppose  $\vec{g}: [a, b] \rightarrow U$  is a smooth parametrized curve. The *line integral* of  $\omega$  over  $C = \vec{g}([a, b])$  is defined by

$$\int_C \omega = \int_{[a, b]} \vec{g}^* \omega$$

If  $\omega = F_1 dx_1 + F_2 dx_2 + \cdots + F_n dx_n$ , then

$$\vec{g}^* \omega = \sum_{i=1}^n (F_i \circ \vec{g}) d(g_i)$$

Since  $d(g_i) = g'_i(t)dt$ , we obtain

$$\vec{g}^* \omega = \left( \sum_{i=1}^n F_i(\vec{g}(t)) g'_i(t) \right) dt$$

Therefore,

$$\int_C \omega = \int_a^b \sum_{i=1}^n F_i(\vec{g}(t)) g'_i(t) dt$$

**Remark 2.1.5.** The definition of the line integral is not a new kind of integral. It is an ordinary single-variable integral after pulling the form back to the parameter interval. The pullback is what converts a 1-form in  $\mathbb{R}^n$  into a 1-form in one variable

$$\vec{g}^* \omega = h(t)dt$$

Then

$$\int_C \omega = \int_a^b h(t)dt$$

**Example 2.1.6.** Let  $C$  be the line segment from  $A(1, -1, 0)$  to  $B(2, 2, 2)$ . Parametrize  $C$  by

$$\vec{g}(t) = A + t(B - A)$$

for  $0 \leq t \leq 1$ . Thus,

$$\vec{g}(t) = (1 + t, -1 + 3t, 2t)$$

Let  $\omega = xydz$ . We compute

$$x \circ \vec{g} = 1 + t, \quad y \circ \vec{g} = -1 + 3t, \quad z \circ \vec{g} = 2t$$

Hence,

$$\vec{g}^*(dz) = d(2) = 2dt$$

Therefore,

$$\vec{g}^*\omega = (x \circ \vec{g})(y \circ \vec{g})\vec{g}^*(dz) = (1+t)(-1+3t)(2dt)$$

Thus,

$$\begin{aligned}\int_C \omega &= \int_0^1 2(1+t)(-1+3t)dt \\ &= 2 \int_0^1 (3t^2 + 2t - 1)dt \\ &= 2[t^3 + t^2 - t]_0^1 \\ &= 2\end{aligned}$$

**Example 2.1.7.** Let  $\omega = -ydx + xdy$  on  $\mathbb{R}^2$ . Let  $C$  be the unit circle traversed counterclockwise, parametrized by

$$\vec{g}(t) = (\cos(t), \sin(t))$$

for  $0 \leq t \leq 2\pi$ . Then

$$\vec{g}^*(dx) = -\sin(t)dt, \quad \vec{g}^*(dy) = \cos(t)dt$$

Also,

$$x \circ \vec{g} = \cos(t), \quad y \circ \vec{g} = \sin(t)$$

Therefore,

$$\begin{aligned}\vec{g}^*\omega &= -(\sin(t))(-\sin(t)dt) + (\cos(t))(\cos(t)dt) \\ &= (\sin^2(t) + \cos^2(t))dt \\ &= dt\end{aligned}$$

Hence,

$$\int_C \omega = \int_0^{2\pi} dt = 2\pi$$

Thus,

$$\int_C -ydx + xdy = 2\pi$$

**Definition 2.1.8.** Let  $U \subseteq \mathbb{R}^n$  be an open set, and let  $\vec{F} : U \rightarrow \mathbb{R}^n$  be a smooth vector field,  $\vec{F} = (F_1, F_2, \dots, F_n)$ . The 1-form associated with  $\vec{F}$  is

$$\omega_{\vec{F}} = F_1dx_1 + F_2dx_2 + \dots + F_ndx_n$$

**Proposition 2.1.9.** Let  $\omega_{\vec{F}} = F_1 dx_1 + F_2 dx_2 + \cdots + F_n dx_n$  be the 1-form associated with a vector field  $\vec{F} = (F_1, F_2, \dots, F_n)$ . If  $\vec{g} : [a, b] \rightarrow U$  parametrizes a curve  $C$ , then

$$\int_C \omega_{\vec{F}} = \int_a^b \vec{F}(\vec{g}(t)) \cdot \vec{g}'(t) dt$$

If  $\vec{T}$  is the unit tangent vector to  $C$ , and  $ds$  denotes arclength, then this is often written in vector-calculus notation as

$$\int_C \omega_{\vec{F}} = \int_C \vec{F} \cdot \vec{T} ds$$

*Proof.* By Definition 2.1.4,

$$\int_C \omega_{\vec{F}} = \int_a^b \sum_{i=1}^n F_i(\vec{g}(t)) g'_i(t) dt$$

But then

$$\sum_{i=1}^n F_i(\vec{g}(t)) g'_i(t) = \vec{F}(\vec{g}(t)) \cdot \vec{g}'(t)$$

Therefore

$$\int_C \omega_{\vec{F}} = \int_a^b \vec{F}(\vec{g}(t)) \cdot \vec{g}'(t) dt$$

□

**Remark 2.1.10.** If  $\vec{F}$  is a force field and a particle moves along a curve  $C$ , then

$$\int_C \vec{F} \cdot \vec{T} ds$$

represents the work done by the force field along the path.

Only the component of  $\vec{F}$  tangent to the curve contributes to the work. The normal component does not move the particle along the curve.

Thus, line integration of 1-forms generalizes the work integral from physics.

**Definition 2.1.11.** Suppose  $\vec{g} : [a, b] \rightarrow U$  parametrizes a curve  $C$ . The same curve with the opposite orientation is parametrized by  $\tilde{g} : [a, b] \rightarrow U$  given by

$$\tilde{g}(t) = \vec{g}(a + b - t)$$

We denote the oppositely oriented curve by  $-C$  or  $C^-$ .

**Proposition 2.1.12.** *Let  $\omega \in \mathcal{F}^1(U)$ . Then*

$$\int_{-C} \omega = - \int_C \omega$$

*Proof.* Let  $C = \vec{g}([a, b])$  and let  $\tilde{g}(t) = \vec{g}(a + b - t)$  parametrize  $-C$ . Write

$$\omega = F_1 dx_1 + F_2 dx_2 + \cdots + F_n dx_n$$

Then

$$\tilde{g}'(t) = -\vec{g}'(a + b - t)$$

Therefore,

$$\int_{-C} \omega = \int_a^b \vec{F}(\tilde{g}(t)) \cdot \tilde{g}'(t) dt = \int_a^b \vec{F}(\vec{g}(a + b - t)) \cdot (-\vec{g}'(a + b - t)) dt$$

Using the substitution  $u = a + b - t$ , we obtain

$$\int_{-C} \omega = - \int_a^b \vec{F}(\vec{g}(u)) \cdot \vec{g}'(u) du$$

Hence,

$$\int_{-C} \omega = - \int_C \omega$$

□

**Remark 2.1.13.** Line integrals of 1-forms depend on orientation. This is analogous to the one-variable identity

$$\int_b^a f(t) dt = - \int_a^b f(t) dt$$

The sign is not a technicality. It is part of the geometric meaning of the integral.

**Proposition 2.1.14.** *Let  $\vec{g} : [a, b] \rightarrow U$  be a smooth parametrization of a curve  $C$ , and let  $\phi : [c, d] \rightarrow [a, b]$  be a smooth increasing bijection with  $\phi(c) = a$  and  $\phi(d) = b$ . Then  $\vec{g} \circ \phi : [c, d] \rightarrow U$  is another parametrization of  $C$  with the same orientation, and*

$$\int_{\vec{g} \circ \phi} \omega = \int_{\vec{g}} \omega$$

*Proof.* Using pullbacks,

$$\int_{\vec{g} \circ \phi} \omega = \int_{[c,d]} (\vec{g} \circ \phi)^* \omega$$

By Theorem 1.3.33,

$$(\vec{g} \circ \phi)^* \omega = \phi^*(\vec{g}^* \omega)$$

Write  $\vec{g}^* \omega = h(t)dt$ . Then

$$\phi^*(\vec{g}^* \omega) = h(\phi(s))\phi'(s)ds$$

Therefore,

$$\int_{\vec{g} \circ \phi} \omega = \int_c^d h(\phi(s))\phi'(s)ds$$

By the ordinary substitution rule,

$$\int_c^d h(\phi(s))\phi'(s)ds = \int_a^b h(t)dt$$

Hence,

$$\int_{\vec{g} \circ \phi} \omega = \int_{\vec{g}} \omega$$

□

**Remark 2.1.15.** A line integral of a 1-form depends on the oriented curve, not on the particular parametrization chosen to describe it. This is one of the reasons differential forms are well suited to integration on geometric objects.

**Definition 2.1.16.** Let  $U \subseteq \mathbb{R}^n$ . A 1-form  $\omega \in \mathcal{F}^1(U)$  is called *exact* if there exists a smooth function  $f : U \rightarrow \mathbb{R}$  such that  $\omega = df$ . The function  $f$  is called a *potential function* for  $\omega$ .

**Remark 2.1.17.** If  $f : U \rightarrow \mathbb{R}$  is smooth, then

$$df = f_{x_1}dx_1 + f_{x_2}dx_2 + \cdots + f_{x_n}dx_n$$

Thus, saying  $\omega = df$  means that if

$$\omega = F_1dx_1 + F_2dx_2 + \cdots + F_n dx_n$$

then  $F_i = f_{x_i}$  for every  $i$ .

In vector-calculus language, the vector field  $\vec{F} = (F_1, F_2, \dots, F_n)$  is the gradient field  $\vec{F} = \nabla f$ .

**Theorem 2.1.18 (The Fundamental Theorem of Calculus for Line Integrals).** *Let  $U \subseteq \mathbb{R}^n$  be open, and suppose  $\omega = df$  for some smooth function  $f : U \rightarrow \mathbb{R}$ . If  $C$  is a piecewise smooth path in  $U$  starting at  $A$  and ending at  $B$ , then*

$$\int_C \omega = f(B) - f(A)$$

*Equivalently, if  $\vec{F} = \nabla f$ , then*

$$\int_C \vec{F} \cdot \vec{T} ds = f(B) - f(A)$$

*Proof.* First suppose  $C$  is parametrized by a single smooth curve  $\vec{g} : [a, b] \rightarrow U$  with  $\vec{g}(a) = A$  and  $\vec{g}(b) = B$ . Since  $\omega = df$ , we have  $\vec{g}^* \omega = \vec{g}^*(df)$ . Because pullback commutes with the exterior derivative,

$$\vec{g}^*(df) = d(\vec{g}^* f)$$

But  $\vec{g}^* f = f \circ \vec{g}$ . Therefore,  $\vec{g}^* \omega = d(f \circ \vec{g})$ . Since  $f \circ \vec{g}$  is an ordinary single-variable function on  $[a, b]$ ,

$$d(f \circ \vec{g}) = (f \circ \vec{g})'(t) dt$$

Hence,

$$\begin{aligned} \int_C \omega &= \int_{[a,b]} \vec{g}^* \omega \\ &= \int_a^b (f \circ \vec{g})'(t) dt \\ &= (f \circ \vec{g})(b) - (f \circ \vec{g})(a) \\ &= f(\vec{g}(b)) - f(\vec{g}(a)) \\ &= f(B) - f(A) \end{aligned}$$

Now suppose  $C$  is piecewise smooth

$$C = C_1 \cup C_2 \cup \cdots \cup C_s$$

Let  $C_j$  start at  $A_j$  and end at  $B_j$  arranged so that

$$A_1 = A, \quad B_j = A_{j+1}, \quad B_s = B$$

Applying the smooth case to each piece,

$$\int_{C_j} \omega = f(B_j) - f(A_j)$$

Therefore,

$$\int_C \omega = \sum_{j=1}^s \int_{C_j} \omega = \sum_{j=1}^s (f(B_j) - f(A_j))$$

This sum telescopes

$$\sum_{j=1}^s (f(B_j) - f(A_j)) = f(B) - f(A)$$

Thus,

$$\int_C \omega = f(B) - f(A)$$

□

**Remark 2.1.19.** This theorem is exactly the ordinary Fundamental Theorem of Calculus in disguise. Indeed, when  $n = 1$ ,  $\omega = df = f'(x)dx$ , and if  $C = [a, b]$ , then

$$\int_C df = \int_a^b f'(x)dx = f(b) - f(a)$$

The line integral version simply allows the interval to be replaced by a curve in  $\mathbb{R}^n$ .

**Example 2.1.20.** Let

$$\omega = 2x dx + 2y dy + 2z dz$$

Observe that

$$\omega = d(x^2 + y^2 + z^2)$$

Let  $C$  be any smooth path from  $A(1, 0, 2)$  to  $B(3, -1, 4)$ . Then by the Fundamental Theorem of Calculus of Line Integrals (Theorem 2.1.18),

$$\int_C \omega = f(B) - f(A)$$

where  $f(x, y, z) = x^2 + y^2 + z^2$ . Compute

$$f(B) = 3^2 + (-1)^2 + 4^2 = 26, \quad f(A) = 1^2 + 0^2 + 2^2 = 5$$

Therefore,

$$\int_C 2x dx + 2y dy + 2z dz = 21$$

The answer is independent of the path.

**Example 2.1.21.** Let  $f(x, y) = e^x + x^2y + \sin(y)$ . Then

$$df = (e^x + 2xy)dx + (x^2 + \cos(y))dy$$

Let

$$\omega = (e^x + 2xy)dx + (x^2 + \cos(y))dy$$

If  $C$  is any path from  $A(0, 0)$  to  $B(1, \pi)$ , then

$$\int_C \omega = f(B) - f(A)$$

Now,  $f(1, \pi) = e + \pi$  and  $f(0, 0) = 1$ . Thus,

$$\int_C \omega = e + \pi - 1$$

Again, the result is independent of the path.

**Example 2.1.22.** Let  $\omega = df$  be exact on  $U$ , and let  $C$  be a closed path in  $U$ . If  $C$  starts and ends at the same point  $A$ , then by the Fundamental Theorem for Line Integrals,

$$\int_C \omega = f(A) - f(A) = 0$$

Therefore,

$$\int_C df = 0$$

for every closed curve  $C$ .

**Definition 2.1.23.** Let  $\omega \in \mathcal{F}^1(U)$ . We say that the line integral

$$\int_C \omega$$

is *path-independent* in  $U$  if whenever  $C_1$  and  $C_2$  are two piecewise smooth paths in  $U$  with the same initial point  $A$  and the same terminal point  $B$ , we have

$$\int_{C_1} \omega = \int_{C_2} \omega$$

In other words, the value of the integral depends only on the endpoints, not on the path taken between them.

**Corollary 2.1.24.** If  $\omega = df$  on  $U$ , then the integral of  $\omega$  is path-independent.



*Proof.* Let  $C_1$  and  $C_2$  be two paths from  $A$  to  $B$ . By Theorem 2.1.18

$$\int_{C_1} \omega = f(B) - f(A), \quad , \quad \int_{C_2} \omega = f(B) - f(A)$$

Therefore,

$$\int_{C_1} \omega = \int_{C_2} \omega$$

□

**Corollary 2.1.25.** *If  $\omega = df$  on  $U$ , then for every closed curve  $C$  in  $U$ ,*

$$\int_C \omega = 0$$

**Remark 2.1.26.** The two conclusions above are closely related:

- Exact forms are path independent
- Exact forms integrate to zero around closed curves.

Indeed, if two paths  $C_1$  and  $C_2$  go from  $A$  to  $B$ , then the path  $C_1 \cup (-C_2)$  is closed. Thus, comparing two path integrals is equivalent to integrating around a closed curve.

**Definition 2.1.27.** A vector field  $\vec{F} : U \rightarrow \mathbb{R}^n$  is called *conservative* if there exists a smooth function  $f : U \rightarrow \mathbb{R}$  such that  $\vec{F} = \nabla f$ . The function  $f$  is called a *potential function* for  $\vec{F}$ . Equivalently, the associated 1-form

$$\omega_{\vec{F}} = F_1 dx_1 + F_2 dx_2 + \cdots + F_n dx_n$$

is exact:  $\omega_{\vec{F}} = df$ .

**Remark 2.1.28.** The word “conservative” comes from physics. If a force field is conservative, then the work done by the field depends only on the initial and final positions. No net work is done around closed paths.

In mechanics, this is connected to conservation of total energy.

**Proposition 2.1.29.** *Suppose a partial of mass  $m$  moves along a curve  $\vec{g} : [a, b] \rightarrow \mathbb{R}^n$  under the influence of a force field  $\vec{F}$ . Assume Newton’s second law*

$$\vec{F}(\vec{g}(t)) = m\vec{g}''(t)$$

*Then the work done by  $\vec{F}$  along the curve is equal to the change in kinetic energy*

$$\int_C \vec{F} \cdot \vec{T} ds = \Delta \left( \frac{1}{2} m \|\vec{g}'(t)\|^2 \right)$$

*Proof.* The work is

$$\int_C \vec{F} \cdot \vec{T} ds = \int_a^b \vec{F}(\vec{g}(t)) \cdot \vec{g}'(t) dt$$

Using Newton's second law,

$$\vec{F}(\vec{g}(t)) = m\vec{g}''(t)$$

we obtain

$$\int_a^b \vec{F}(\vec{g}(t)) \vec{g}'(t) dt = m \int_a^b \vec{g}''(t) \cdot \vec{g}'(t) dt$$

But

$$\frac{d}{dt} \left( \frac{1}{2} \|\vec{g}'(t)\|^2 \right) = \vec{g}''(t) \cdot \vec{g}'(t)$$

Therefore,

$$\begin{aligned} m \int_a^b \vec{g}''(t) \cdot \vec{g}'(t) dt &= m \int_a^b \frac{d}{dt} \left( \frac{1}{2} \|\vec{g}'(t)\|^2 \right) dt \\ &= \frac{1}{2} m \|\vec{g}'(b)\|^2 - \frac{1}{2} m \|\vec{g}'(a)\|^2 \end{aligned}$$

This is precisely the change in kinetic energy.  $\square$

The Fundamental Theorem for Line Integrals gives one direction immediately: exact forms are path independent, and integrate to zero around closed curves. Under mild connectedness assumptions, the converses are also true.

**Theorem 2.1.30.** *Let  $U \subseteq \mathbb{R}^n$  be open and path-connected. Let*

$$\omega = F_1 dx_1 + F_2 dx_2 + \cdots + F_n dx_n$$

*be a smooth 1-form on  $U$ . The following are equivalent*

1. *For every closed piecewise smooth curve  $C$  in  $U$ ,*

$$\int_C \omega = 0$$

2. *The line integral of  $\omega$  is path-independent in  $U$ .*

3. *There exists a smooth function  $f : U \rightarrow \mathbb{R}$  such that  $\omega = df$ .*

*Equivalently, the associated vector field  $\vec{F} = (F_1, F_2, \dots, F_n)$  is conservative.*

*Proof.* Assume that (1) holds. Let  $C_1$  and  $C_2$  be two paths from  $A$  to  $B$ . Then for any closed curve  $C$  in  $U$ ,

$$C = C_1 \cup (-C_2)$$

is a closed curve. Therefore,

$$0 = \int_C \omega = \int_{C_1} \omega + \int_{-C_2} \omega$$

Since reversing orientation changes sign of the integral,

$$\int_{-C_2} \omega = - \int_{C_2} \omega$$

Therefore,

$$0 = \int_{C_1} \omega - \int_{C_2} \omega$$

Hence,

$$\int_{C_1} \omega = \int_{C_2} \omega$$

So the integral is path independent, proving (2).

Assume that (2) holds. Fix a base point  $\vec{a} \in U$ . For each  $\vec{x} \in U$ , define

$$f(\vec{x}) = \int_{\vec{a}}^{\vec{x}} \omega$$

where the integral is taken along any piecewise smooth path in  $U$  from  $\vec{a}$  to  $\vec{x}$ . Because the integral is path independent, the value of  $f(\vec{x})$  is well-defined. We now show that  $df = \omega$ . Let  $\vec{x} = (x_1, x_2, \dots, x_n) \in U$ . Since  $U$  is open, for  $\delta > 0$ , the line segment from  $\vec{x}$  to  $\vec{x} + \delta \vec{e}_i$  lies in  $U$ . Then

$$f(\vec{x} + \delta \vec{e}_i) - f(\vec{x}) = \int_{\vec{x}}^{\vec{x} + \delta \vec{e}_i} \omega$$

Parametrize this short line segment by

$$\gamma(t) = \vec{x} + t\vec{e}_i$$

for  $0 \leq t \leq \delta$ . Then  $\gamma'(t) = \vec{e}_i$  and

$$\int_{\vec{x}}^{\vec{x} + \delta \vec{e}_i} \omega = \int_0^\delta \omega(\gamma(t))\gamma'(t)dt$$

Since  $\omega = F_1dx_1 + F_2dx_2 + \cdots + F_n dx_n$ , we have

$$\omega(\gamma(t))(\vec{e}_i) = F_i(\vec{x} + t\vec{e}_i)$$

Thus,

$$f(\vec{x} + \delta\vec{e}_i) - f(\vec{x}) = \int_0^\delta F_i(\vec{x} + t\vec{e}_i)dt$$

Therefore,

$$\frac{f(\vec{x} + \delta\vec{e}_i) - f(\vec{x})}{\delta} = \frac{1}{\delta} \int_0^\delta F_i(\vec{x} + t\vec{e}_i)dt$$

Taking the limit as  $\delta \rightarrow 0$ , we have

$$f_{x_i}(\vec{x}) = F_i(\vec{x})$$

Since this holds for every  $i$ ,

$$df = F_1dx_1 + F_2dx_2 + \cdots + F_n dx_n = \omega$$

Therefore,  $\omega$  is exact, proving (3).

Finally, assume (3) holds. If  $C$  is a closed curve beginning and ending at  $A$ , then by the Fundamental Theorem for Line Integrals (Theorem 2.1.18),

$$\int_C \omega = f(A) - f(A) = 0$$

Thus

$$\int_C \omega = 0$$

for every closed curve  $C$ , proving (1).  $\square$

**Remark 2.1.31.** If  $U$  is not path connected, then the same theorem holds on each path-connected component of  $U$ . That is, one constructs a potential function separately on each connected piece.

## 2.2 Finding a Potential Function

In Section 2.1, we proved that if a 1-form  $\omega$  is exact, say  $\omega = df$ , then its line integral depends only on the endpoints

$$\int_C \omega = f(B) - f(A)$$

Thus, finding a potential function can turn a difficult line integral into a simpler endpoint calculation. In this section, we discuss practical methods for finding potential functions. We also prove an important result: on a star-shaped region, every closed 1-form is exact.

**Proposition 2.2.1.** *Let  $\omega = P(x, y)dx + Q(x, y)dy$  be a smooth 1-form on an open set  $U \subseteq \mathbb{R}^2$ . Then  $\omega$  is closed if and only if  $Q_x = P_y$ .*

*Proof.* We compute  $d\omega = dP \wedge dx + dQ \wedge dy$ . Since

$$dP = P_x dx + P_y dy, \quad dQ = Q_x dx + Q_y dy$$

we have

$$\begin{aligned} dP \wedge dx &= (P_x dx + P_y dy) \wedge dx = -P_y dx \wedge dy, \\ dQ \wedge dy &= (Q_x dx + Q_y dy) \wedge dy = Q_x dx \wedge dy \end{aligned}$$

Therefore,

$$d\omega = (Q_x - P_y)dx \wedge dy$$

Thus,  $d\omega = 0$  if and only if  $Q_x - P_y = 0$  if and only if  $Q_x = P_y$ .  $\square$

**Proposition 2.2.2.** *Let  $\omega = Pdx + Qdy + Rdz$  be a smooth 1-form on an open set  $U \subseteq \mathbb{R}^3$ . Then  $\omega$  is closed if and only if  $R_y = Q_z$ ,  $P_z = R_x$ , and  $Q_x = P_y$ . Equivalently, the corresponding vector field  $\vec{F} = (P, Q, R)$  satisfies*

$$\nabla \times \vec{F} = \vec{0}_3$$

*Proof.* From the coordinate formula for the exterior derivative

$$d\omega = (R_y - Q_z)dy \wedge dz + (P_z - R_x)dz \wedge dx + (Q_x - P_y)dx \wedge dy$$

Therefore,  $d\omega = 0$  if and only if each coefficient is zero. Thus,  $d\omega = 0$  if and only if  $R_y = Q_z$ ,  $P_z = R_x$ , and  $Q_x = P_y$ .  $\square$

If we already know that a 1-form is exact, or that its line integrals are path independent, then a potential function can be constructed by integration.

**Example 2.2.3.** Let

$$\omega = (e^x + 2xy)dx + (x^2 + \cos(y))dy$$

on  $\mathbb{R}^2$ . First check that  $\omega$  is closed. Here,  $P = e^x + 2xy$  and  $Q = x^2 + \cos(y)$ , so

$$P_y = 2x, \quad Q_x = 2x$$

Thus, we have  $P_y = Q_x$ , so  $d\omega = 0$ . Since  $\mathbb{R}^2$  is star-shaped, this form is exact. We construct a potential function using the base point  $\vec{0}_2 = (0, 0)$ . Let  $\vec{x}_0 = (x_0, y_0) \in \mathbb{R}^2$ . Parametrize the line segment from  $\vec{0}_2$  to  $\vec{x}_0$  by

$$\vec{r}(t) = t\vec{x}_0 = (tx_0, ty_0)$$

for  $0 \leq t \leq 1$ . Then  $dx = x_0 dt$  and  $dy = y_0 dt$ . Along the path,  $x = tx_0$  and  $y = ty_0$ . Therefore,

$$\vec{r}^*\omega = (e^{tx_0} + 2(tx_0)(ty_0))x_0 dt + ((tx_0)^2 + \cos(ty_0))y_0 dt$$

Thus,

$$\begin{aligned} f(\vec{x}_0) &= \int_0^1 [x_0 e^{tx_0} + 2t^2 x_0^2 y_0 + t^2 x_0^2 y_0 + y_0 \cos(ty_0)] dt \\ &= \int_0^1 [x_0 e^{tx_0} + 3t^2 x_0^2 y_0 + y_0 \cos(ty_0)] dt \end{aligned}$$

Now,

$$\begin{aligned} \int_0^1 x_0 e^{tx_0} dt &= e^{x_0} - 1, \\ \int_0^1 3t^2 x_0^2 y_0 dt &= x_0^2 y_0, \\ \int_0^1 y_0 \cos(ty_0) dt &= \sin(y_0) \end{aligned}$$

Therefore,

$$f(\vec{x}_0) = e^{x_0} - 1 + x_0^2 y_0 + \sin(y_0)$$

Dropping the subscript on the coordinates, we obtain

$$f(x, y) = e^x + x^2 y + \sin(y) - 1$$

The constant  $-1$  is not important. We could also use

$$f(x, y) = e^x + x^2 y + \sin(y)$$

**Remark 2.2.4.** A potential function is never unique. If  $df = \omega$ , then  $d(f + C) = df$  for every constant  $C \in \mathbb{R}$ . Thus, potential functions are unique up to an additive constant on each path connected component of the domain.

Sometimes it is easier to choose a path made of coordinate line segments.

**Example 2.2.5.** Again, consider

$$\omega = (e^x + 2xy)dx + (x^2 + \cos(y))dy$$

on  $\mathbb{R}^2$ . We construct a potential function from the base point  $\vec{0}_2 = (0, 0)$  to a general point  $\vec{x}_0 = (x_0, y_0)$  using the path that first moves horizontally, then vertically.

The first segment goes from  $(0, 0)$  to  $(x_0, 0)$ . On this segment,  $y = 0$  and  $dy = 0$ . Thus,

$$\omega = e^x dx$$

So the contribution is

$$\int_0^{x_0} e^x dx = e^{x_0} - 1$$

The second segment goes from  $(x_0, 0)$  to  $(x_0, y_0)$ . On this segment,  $x = x_0$  and  $dx = 0$ . Thus,

$$\omega = (x_0^2 + \cos(y))dy$$

So the contribution is

$$\int_0^{y_0} (x_0^2 + \cos(y))dy = x_0^2 y_0 + \sin(y_0)$$

Adding the two contributions gives

$$f(\vec{x}_0) = e^{x_0} - 1 + x_0^2 y_0 + \sin(y_0)$$

Hence,

$$f(x, y) = e^x + x^2 y + \sin(y) - 1$$

Again, the constant may be omitted.

**Remark 2.2.6.** The rectangular path method is often faster than using a line segment. However, it assumes that the relevant coordinate paths stay inside the domain. For example, if the domain has holes or restrictions, then a horizontal-then-vertical path may not always lie inside the domain. In that case, one must choose paths more carefully.

The most common computational method is to integrate one coefficient at a time.

Let  $\omega = Pdx + Qdy$  be exact on an open set  $U \subseteq \mathbb{R}^2$ . To find  $f$  with  $df = \omega$ , we solve

$$f_x = P, \quad f_y = Q$$

A practical method is

1. Integrate  $P$  with respect to  $x$

$$f(x, y) = \int P dx + h(y)$$

2. Differentiate this expression with respect to  $y$ .
3. Compare with  $Q$  to determine  $h'(y)$ .
4. Integrate  $h'(y)$  to find  $h(y)$ .

**Example 2.2.7.** Let

$$\omega = (e^x + 2xy)dx + (x^2 + \cos(y))dy$$

We want  $df = \omega$ . Thus,

$$f_x = e^x + 2xy, \quad f_y = x^2 + \cos(y)$$

Start by integrating the first equation with respect to  $x$

$$f(x, y) = \int (e^x + 2xy)dx = e^x + x^2y + h(y)$$

where  $h(y)$  is the “constant” of integration.

Now we differentiate with respect to  $y$

$$f_y = x^2 + h'(y)$$

Compare with the required equation  $f_y = x^2 + \cos(y)$ . Thus,  $h'(y) = \cos(y)$ . Therefore,  $h(y) = \sin(y) + C$ . So

$$f(x, y) = e^x + x^2y + \sin(y) + C$$

Hence, a potential function is

$$f(x, y) = e^x + x^2y + \sin(y)$$

**Remark 2.2.8.** In the expression  $f(x, y) = e^x + x^2y + h(y)$ , the term  $h(y)$  appears because when integrating with respect to  $x$ , anything depending only on  $y$  behaves like a constant. This is the multivariable version of the constant of integration.

Let  $\omega = Pdx + Qdy + Rdz$  be exact on an open set  $U \subseteq \mathbb{R}^3$ . To find  $f$  such that  $df = \omega$ , we solve

$$f_x = P, \quad f_y = Q, \quad f_z = R$$

A practical method is



1. Integrate  $P$  with respect to  $x$

$$f(x, y, z) = \int P(x, y, z) dx + g(y, z)$$

2. Differentiate this expression with respect to  $y$  and compare  $Q$  to determine the partial information about  $g$ .
3. Differentiate with respect to  $z$  and compare with  $R$  to finish determining  $g$ .

**Example 2.2.9.** Let

$$\omega = \left( \frac{z}{x} + y \right) dx + (x + z) dy + (\log(x) + y + 2z) dz$$

on the region

$$U = \{(x, y, z) \in \mathbb{R}^3 : x > 0\}$$

We want to find  $f$  such that  $df = \omega$ . Thus,

$$f_x = \frac{z}{x} + y, \quad f_y = x + z, \quad f_z = \log(x) + y + 2z$$

Start by integrating  $f_x$  with respect to  $x$

$$f(x, y, z) = z \log(x) + xy + g(y, z)$$

Now differentiate with respect to  $y$

$$f_y = x + g_y(y, z)$$

Compare with  $f_y = x + z$  to obtain  $g_y(y, z) = z$ . Therefore,  $g(y, z) = yz + h(z)$ , so

$$f(x, y, z) = z \log(x) + xy + yz + h(z)$$

Now differentiate with respect to  $z$

$$f_z = \log(x) + y + h'(z)$$

Compare with  $f_z = \log(x) + y + 2z$  to obtain  $h'(z) = 2z$ . Therefore,  $h(z) = z^2 + C$ . Hence,

$$f(x, y, z) = z \log(x) + xy + yz + z^2 + C$$

A convenient potential function is

$$f(x, y, z) = z \log(x) + xy + yz + z^2$$

**Example 2.2.10.** Let

$$\omega = \left( \frac{z}{x} + y \right) dx + (x + z)dy + (\log(x) + y + 2z)dz$$

on the region

$$U = \{(x, y, z) \in \mathbb{R}^3 : x > 0\}$$

From Example 2.2.9, a potential function is

$$f(x, y, z) = z \log(x) + xy + yz + z^2$$

Let  $C$  be the curve

$$\vec{r}(t) = (e^t, t^6 + 4t^3 - 1, t^4 + (t - t^2)e^{\sin(t)})$$

for  $0 \leq t \leq 1$ . The initial point is  $\vec{r}(0) = (1, -1, 0)$ , and the terminal point is  $\vec{r}(1) = (e, 4, 1)$ .

Since  $\omega = df$ , the Fundamental Theorem for Line Integrals (Theorem 2.1.18),

$$\int_C \omega = f(\vec{r}(1)) - f(\vec{r}(0)) = (4e + 6) - (-1) = 4e + 7$$

The complicated parametrization of  $C$  does not matter once a potential function is known.

The next result gives an important condition under which closedness is enough to guarantee exactness.

**Definition 2.2.11.** Let  $U \subseteq \mathbb{R}^n$  be an open set. We say that  $U$  is *star-shaped with respect to a point*  $\vec{a} \in U$  if for every  $\vec{x} \in U$ , the line segment from  $\vec{a}$  to  $\vec{x}$  lies entirely in  $U$ .

Equivalently,

$$\vec{a} + t(\vec{x} - \vec{a}) \in U$$

for every  $\vec{x} \in U$  and every  $0 \leq t \leq 1$ .

A set is called *star-shaped* if it is star-shaped with respect to at least one point.

**Example 2.2.12.** The following sets are star-shaped:

1. Every open ball in  $\mathbb{R}^n$ .
2. Every convex open set.
3. The whole space  $\mathbb{R}^n$ .

4. Any open rectangle in  $\mathbb{R}^n$ .

For a convex set, every line segment between any two points lies in the set. Therefore, a convex set is star-shaped with respect to every point in the set.

**Example 2.2.13.** The punctured plane  $\mathbb{R}^2 \setminus \{\vec{0}_2\}$  is not star-shaped. No matter what point  $\vec{a} \in \mathbb{R}^2 \setminus \{\vec{0}_2\}$  we choose, there are points  $\vec{x}$  such that the line segment from  $\vec{a}$  to  $\vec{x}$  passes through the origin, which is not in the domain.

**Theorem 2.2.14.** Let  $U \subseteq \mathbb{R}^n$  be star-shaped with respect to  $\vec{a} \in U$ . Let  $\omega \in \mathcal{F}^1(U)$  be closed. Then  $\omega$  is exact. More explicitly, a potential function is given by

$$f(\vec{x}) = \int_0^1 \sum_{i=1}^n F_i(\vec{a} + t(\vec{x} - \vec{a}))(x_i - a_i) dt$$

*Proof.* For each  $\vec{x} \in U$ , the line segment from  $\vec{a}$  to  $\vec{x}$  lies in  $U$ . Parametrize this line segment by

$$\vec{r}_{\vec{x}}(t) = \vec{a} + t(\vec{x} - \vec{a})$$

for  $0 \leq t \leq 1$ . Define

$$f(\vec{x}) = \int_{\vec{r}_{\vec{x}}} \omega$$

Since  $\vec{r}'_{\vec{x}}(t) = \vec{x} - \vec{a}$ , we have

$$\omega(\vec{r}_{\vec{x}}(t))(\vec{r}'_{\vec{x}}(t)) = \sum_{i=1}^n F_i(\vec{a} + t(\vec{x} - \vec{a}))(x_i - a_i)$$

Therefore,

$$f(\vec{x}) = \int_0^1 \sum_{i=1}^n F_i(\vec{a} + t(\vec{x} - \vec{a}))(x_i - a_i) dt$$

We now show that  $df = \omega$ . Fix  $1 \leq j \leq n$ . Differentiating under the integral sign gives

$$f_{x_j}(\vec{x}) = \int_0^1 \left[ \sum_{i=1}^n F_i(\vec{a} + t(\vec{x} - \vec{a}))(x_i - a_i) \right]_{x_j} dt$$

Using the product rule and the chain rule,

$$f_{x_j}(\vec{x}) = \int_0^1 \left[ F_j(\vec{r}_{\vec{x}}(t)) + t \sum_{i=1}^n (F_i)_{x_j}(\vec{r}_{\vec{x}}(t))(x_i - a_i) \right] dt$$

Since  $\omega$  is closed,  $(F_i)_{x_j} = (F_j)_{x_i}$  for all  $i, j$ . Thus,

$$f_{x_j}(\vec{x}) = \int_0^1 \left[ F_j(\vec{r}_{\vec{x}}(t) + t \sum_{i=1}^n (F_j)_{x_i}(\vec{r}_{\vec{x}}(t))(x_i - a_i) \right] dt$$

By the chain rule,

$$F_j'(\vec{r}_{\vec{x}}(t)) = \sum_{i=1}^n (F_j)_{x_i}(\vec{r}_{\vec{x}}(t))(x_i - a_i)$$

Therefore,

$$f_{x_j}(\vec{x}) = \int_0^1 [F_j(\vec{r}_{\vec{x}}(t) + t F_j'(\vec{r}_{\vec{x}}(t))] dt$$

But

$$F_j(\vec{r}_{\vec{x}}(t) + t F_j'(\vec{r}_{\vec{x}}(t))) = (t F_j(\vec{r}_{\vec{x}}(t)))'$$

Hence,

$$f_{x_j}(\vec{x}) = \int_0^1 (t F_j(\vec{r}_{\vec{x}}(t)))' dt = F_j(\vec{r}_{\vec{x}}(1))$$

Since  $\vec{r}_{\vec{x}}(1) = \vec{x}$ , we get

$$f_{x_j}(\vec{x}) = F_j(\vec{x})$$

This holds for every  $j$ , so

$$df = F_1 dx_1 + F_2 dx_2 + \cdots + F_n dx_n = \omega$$

Therefore,  $\omega$  is exact. □

**Corollary 2.2.15.** *Let  $U \subseteq \mathbb{R}^n$  be open and convex. If  $\omega \in \mathcal{F}^1(U)$  is closed, then  $\omega$  is convex.*

*Proof.* Every convex open set is star-shaped with respect to every point in the set. Therefore, the result follows from Theorem 2.2.14. □

**Corollary 2.2.16.** *If  $\omega \in \mathcal{F}^1(\mathbb{R}^n)$  is closed, then  $\omega$  is exact.*

*Proof.* The whole space  $\mathbb{R}^n$  is convex, hence star-shaped. Therefore, the result follows from Corollary 2.2.15. □

**Example 2.2.17.** Consider  $U = \mathbb{R}^2 \setminus \{\vec{0}_2\}$  and the 1-form

$$\omega = -\frac{y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$$

First, note that  $\omega$  is closed. However, let  $C$  be the unit circle parametrized counterclockwise by

$$\vec{r}(t) = (\cos(t), \sin(t))$$

for  $0 \leq t \leq 2\pi$ . Then

$$\vec{r}^*(dx) = -\sin(t)dt, \quad \vec{r}^*(dy) = \cos(t)dt$$

Also,  $x^2 + y^2 = 1$  on  $C$ . Therefore,

$$\vec{r}^*\omega = -(\sin(t))(-\sin(t)dt) + (\cos(t))(\cos(t)dt) = dt$$

Thus,

$$\int_C \omega = \int_0^{2\pi} dt = 2\pi$$

If  $\omega$  were exact, then its integral around every closed curve would be zero. Since

$$\int_C \omega = 2\pi \neq 0$$

$\omega$  is not exact on the punctured plane.

This example shows that closed does not always imply exact.

**Remark 2.2.18.** The obstruction is not algebraic; it is topological. The punctured plane has a hole at the origin. The form

$$\omega = -\frac{y}{x^2 + y^2}dx + \frac{x}{x^2 + y^2}dy$$

measures angular change around that hole.

Locally, away from a branch cut, it behaves like  $d\theta$ . But there is no globally defined smooth angle function  $\theta : \mathbb{R}^2 \setminus \{\vec{0}_2\} \rightarrow \mathbb{R}$  whose differential is  $\omega$ .

**Example 2.2.19.** Let  $\vec{x} = (x, y, z) \in \mathbb{R}^3 \setminus \{\vec{0}_3\}$ . Consider the inverse-square vector field

$$\vec{F}(\vec{x}) = -GM \frac{\vec{x}}{\|\vec{x}\|^3}$$

The associated 1-form is

$$\omega = -GM \frac{xdx + ydy + zdz}{(x^2 + y^2 + z^2)^{3/2}}$$

on  $\mathbb{R}^3 \setminus \{\vec{0}_3\}$ .

Now,  $\|\vec{x}\| = \sqrt{x^2 + y^2 + z^2}$ . We compute

$$d\|\vec{x}\| = \frac{xdx + ydy + zdz}{\|\vec{x}\|}$$

Also,

$$d\left(\frac{1}{\|\vec{x}\|}\right) = -\frac{1}{\|\vec{x}\|^2}d\|\vec{x}\| = -\frac{xdx + ydy + zdz}{\|\vec{x}\|^3}$$

Therefore,

$$\omega = GMd\left(\frac{1}{\|\vec{x}\|}\right)$$

Hence, a potential function is

$$f(\vec{x}) = \frac{GM}{\|\vec{x}\|} = \frac{GM}{\sqrt{x^2 + y^2 + z^2}}$$

Thus,  $df = \omega$ .

**Remark 2.2.20.** Physicists often define gravitational potential energy with the opposite sign, depending on convention. In our differential-form convention, the potential function  $f$  is chosen so that  $df = \omega$  where  $\omega$  is the 1-form associated with the force field.

## 2.3 Green's Theorem

In previous sections, we studied line integrals of 1-forms and saw that exact forms integrate to zero around closed curves. Green's Theorem explains what happens when a 1-form is not exact.

The main idea is that the integral of a 1-form around the boundary of a planar region is controlled by the exterior derivative of the form inside the region. If  $\omega = Pdx + Qdy$ , then  $d\omega = (Q_x - P_y)dx \wedge dy$ . Green's Theorem says that, under suitable hypotheses

$$\int_{\partial S} \omega = \int_S d\omega$$

In coordinates, this becomes the familiar formula

$$\int_{\partial S} Pdx + Qdy = \iint_S (Q_x - P_y)dA$$

This theorem is the two-dimensional prototype of Stokes's Theorem.

**Definition 2.3.1.** Let  $S \subseteq \mathbb{R}^2$  be a bounded region with piecewise smooth boundary. The boundary  $\partial S$  is said to be *positively oriented* if it is traversed so that the region  $S$  stays on the left. For a simple region with no holes, this means that the boundary is traversed counterclockwise.

**Remark 2.3.2.** If  $S$  has holes, then positive orientation means

- the outer boundary is traversed counterclockwise,
- each inner boundary component is traversed clockwise.

This convention ensures that the region always remains on the left as one travels along the boundary.

For example, if  $S$  is an annulus, then

$$\partial S = C_{\text{outer}} - C_{\text{inner}}$$

where both circles are geometrically counterclockwise, but the inner circle receives the opposite orientation as part of the boundary of  $S$ .

We first prove Green's Theorem for rectangles. The general version follows by decomposing more complicated regions into smaller rectangles or by a limiting argument.

**Theorem 2.3.3 (Green's Theorem, Rectangles).** Let  $R = [a, b] \times [c, d] \subseteq \mathbb{R}^2$  be a rectangle, oriented by the standard orientation  $dx \wedge dy$ . Let  $\omega = P(x, y)dx + Q(x, y)dy$  be a smooth 1-form on an open set containing  $R$ . Then

$$\int_{\partial R} \omega = \int_R d\omega$$

Equivalently,

$$\int_{\partial R} Pdx + Qdy = \int_a^b \int_c^d (Q_x - P_y)dydx$$

Here,  $\partial R$  is oriented counterclockwise.

*Proof.* Write

$$\partial R = C_1 + C_2 + C_3 + C_4$$

where the four sides are oriented counterclockwise:  $C_1$  is the bottom edge from  $(a, c)$  to  $(b, c)$ ,  $C_2$  is the right edge from  $(b, c)$  to  $(b, d)$ ,  $C_3$  is the top edge from  $(b, d)$  to  $(a, d)$ , and  $C_4$  is the left edge from  $(a, d)$  to  $(a, c)$ .

We first compute the contribution from  $Pdx$ . On the edge of  $C_1$ , we have

$$y = c, \quad dy = 0, \quad x : a \rightarrow b$$

Thus,

$$\int_{C_1} P dx = \int_a^b P(x, c) dx$$

On the top edge  $C_3$ , we have

$$y = d, \quad dy = 0, \quad x : b \rightarrow a$$

Thus,

$$\int_{C_3} P dx = \int_b^a P(x, d) dx = - \int_a^b P(x, d) dx$$

On the vertical edges  $C_2$  and  $C_4$ , we have  $dx = 0$ , so

$$\int_{C_2} P dx = \int_{C_4} P dx = 0$$

Therefore,

$$\int_{\partial R} P dx = \int_a^b P(x, c) dx - \int_a^b P(x, d) dx = - \int_a^b (P(x, d) - P(x, c)) dx$$

By the Fundamental Theorem of Calculus,

$$P(x, d) - P(x, c) = \int_c^d P_y(x, y) dy$$

Hence,

$$\int_{\partial R} P dx = - \int_a^b \int_c^d P_y(x, y) dy dx$$

Now compute the contribution from  $Q dy$ . On the right edge  $C_2$ , we have

$$x = b, \quad dx = 0, \quad y : c \rightarrow d$$

Thus,

$$\int_{C_2} Q dy = \int_c^d Q(b, y) dy$$

On the left edge  $C_4$ , we have

$$x = a, \quad dx = 0, \quad y : d \rightarrow c$$

Thus,

$$\int_{C_4} Q dy = \int_d^c Q(a, y) dy = - \int_c^d Q(a, y) dy$$



On the horizontal edges  $C_1$  and  $C_3$ , we have  $dy = 0$ , so

$$\int_{C_1} Q dy = \int_{C_3} Q dy = 0$$

Therefore,

$$\int_{\partial R} Q dy = \int_c^d Q(b, y) dy - \int_c^d Q(a, y) dy = \int_c^d (Q(b, y) - Q(a, y)) dy$$

By the Fundamental Theorem of Calculus,

$$Q(b, y) - Q(a, y) = \int_a^b Q_x(x, y) dx$$

Hence,

$$\int_{\partial R} Q dy = \int_c^d \int_a^b Q_x(x, y) dx dy$$

By Fubini's Theorem,

$$\int_c^d \int_a^b Q_x(x, y) dx dy = \int_a^b \int_c^d Q_x(x, y) dy dx$$

Combining the two parts

$$\begin{aligned} \int_{\partial R} \omega &= \int_{\partial R} P dx + \int_{\partial R} Q dy \\ &= - \int_a^b \int_c^d P_y(x, y) dy dx + \int_a^b \int_c^d Q_x(x, y) dy dx \\ &= \int_a^b \int_c^d (Q_x - P_y) dy dx \end{aligned}$$

Since  $d\omega = (Q_x - P_y)dx \wedge dy$ , we have

$$\int_R d\omega = \int_a^b \int_c^d (Q_x - P_y) dy dx$$

Therefore,

$$\int_{\partial R} \omega = \int_R d\omega$$

□

**Remark 2.3.4.** The proof shows why the signs of Green's Theorem appears naturally. The term  $Q_x$  comes from the difference between the left and right sides of the rectangle, while the term  $-P_y$  comes from the difference between the top and bottom sides. Thus,  $Q_x - P_y$  measures the net local tendency of the form  $Pdx + Qdy$  to circulate counterclockwise.

**Definition 2.3.5.** A bounded subset  $S \subseteq \mathbb{R}^2$  is called a *rectangular planar region* if its boundary is a finite union of piecewise smooth curves and if  $S$  can be decomposed into finitely many simpler regions for which iterated integration is valid. In this course, our regions will usually be rectangles, disks, triangles, annuli, or regions bounded by finitely many smooth curves.

**Theorem 2.3.6 (Green's Theorem).** *Let  $S \subseteq \mathbb{R}^2$  be a regular planar region, oriented by  $dx \wedge dy$ . Let  $\omega = Pdx + Qdy$  be a smooth 1-form on an open set containing  $S$ . Then*

$$\int_{\partial S} \omega = \int_S d\omega$$

*Equivalently,*

$$\int_{\partial S} Pdx + Qdy = \iint_S (Q_x - P_y) dA$$

*The boundary  $\partial S$  is given the positive orientation.*

*Proof.* The proof for a rectangle is the essential case.

For more general regions, one decomposes  $S$  into many small rectangles or simple pieces. Green's Theorem (Theorem 2.3.3) applies to each piece. When the boundary integrals are added together, the contributions from interior edges cancel because each interior edge is traversed once in each direction.

Only the outer boundary remains. Thus,

$$\sum_{\text{pieces}} \int_{\partial S_j} \omega = \int_{\partial S} \omega$$

while

$$\sum_{\text{pieces}} \int_{S_j} d\omega = \int_S d\omega$$

Therefore,

$$\int_{\partial S} \omega = \int_S d\omega$$

□

**Remark 2.3.7.** Green's Theorem (Theorem 2.3.6) is the first major example of the general principle: The integral of a form over a boundary equals the integral of its derivative over the region.

In dimension 1, this principle is the ordinary Fundamental Theorem of Calculus

$$\int_{\partial[a,b]} f = \int_{[a,b]} df$$

In dimension 2, it becomes Green's Theorem

$$\int_{\partial S} \omega = \int_S d\omega$$

Later, in higher dimensions, this becomes Stokes's Theorem.

**Example 2.3.8.** Let  $\omega = (-y^2)dx + x^2dy$ . Let  $C$  be the positively oriented boundary of the square  $S = [0, 1]^2$ .

Here,  $P = -y^2$  and  $Q = x^2$ , so  $Q_x = 2x$  and  $P_y = -2y$ . So

$$Q_x - P_y = 2x - (-2y) = 2x + 2y$$

By Green's Theorem (Theorem 2.3.6),

$$\int_C \omega = \iint_S (2x + 2y) dA$$

Thus,

$$\begin{aligned} \int_C \omega &= \int_0^1 \int_0^1 (2x + 2y) dy dx \\ &= \int_0^1 [2xy + y^2]_{y=0}^{y=1} dx \\ &= \int_0^1 (2x + 1) dx \\ &= [x^2 + x]_0^1 \\ &= 2 \end{aligned}$$

Therefore,

$$\int_C -y^2 dx + x^2 dy = 2$$

**Remark 2.3.9.** Without Green's Theorem (Theorem 2.3.6), we would need to compute four separate line integrals, one for each side of the square. Green's Theorem turns the boundary integral into a single double integral over the region.

**Example 2.3.10.** Let  $\omega = -ydx + xdy$ . Let  $C$  be the circle of radius  $a > 0$  centered at the origin, oriented counterclockwise. Let

$$S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq a^2\}$$

Here,  $P = -y$  and  $Q = x$ . Then  $Q_x = 1$  and  $P_y = -1$ , so

$$Q_x - P_y = 1 - (-1) = 2$$

By Green's Theorem (Theorem 2.3.6),

$$\int_C \omega = \iint_S 2dA$$

Since the area of  $S$  is  $\pi a^2$ , we get

$$\int_C \omega = 2\pi a^2$$

Therefore,

$$\int_C -ydx + xdy = 2\pi a^2$$

**Remark 2.3.11.** The form  $-ydx + xdy$  measures counterclockwise rotation around the origin. Its exterior derivative is

$$d(-ydx + xdy) = 2dx \wedge dy$$

Thus, the total circulation around the circle is twice the area enclosed by the circle.

Green's Theorem can also be used to compute area by line integration.

**Proposition 2.3.12.** Let  $S \subseteq \mathbb{R}^2$  be a regular planar region with positively oriented boundary  $C = \partial S$ . Then

$$\text{Area}(S) = \int_C xdy, \quad \text{Area}(S) = - \int_C ydx$$

Equivalently,

$$\text{Area}(S) = \frac{1}{2} \int_C -ydx + xdy$$

*Proof.* First, observe that  $d(xdy) = dx \wedge dy$ . Therefore, by Green's Theorem (Theorem 2.3.6)

$$\int_C xdy = \int_S dx \wedge dy = \text{Area}(S)$$

Similarly,

$$d(-ydx) = -dy \wedge dx = dx \wedge dy$$

Therefore,

$$-\int_C ydx = \int_C -ydx = \int_S dx \wedge dy = \text{Area}(S)$$

Finally,

$$d\left(\frac{1}{2}(-ydx + xdy)\right) = \frac{1}{2}(dx \wedge dy + dx \wedge dy) = dx \wedge dy$$

Thus,

$$\frac{1}{2} \int_C -ydx + xdy = \int_S dx \wedge dy = \text{Area}(S)$$

□

**Example 2.3.13.** Let  $S$  be the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$$

where  $a, b > 0$ .

We compute its area using Green's Theorem (Theorem 2.3.6). Parametrize the positively oriented boundary by

$$\vec{r}(t) = (a \cos(t), b \sin(t))$$

for  $0 \leq t \leq 2\pi$ . Using the formula,

$$\text{Area}(S) = \int_{\partial S} xdy$$

we have  $x = a \cos(t)$  and  $dy = b \cos(t)dt$ . Therefore,

$$\text{Area}(S) = \int_0^{2\pi} (a \cos(t))(b \cos(t))dt = ab \int_0^{2\pi} \cos^2(t)dt = \pi ab$$

**Example 2.3.14.** Suppose  $C$  is a positively oriented simple closed curve parametrized by

$$\vec{r}(t) = (x(t), y(t))$$

for  $\alpha \leq t \leq \beta$ . Then  $dy = y'(t)dt$  and  $dx = x'(t)dt$ . Using

$$\text{Area}(S) = \frac{1}{2} \int_C -ydx + xdy$$

we get

$$\text{Area}(S) = \frac{1}{2} \int_{\alpha}^{\beta} (x(t)y'(t) - y(t)x'(t))dt$$

This formula is often useful when a region is described by a parametrized boundary curve.

**Proposition 2.3.15.** Let  $U \subseteq \mathbb{R}^2$  be a simply connected region. Suppose  $\omega \in \mathcal{F}^1(U)$  is a smooth and closed. Then  $\omega$  is exact.

*Proof.* Since  $d\omega = 0$ , Green's Theorem (Theorem 2.3.6) gives

$$\int_{\partial S} \omega = \int_S d\omega = 0$$

for every regular region  $S \subseteq U$ . If  $U$  is simply connected, then every closed curve can be regarded as the boundary of a region lying inside  $U$ . Therefore

$$\int_C \omega = 0$$

for every closed curve  $C$  in  $U$ . By Theorem 2.1.30,  $\omega$  is exact.  $\square$

**Remark 2.3.16.** This result explains why the condition  $d\omega = 0$  is enough on nice planar domains with no holes. However, if the domain has a hole, then a closed curve fail to bound a region inside the domain. In that case, Green's Theorem (Theorem 2.3.6) cannot be applied directly to the missing region. This is exactly what happens in the punctured plane.

**Remark 2.3.17.** There is no contradiction with Green's Theorem. The unit circle is the boundary of the disk  $x^2 + y^2 \leq 1$ . But the form

$$\omega = -\frac{y}{x^2 + y^2}dx + \frac{x}{x^2 + y^2}dy$$

is not defined at the origin. Therefore, Green's Theorem cannot be applied to the entire disk. The missing point is exactly the obstruction.

**Example 2.3.18.** Let

$$\omega = -\frac{y}{x^2 + y^2}dx + \frac{x}{x^2 + y^2}dy$$

on  $U = \mathbb{R}^2 \setminus \{\vec{0}_2\}$ . Let  $C$  be any positively oriented simple closed curve enclosing the origin. Let  $\Gamma$  be a small circle centered at the origin lying inside  $C$ , also oriented counterclockwise.

Let  $S$  be the annular region between  $C$  and  $\Gamma$ . As the boundary of  $S$ , the curve  $C$  has positive orientation, while the inner circle has negative orientation. Thus,

$$\partial S = C - \Gamma$$

Since  $d\omega = 0$  on  $S$ , Green's Theorem (Theorem 2.3.6) gives

$$0 = \int_S d\omega = \int_{\partial S} \omega$$

Therefore,

$$0 = \int_C \omega - \int_{\Gamma} \omega$$

Hence,

$$\int_C \omega = \int_{\Gamma} \omega$$

But from the unit circle computation, every positively oriented circle centered at the origin gives

$$\int_{\Gamma} \omega = 2\pi$$

Therefore,

$$\int_C \omega = 2\pi$$

**Remark 2.3.19.** The integral

$$\frac{1}{2\pi} \int_C \left( -\frac{y}{x^2 + y^2}dx + \frac{x}{x^2 + y^2}dy \right)$$

counts how many times the closed curve  $C$  winds around the origin. This integer is called the *winding number* of  $C$  around the origin. Thus, Green's Theorem (Theorem 2.3.6) connects calculus with topology: the value of a line integral can detect whether a curve surrounds a hole.

Green's Theorem also has a flux interpretation.

**Proposition 2.3.20 (Green's Theorem, Flux Form).** *Let  $\vec{F} = (P, Q)$  be a smooth vector field on a region  $S \subseteq \mathbb{R}^2$ . Then*

$$\int_{\partial S} \vec{F} \cdot \vec{n} ds = \iint_S \operatorname{div}(\vec{F}) dA$$

where  $\vec{n}$  is the outward unit normal vector along  $\partial S$ .

In coordinates,

$$\int_{\partial S} P dy - Q dx = \iint_S (P_x + Q_y) dA$$

*Proof.* Consider the 1-form

$$\eta = -Q dx + P dy$$

Then

$$d\eta = d(-Q) \wedge dx + dP \wedge dy$$

Compute

$$\begin{aligned} d(-Q) \wedge dx &= -(Q_x dx + Q_y dy) \wedge dx = Q_y dx \wedge dy, \\ dP \wedge dy &= (P_x dx + P_y dy) \wedge dy = P_x dx \wedge dy \end{aligned}$$

Thus,

$$d\eta = (P_x + Q_y) dx \wedge dy$$

By Green's Theorem,

$$\int_{\partial S} \eta = \int_S d\eta$$

Therefore,

$$\int_{\partial S} -Q dx + P dy = \iint_S (P_x + Q_y) dA$$

This is the flux form of Green's Theorem. □

**Remark 2.3.21.** The circulation form of Green's Theorem uses

$$\omega = P dx + Q dy$$

and gives

$$\int_{\partial S} P dx + Q dy = \iint_S (Q_x - P_y) dA$$

The flux form uses

$$\eta = -Q dx + P dy$$



and gives

$$\int_{\partial S} -Qdx + Pdy = \iint_S (P_x + Q_y)dA$$

The first detects rotation, and the second detects expansion or compression.



## Chapter 3

# Surface Integrals and Flux

In Chapter 2, we studied line integrals of 1-forms over curves. We now move up one dimension and study integrals of 2-forms over surfaces in  $\mathbb{R}^3$ .

The main new idea is orientation. A curve is oriented by choosing a direction of travel, while a surface is oriented by choosing a continuous unit normal vector field. Once an orientation is chosen, we can integrate differential 2-forms over the surface using a parametrization

$$\vec{r}(u, v) = (x(u, v), y(u, v), z(u, v))$$

As before, the pullback converts the surface integral into an ordinary double integral over a parameter region.

Surface integrals naturally describe flux. If  $\vec{F} = (P, Q, R)$  is a vector field in  $\mathbb{R}^3$ , then the corresponding flux 2-form is

$$\omega = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$$

Integrating this form over an oriented surface measures how much of the vector field passes through the surface.

We will also discuss surface area. While flux integrals measure the signed flow of a vector field through a surface, surface area measures the geometric size of the surface itself. For a parametrized surface, the area element is determined by

$$\|\vec{r}_u \times \vec{r}_v\| du dv$$

This chapter prepares us for Stokes's Theorem in Chapter 4, where the line integrals around the boundary of a surface will be related to surface integrals over the surface itself.

### 3.1 Oriented Surfaces in $\mathbb{R}^3$ , Area Forms, and Flux

In Chapter 2, we integrated 1-forms over oriented surfaces. We now move up one dimension and integrate 2-forms over oriented surfaces in  $\mathbb{R}^3$ . The guiding idea is the same as before. If a surface  $S$  is parametrized by  $\vec{r}: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ , then a 2-form  $\omega$  on  $\mathbb{R}^3$  can be pulled back to a 2-form on  $D$   $\vec{r}^*\omega$ .

Since  $D$  is a region in  $\mathbb{R}^2$ , this pullback can be integrated using an ordinary double integral. Thus, surface integration is again based on pullback.

**Definition 3.1.1.** Let  $D \subseteq \mathbb{R}^2$  be a region, and let  $\vec{r}: D \rightarrow \mathbb{R}^3$  be a smooth vector-valued function. We write

$$\vec{r}(u, v) = (x(u, v), y(u, v), z(u, v))$$

The image  $S = \vec{r}(D)$  is called a *parametrized surface* if

$$\vec{r}_u(u, v) \times \vec{r}_v(u, v) \neq \vec{0}_3$$

for every  $(u, v) \in D$ .

The condition  $\vec{r}_u \times \vec{r}_v \neq \vec{0}_3$  means that the two tangent vectors  $\vec{r}_u$  and  $\vec{r}_v$  are linearly independent. Therefore, they span a tangent plane to the surface.

**Remark 3.1.2.** The vectors  $\vec{r}_u(u, v)$  and  $\vec{r}_v(u, v)$  are tangent to the surface at the point  $\vec{r}(u, v)$ . The cross product

$$\vec{r}_u(u, v) \times \vec{r}_v(u, v)$$

is perpendicular to the tangent plane. Its direction determines an orientation, and its magnitude measures the local area-scaling factor of the parametrization.

**Example 3.1.3.** Let  $z = f(x, y)$  be the graph of a smooth scalar-valued function  $f: U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ . A natural parametrization is

$$\vec{r}(x, y) = (x, y, f(x, y))$$

Then  $\vec{r}_x = (1, 0, f_x)$  and  $\vec{r}_y = (0, 1, f_y)$ . Therefore,

$$\vec{r}_x \times \vec{r}_y = \det \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{bmatrix} = (-f_x, -f_y, 1)$$

Since the third component is positive, this parametrization gives the graph the upward orientation.

**Definition 3.1.4.** An *orientation* on a smooth surface  $S \subseteq \mathbb{R}^3$  is a continuous choice of unit normal vector  $\vec{n}(\vec{p})$  at every point  $\vec{p} \in S$ . That is,  $\|\vec{n}(\vec{p})\| = 1$  and  $\vec{n}(\vec{p})$  is perpendicular to the tangent plane of  $S$  at  $\vec{p}$ . A surface together with a chosen orientation is called an *oriented surface*.

**Remark 3.1.5.** Choosing a unit normal vector field is equivalent to deciding which ordered bases of each tangent plane are positively oriented. If  $(\vec{u}, \vec{v})$  is an ordered basis for the tangent plane at  $\vec{p}$ , then it is positively oriented when  $\vec{u} \times \vec{v}$  points in the same direction as the chosen vector  $\vec{n}(\vec{p})$ . Equivalently,  $(\vec{u}, \vec{v})$  is positively oriented if

$$\vec{n}(\vec{p}) \cdot (\vec{u} \times \vec{v}) > 0$$

**Definition 3.1.6.** Let  $\vec{r} : D \rightarrow S \subseteq \mathbb{R}^3$  be a parametrization satisfying  $\vec{r}_u \times \vec{r}_v \neq \vec{0}_3$ . The parametrization induces the unit normal vector field

$$\vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|}$$

This is the orientation determined by the ordered pair  $(\vec{r}_u, \vec{r}_v)$ .

**Remark 3.1.7.** If we switch the order of the parameters, then the orientation reverses. Indeed,

$$\vec{r}_v \times \vec{r}_u = -(\vec{r}_u \times \vec{r}_v)$$

Thus, the parametrizations  $\vec{r}(u, v)$  and  $\vec{r}(v, u)$  may trace out the same surface, but they induce opposite orientations.

**Example 3.1.8.** Let

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = a^2\}$$

be the sphere of radius  $a > 0$  centered at the origin. The outward unit normal at a point  $\vec{x} = (x, y, z) \in S$  is

$$\vec{n}(\vec{x}) = \frac{\vec{x}}{\|\vec{x}\|} = \frac{1}{a}(x, y, z)$$

The inward unit normal is

$$-\vec{n}(\vec{x}) = -\frac{1}{a}(x, y, z)$$

Thus, the same sphere has two possible orientations: outward and inward.

**Example 3.1.9.** Let  $S = \vec{r}(D)$  be an oriented parametrized surface in  $\mathbb{R}^3$ , where  $\vec{r}: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$  is an orientation-preserving parametrization.

Let  $\omega \in \mathcal{F}^2(U)$  be a smooth 2-form on an open set  $U \subseteq \mathbb{R}^3$  containing  $S$ . The integral of  $\omega$  over  $S$  is defined by

$$\int_S \omega = \int_D \vec{r}^* \omega$$

**Remark 3.1.10.** The definition is exactly analogous to line integrals. For a curve  $\vec{r}: [a, b] \rightarrow \mathbb{R}^n$ , we integrate a 1-form by pulling it back to an interval.

For a surface,  $\vec{r}: D \rightarrow \mathbb{R}^3$ , we integrate a 2-form by pulling it back to a region in the parameter plane.

**Proposition 3.1.11.** Let  $\vec{r}(u, v) = (x(u, v), y(u, v), z(u, v))$ . Then

$$\begin{aligned}\vec{r}^*(dy \wedge dz) &= (y_u z_v - y_v z_u) du \wedge dv, \\ \vec{r}^*(dz \wedge dx) &= (z_u x_v - z_v x_u) du \wedge dv, \\ \vec{r}^*(dx \wedge dy) &= (x_u y_v - x_v y_u) du \wedge dv.\end{aligned}$$

*Proof.* We compute the first formula. Since

$$\vec{r}^*(dy) = d(y \circ \vec{r}) = y_u du + y_v dv$$

and

$$\vec{r}^*(dz) = d(z \circ \vec{r}) = z_u du + z_v dv$$

we have

$$\begin{aligned}\vec{r}^*(dy \wedge dz) &= \vec{r}^*(dy) \wedge \vec{r}^*(dz) \\ &= (y_u du + y_v dv) \wedge (z_u du + z_v dv) \\ &= y_u z_v du \wedge dv + y_v z_u dv \wedge du \\ &= y_u z_v du \wedge dv - y_v z_u du \wedge dv \\ &= (y_u z_v - y_v z_u) du \wedge dv\end{aligned}$$

□

**Remark 3.1.12.** The coefficients in Proposition 3.1.11 are precisely the components of the cross product

$$\vec{r}_u \times \vec{r}_v$$

Indeed,

$$\vec{r}_u \times \vec{r}_v = \begin{bmatrix} y_u z_v - z_u y_v \\ z_u x_v - x_u z_v \\ x_u y_v - y_u x_v \end{bmatrix}$$

Since scalar multiplication commutes, this is the same as

$$\vec{r}_u \times \vec{r}_v = \begin{bmatrix} y_u z_v - y_v z_u \\ z_u x_v - z_v x_u \\ x_u y_v - x_v y_u \end{bmatrix}$$

Thus, pulling back the standard 2-form records the components of the oriented normal area vector.

**Definition 3.1.13.** Let  $S \subseteq \mathbb{R}^3$  be an oriented surface. The *oriented area form* on  $S$  is the 2-form  $\sigma$  that assigns to each ordered pair of tangent vectors  $(\vec{u}, \vec{v})$  the signed area of the parallelogram spanned by  $\vec{u}$  and  $\vec{v}$ . Thus,

$$\sigma(\vec{p})(\vec{u}, \vec{v}) > 0$$

when  $(\vec{u}, \vec{v})$  is a positively oriented basis for the tangent plane, and

$$\sigma(\vec{p})(\vec{u}, \vec{v}) < 0$$

when  $(\vec{u}, \vec{v})$  has the opposite orientation.

**Theorem 3.1.14.** Let  $S \subseteq \mathbb{R}^3$  be an oriented surface with unit normal vector field  $\vec{n} = (n_1, n_2, n_3)$ . Then the oriented area on  $S$  is

$$\sigma = n_1 dy \wedge dz + n_2 dz \wedge dx + n_3 dx \wedge dy$$

*Proof.* Let  $\vec{u}$  and  $\vec{v}$  be tangent vectors to  $S$  at a point  $\vec{p} \in S$ . Since  $\vec{n}$  is perpendicular to the tangent plane and has length 1, the scalar triple product  $\vec{n} \cdot (\vec{u} \times \vec{v})$  is the signed area of the parallelogram spanned by  $\vec{u}$  and  $\vec{v}$ .

Now write  $\vec{n} = (n_1, n_2, n_3)$ . The scalar triple product expands as

$$\vec{n} \cdot (\vec{u} \times \vec{v}) = n_1(dy \wedge dz)(\vec{u}, \vec{v}) + n_2(dz \wedge dx)(\vec{u}, \vec{v}) + n_3(dx \wedge dy)(\vec{u}, \vec{v})$$

Therefore,

$$\sigma(\vec{u}, \vec{v}) = (n_1 dy \wedge dz + n_2 dz \wedge dx + n_3 dx \wedge dy)(\vec{u}, \vec{v})$$

Hence,

$$\sigma = n_1 dy \wedge dz + n_2 dz \wedge dx + n_3 dx \wedge dy$$

□

**Remark 3.1.15.** The formula

$$\sigma = n_1 dy \wedge dz + n_2 dz \wedge dx + n_3 dx \wedge dy$$

says that the oriented area form is built from the components of the unit normal vector. Each standard 2-form measures signed projected area:  $dy \wedge dz$  measures the signed area projected onto the  $yz$ -plane,  $dz \wedge dx$  measures the signed area projected onto the  $zx$ -plane, and  $dx \wedge dy$  measures signed area projected onto the  $xy$ -plane.

The normal vector tells us how to combine these projected areas into true oriented surface area.

**Proposition 3.1.16.** *Let  $S = \vec{r}(D)$  be an oriented parametrized surface, and suppose the parametrization induces the chosen orientation. Then*

$$\vec{r}^* \sigma = \|\vec{r}_u \times \vec{r}_v\| du \wedge dv$$

*Proof.* Since the parametrization induces the orientation, the unit normal is

$$\vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|}$$

By the formula for the area form, we have

$$\sigma(\vec{r}_u, \vec{r}_v) = \vec{n} \cdot (\vec{r}_u \times \vec{r}_v)$$

Substituting the expression for  $\vec{n}$  gives

$$\sigma(\vec{r}_u, \vec{r}_v) = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|} \cdot (\vec{r}_u \times \vec{r}_v) = \|\vec{r}_u \times \vec{r}_v\|$$

Hence,

$$\vec{r}^* \sigma = \|\vec{r}_u \times \vec{r}_v\| du \wedge dv$$

□

**Corollary 3.1.17.** *If  $S = \vec{r}(D)$  is a parametrized surface, then its surface area is*

$$\text{Area}(S) = \int_D \|\vec{r}_u \times \vec{r}_v\| du dv$$

*Equivalently, if  $\sigma$  is the oriented area form and  $\vec{r}$  is orientation-preserving, then*

$$\text{Area}(S) = \int_S \sigma$$



**Example 3.1.18.** Let  $S$  be the graph  $z = f(x, y)$  oriented upward. A parametrization is

$$\vec{r}(x, y) = (x, y, f(x, y))$$

We computed

$$\vec{r}_x \times \vec{r}_y = (-f_x, -f_y, 1)$$

Thus, the upward unit normal is

$$\vec{n} = \frac{(-f_x, -f_y, 1)}{\sqrt{1 + f_x^2 + f_y^2}}$$

Therefore, the oriented area form is

$$\sigma = \frac{-f_x}{\sqrt{1 + f_x^2 + f_y^2}} dy \wedge dz + \frac{-f_y}{\sqrt{1 + f_x^2 + f_y^2}} dz \wedge dx + \frac{1}{\sqrt{1 + f_x^2 + f_y^2}} dx \wedge dy$$

Pulling back by  $\vec{r}(x, y)$ , we get

$$\vec{r}^* \sigma = \sqrt{1 + f_x^2 + f_y^2} dx \wedge dy$$

Therefore,

$$\text{Area}(S) = \iint_D \sqrt{1 + f_x^2 + f_y^2} dA$$

**Definition 3.1.19.** Let  $\vec{F} = (P, Q, R)$  be a smooth vector field on an open set  $U \subseteq \mathbb{R}^3$ . The *flux 2-form* associated with  $\vec{F}$  is

$$\omega_{\vec{F}} = P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy$$

**Remark 3.1.20.** This is the surface integral analogue of the 1-form associated with a vector field. For work integrals, a vector field  $\vec{F} = (P, Q, R)$  corresponds to the 1-form

$$P dx + Q dy + R dz$$

For flux integrals, the same vector field corresponds to the 2-form

$$P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy$$

The first form measures the component of  $\vec{F}$  along a curve. The second form measures the component of  $\vec{F}$  through a surface.

**Theorem 3.1.21.** *Let  $S = \vec{r}(D)$  be an oriented parametrized surface in  $\mathbb{R}^3$ , where  $\vec{r}: D \rightarrow \mathbb{R}^3$  is orientation preserving. Let  $\vec{F} = (P, Q, R)$  be a smooth vector field, and let*

$$\omega_{\vec{F}} = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$$

*be its flux 2-form. Then*

$$\int_S \omega_{\vec{F}} = \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) du dv$$

*Equivalently, if  $\vec{n}$  is the chosen unit normal to  $S$ , then*

$$\int_S \omega_{\vec{F}} = \int_S \vec{F} \cdot \vec{n} dS$$

*Proof.* Pull back the flux 2-form

$$\vec{r}^* \omega_{\vec{F}} = (P \circ \vec{r}) \vec{r}^*(dy \wedge dz) + (Q \circ \vec{r}) \vec{r}^*(dz \wedge dx) + (R \circ \vec{r}) \vec{r}^*(dx \wedge dy)$$

Using Proposition 3.1.11, we have

$$\begin{aligned} \vec{r}^* \omega_{\vec{F}} &= (P \circ \vec{r})(y_u z_v - y_v z_u) du \wedge dv \\ &\quad + (Q \circ \vec{r})(z_u x_v - z_v x_u) du \wedge dv \\ &\quad + (R \circ \vec{r})(x_u y_v - x_v y_u) du \wedge dv \end{aligned}$$

The coefficient is exactly

$$\vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v)$$

Therefore,

$$\vec{r}^* \omega_{\vec{F}} = \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) du \wedge dv$$

Integrating over  $D$  gives

$$\int_S \omega_{\vec{F}} = \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) du dv$$

Now if

$$\vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|}$$

then

$$\vec{r}_u \times \vec{r}_v = \vec{n} \|\vec{r}_u \times \vec{r}_v\|$$

Since  $dS = \|\vec{r}_u \times \vec{r}_v\| du dv$ , we obtain

$$\vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) du dv = \vec{F} \cdot \vec{n} dS$$

Thus,

$$\int_S \omega_{\vec{F}} = \int_S \vec{F} \cdot \vec{n} dS$$

□

**Remark 3.1.22.** The sign of a flux integral depends on the orientation of the surface. If the orientation is reversed, then  $\vec{n}$  is replaced by  $-\vec{n}$  and the flux changes sign

$$\int_{-S} \omega_{\vec{F}} = - \int_S \omega_{\vec{F}}$$

This is because flux measures signed flow through the surface.

**Example 3.1.23.** Let  $S$  be the graph  $z = f(x, y)$  over a region  $D \subseteq \mathbb{R}^2$  oriented upward. Let  $\vec{F} = (P, Q, R)$  be a vector field on  $\mathbb{R}^3$ . The parametrization is

$$\vec{r}(x, y) = (x, y, f(x, y))$$

Then

$$\vec{r}_x \times \vec{r}_y = (-f_x, -f_y, 1)$$

Therefore,

$$\int_S \omega_{\vec{F}} = \iint_D \vec{F}(x, y, f(x, y)) \cdot (-f_x, -f_y, 1) dx dy$$

In coordinates

$$\int_S \omega_{\vec{F}} = \iint_D [-P(x, y, f(x, y))f_x - Q(x, y, f(x, y))f_y + R(x, y, f(x, y))] dx dy$$

**Example 3.1.24.** Let  $\vec{F} = (0, 0, 1)$ . Let  $S$  be the graph  $z = f(x, y)$  over a region  $D \subseteq \mathbb{R}^2$  oriented upward. Using Example 3.1.23, we have

$$\vec{F} \cdot (-f_x, -f_y, 1) = 1$$

Therefore,

$$\int_S \omega_{\vec{F}} = \iint_D dx dy = \text{Area}(D)$$

This says that the upward flux of a constant vertical field through an upward-oriented graph equals the area of the projection of the surface onto the  $xy$ -plane.

**Example 3.1.25.** Let  $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$  be the unit sphere oriented outward. Let  $\vec{F}(x, y, z) = (x, y, z)$ . On the unit sphere,  $\vec{n}(x, y, z) = (x, y, z)$ . Therefore

$$\vec{F} \cdot \vec{n} = (x, y, z) \cdot (x, y, z) = x^2 + y^2 + z^2 = 1$$

Thus,

$$\int_S \omega_{\vec{F}} = \int_S \vec{F} \cdot \vec{n} dS = \int_S dS$$

Since the surface area of the unit sphere is  $4\pi$ , we get

$$\int_S \omega_{\vec{F}} = 4\pi$$

**Example 3.1.26.** Let  $S$  be the lateral surface of the cylinder  $x^2 + y^2 = a^2$ ,  $0 \leq z \leq h$  oriented outward. A parametrization is

$$\vec{r}(\theta, z) = (a \cos(\theta), a \sin(\theta), z)$$

for  $0 \leq \theta \leq 2\pi$  and  $0 \leq z \leq h$ . Then

$$\vec{r}_\theta = (-a \sin(\theta), a \cos(\theta), 0), \quad \vec{r}_z = (0, 0, 1)$$

Thus,

$$\vec{r}_\theta \times \vec{r}_z = (a \cos(\theta), a \sin(\theta), 0)$$

This points outward.

Let  $\vec{F}(x, y, z) = (x, y, 0)$ . Then

$$\vec{F}(\vec{r}(\theta, z)) = (a \cos(\theta), a \sin(\theta), 0)$$

Therefore,

$$\vec{F}(\vec{r}(\theta, z)) \cdot (\vec{r}_\theta \times \vec{r}_z) = a^2 \cos(\theta) + a^2 \sin^2(\theta) = a^2$$

Hence,

$$\int_S \omega_{\vec{F}} = \int_0^{2\pi} \int_0^h a^2 dz d\theta = 2\pi a^2 h$$

Therefore,

$$\int_S \omega_{\vec{F}} = 2\pi a^2 h$$

**Proposition 3.1.27.** *Let  $S \subseteq \mathbb{R}^3$  be an oriented surface with unit normal vector field  $\vec{n} = (n_1, n_2, n_3)$ . The flux 2-form associated with the vector field  $\vec{F} = \vec{n}$  is*

$$\omega_{\vec{n}} = n_1 dy \wedge dz + n_2 dz \wedge dx + n_3 dx \wedge dy$$

*Therefore,  $\omega_{\vec{n}} = \sigma$ . That is, the oriented area form is the flux of the unit normal vector field.*

**Remark 3.1.28.** This explains why flux is naturally expressed using 2-forms. The area form measures how much oriented area the surface has. A flux form weights that oriented area by the amount of the vector field passing through it. In other words,

$$\omega_{\vec{F}} = (\vec{F} \cdot \vec{n})\sigma$$

when restricted to the oriented surface  $S$ . Thus,

$$\int_S \omega_{\vec{F}} = \int_S (\vec{F} \cdot \vec{n}) dS$$

## 3.2 Surface Area

In Section 3.1, we studied oriented surfaces, area forms, and flux. The main object there was a 2-form, which changes sign when the orientation of the surface is reversed.

Surface area is different. It is a nonnegative geometric quantity. It should not depend on whether a surface is oriented upward or downward, inward or outward. Thus, surface area is not itself the integral of an arbitrary 2-form. Instead, it is computed using the magnitude of the oriented normal area vector.

If a surface is parametrized by  $\vec{r}: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ , then the surface area is given by

$$\text{Area}(S) = \iint_D \|\vec{r}_u \times \vec{r}_v\| du dv$$

The factor  $\|\vec{r}_u \times \vec{r}_v\|$  measures how much of the parametrization stretches small areas in the parameter domain.

**Definition 3.2.1.** Let  $S = \vec{r}(D)$  be a parametrized surface in  $\mathbb{R}^3$ , where  $\vec{r}: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$  is smooth and satisfies

$$\vec{r}_u \times \vec{r}_v \neq \vec{0}_3$$

The *surface area element* is denoted by  $dS$  and is defined by

$$dS = \|\vec{r}_u \times \vec{r}_v\| du dv$$

Thus, the surface area of  $S$  is

$$\text{Area}(S) = \iint_D dS = \iint_D \|\vec{r}_u \times \vec{r}_v\| du dv$$

**Remark 3.2.2.** The symbol  $dS$  should not be interpreted as the exterior derivative of a function  $S$ . It is notation for a nonoriented surface area element, just as  $ds$  denotes arclength element for curves. Both  $ds$  and  $dS$  measure geometric length or area, not signed orientation.

**Remark 3.2.3.** The vector  $\vec{r}_u \times \vec{r}_v$  is called the oriented normal area vector. Its direction is normal to the surface, and its magnitude is the area of the parallelogram spanned by the tangent vectors  $\vec{r}_u$  and  $\vec{r}_v$ . Therefore,

$$\|\vec{r}_u \times \vec{r}_v\|$$

measures how much of an infinitesimal rectangle in the  $uv$ -plane is stretched into an infinitesimal parallelogram on the surface.

**Theorem 3.2.4.** Let  $S = \vec{r}(D)$  be a parametrized surface in  $\mathbb{R}^3$  where  $\vec{r} : D \rightarrow \mathbb{R}^3$  is a smooth and one-to-one, except possibly along boundary curves. Suppose  $\vec{r}_u \times \vec{r}_v \neq \vec{0}_3$  on  $D$ . Then

$$\text{Area}(S) = \iint_D \|\vec{r}_u \times \vec{r}_v\| du dv$$

A small rectangle in the parameter domain with side lengths  $\Delta u$  and  $\Delta v$  is mapped approximately to a parallelogram on the surface spanned by  $\vec{r}_u \Delta u$  and  $\vec{r}_v \Delta v$ . The area of this parallelogram is

$$\|(\vec{r}_u \Delta u) \times (\vec{r}_v \Delta v)\| = \|\vec{r}_u \times \vec{r}_v\| \Delta u \Delta v$$

Adding these small area contributions and taking a limit gives

$$\text{Area}(S) = \iint_D \|\vec{r}_u \times \vec{r}_v\| du dv$$

**Remark 3.2.5.** The formula uses the magnitude  $\|\vec{r}_u \times \vec{r}_v\|$  rather than the vector  $\vec{r}_u \times \vec{r}_v$ . This is because surface area is not signed. If we reverse the orientation by switching  $u$  and  $v$ , then

$$\vec{r}_v \times \vec{r}_u = -(\vec{r}_u \times \vec{r}_v)$$

but the magnitudes are the same. Thus, the surface area is unchanged by reversing orientation.

**Proposition 3.2.6.** *Let  $S$  be an oriented surface with unit normal vector field  $\vec{n}$ . The oriented area form is*

$$\sigma = n_1 dy \wedge dz + n_2 dz \wedge dx + n_3 dx \wedge dy$$

*If  $\vec{r}: D \rightarrow S$  is an orientation-preserving parametrization, then*

$$\vec{r}^* \sigma = \|\vec{r}_u \times \vec{r}_v\| du \wedge dv$$

*Therefore,*

$$\int_S \sigma = \text{Area}(S)$$

*Proof.* Since  $\vec{r}$  is orientation-preserving, the chosen unit normal is

$$\vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|}$$

The area form satisfies

$$\sigma(\vec{r}_u, \vec{r}_v) = \vec{n} \cdot (\vec{r}_u \times \vec{r}_v)$$

Substituting the formula for  $\vec{n}$  gives

$$\sigma(\vec{r}_u, \vec{r}_v) = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|} \cdot (\vec{r}_u \times \vec{r}_v) = \|\vec{r}_u \times \vec{r}_v\|$$

Thus,

$$\vec{r}^* \sigma = \|\vec{r}_u \times \vec{r}_v\| du \wedge dv$$

Integrating gives

$$\int_S \sigma = \iint_D \|\vec{r}_u \times \vec{r}_v\| du dv = \text{Area}(S)$$

□

**Remark 3.2.7.** If the parametrization is orientation-reversing, then

$$\vec{r}^* \sigma = -\|\vec{r}_u \times \vec{r}_v\| du \wedge dv$$

Thus, the oriented integral

$$\int_S \sigma$$

depends on the orientation of the surface and the parametrization. But the geometric surface area always uses the nonnegative quantity  $\|\vec{r}_u \times \vec{r}_v\|$ .

The surface area can also be written using dot products instead of cross products.

**Definition 3.2.8.** Let  $\vec{r} : D \rightarrow \mathbb{R}^3$  be a parametrized surface. Define  $E = \vec{r}_u \cdot \vec{r}_u$ ,  $F = \vec{r}_u \cdot \vec{r}_v$ , and  $G = \vec{r}_v \cdot \vec{r}_v$ . The expression

$$Edu^2 + 2Fdudv + Gdv^2$$

is called the *first fundamental form* of the surface.

**Proposition 3.2.9.** Let  $\vec{r} : D \rightarrow \mathbb{R}^3$  be a parametrized surface. Then

$$\|\vec{r}_u \times \vec{r}_v\| = \sqrt{EG - F^2}$$

Therefore,

$$dS = \sqrt{EG - F^2}dudv$$

*Proof.* Recall the vector identity

$$\|\vec{a} \times \vec{b}\|^2 = \|\vec{a}\|^2\|\vec{b}\|^2 - (\vec{a} \cdot \vec{b})^2$$

Apply this with  $\vec{a} = \vec{r}_u$  and  $\vec{b} = \vec{r}_v$ . Then

$$\|\vec{r}_u \times \vec{r}_v\|^2 = \|\vec{r}_u\|^2\|\vec{r}_v\|^2 - (\vec{r}_u \cdot \vec{r}_v)^2$$

Using  $E = \vec{r}_u \cdot \vec{r}_u$ ,  $F = \vec{r}_u \cdot \vec{r}_v$ , and  $G = \vec{r}_v \cdot \vec{r}_v$ , we get

$$\|\vec{r}_u \times \vec{r}_v\|^2 = EG - F^2 \geq 0$$

Thus,

$$\|\vec{r}_u \times \vec{r}_v\| = \sqrt{EG - F^2}$$

□

**Remark 3.2.10.** The quantity  $EG - F^2$  is the determinant of the *Gram matrix*

$$\begin{bmatrix} \vec{r}_u \cdot \vec{r}_u & \vec{r}_u \cdot \vec{r}_v \\ \vec{r}_v \cdot \vec{r}_u & \vec{r}_v \cdot \vec{r}_v \end{bmatrix}$$

Thus,  $\sqrt{EG - F^2}$  is the area of the parallelogram spanned by  $\vec{r}_u$  and  $\vec{r}_v$ . This formula is useful when the cross product is computationally messy.



**Example 3.2.11.** Let  $S$  be the graph  $z = 2x - y$  over the rectangle  $D = [0, 1] \times [0, 2]$ . Here,  $f(x, y) = 2x - y$ , so

$$f_x = 2, \quad f_y = -1$$

Therefore,

$$\sqrt{1 + f_x^2 + f_y^2} = \sqrt{1 + 4 + 1} = \sqrt{6}$$

Thus,

$$\text{Area}(S) = \iint_D \sqrt{6} dx dy$$

Since  $\text{Area}(D) = 2$ , we get

$$\text{Area}(S) = 2\sqrt{6}$$

**Example 3.2.12.** Let  $S$  be the graph  $z = x^2 + y^2$  over the disk

$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq a^2\}$$

Here,  $f(x, y) = x^2 + y^2$ , so  $f_x = 2x$  and  $f_y = 2y$ . Therefore,

$$\sqrt{1 + f_x^2 + f_y^2} = \sqrt{1 + 4x^2 + 4y^2}$$

Thus,

$$\text{Area}(S) = \iint_D \sqrt{1 + 4x^2 + 4y^2}$$

Using polar coordinates, let  $x = r \cos(\theta)$  and  $y = r \sin(\theta)$ . Then  $dA = r dr d\theta$ . Hence,

$$\text{Area}(S) = \int_0^{2\pi} \int_0^a \sqrt{1 + 4r^2} r dr d\theta$$

Let  $s = 1 + 4r^2$ . Then  $ds = 8r dr$ . Therefore,

$$\int r \sqrt{1 + 4r^2} dr = \frac{1}{8} \int s^{\frac{1}{2}} ds = \frac{1}{12} s^{\frac{3}{2}}$$

Thus,

$$\text{Area}(S) = 2\pi \left[ \frac{1}{12} (1 + 4r^2)^{\frac{3}{2}} \right]_0^a = \frac{\pi}{6} \left( (1 + 4a^2)^{\frac{3}{2}} - 1 \right)$$

**Proposition 3.2.13.** Let  $z = f(r)$  for  $0 \leq r \leq a$  be a curve in the  $rz$ -plane revolved around the  $z$ -axis. A parametrization of the resulting surface is

$$\vec{r}(r, \theta) = (r \cos \theta, r \sin(\theta), f(r))$$

for  $0 \leq r \leq a$  and  $0 \leq \theta \leq 2\pi$ . Then  $dS = r\sqrt{1 + f'(r)^2}drd\theta$ . Therefore,

$$\text{Area}(S) = 2\pi \int_0^a r\sqrt{1 + f'(r)^2}dr$$

*Proof.* Compute  $\vec{r}_r = (\cos(\theta), \sin(\theta), f'(r))$  and  $\vec{r}_\theta = (-r \sin(\theta), r \cos(\theta), 0)$ . Then

$$\vec{r}_r \times \vec{r}_\theta = (-rf'(r) \cos(\theta), -rf'(r) \sin(\theta), r)$$

Therefore,

$$\begin{aligned} \|\vec{r}_r \times \vec{r}_\theta\| &= \sqrt{r^2 f'(r)^2 \cos^2(\theta) + r^2 f'(r)^2 \sin^2(\theta) + r^2} \\ &= r\sqrt{1 + f'(r)^2} \end{aligned}$$

Hence,  $dS = r\sqrt{1 + f'(r)^2}drd\theta$ . Integrating over  $0 \leq \theta \leq 2\pi$  gives

$$\text{Area}(S) = 2\pi \int_0^a r\sqrt{1 + f'(r)^2}dr$$

□

**Example 3.2.14.** Consider the cone  $z = \frac{h}{a}r$  for  $0 \leq r \leq a$ , obtained by joining the origin to the circle of radius  $a$  at height  $h$ . Here,  $f(r) = \frac{h}{a}r$ , so  $f'(r) = \frac{h}{a}$ . Therefore,

$$\text{Area}(S) = 2\pi \int_0^a r\sqrt{1 + \frac{h^2}{a^2}}dr = 2\pi \int_0^a r \frac{\sqrt{a^2 + h^2}}{a}dr$$

Therefore,

$$\text{Area}(S) = 2\pi \frac{\sqrt{a^2 + h^2}}{a} \int_0^a r dr = 2\pi \frac{\sqrt{a^2 + h^2}}{a} \cdot \frac{a^2}{2} = \pi a \sqrt{a^2 + h^2}$$

This is the lateral surface area of a cone.

**Example 3.2.15.** Let  $S$  be the sphere of radius  $a > 0$  centered at the origin. A spherical parametrization is

$$\vec{r}(\phi, \theta) = (a \sin(\phi) \cos(\theta), a \sin(\phi) \sin(\theta), a \cos(\phi))$$

where  $0 \leq \phi \leq \pi$  and  $0 \leq \theta \leq 2\pi$ . Compute

$$\begin{aligned}\vec{r}_\phi &= (a \cos(\phi) \cos(\theta), a \cos(\phi) \sin(\theta), -a \sin(\phi)), \\ \vec{r}_\theta &= (-a \sin(\phi) \sin(\theta), a \sin(\phi) \cos(\theta), 0)\end{aligned}$$

A direct calculation gives

$$\|\vec{r}_\phi \times \vec{r}_\theta\| = a^2 \sin(\phi)$$

Therefore,

$$\begin{aligned}\text{Area}(S) &= \int_0^{2\pi} \int_0^\pi a^2 \sin(\phi) d\phi d\theta \\ &= a^2 \left( \int_0^{2\pi} d\theta \right) \left( \int_0^\pi \sin(\phi) d\phi \right) \\ &= a^2 (2\pi)(2) \\ &= 4\pi a^2\end{aligned}$$

**Proposition 3.2.16.** *Let  $S$  be the plane*

$$n_1x + n_2y + n_3z = c$$

where  $\vec{n} = (n_1, n_2, n_3)$  is a unit normal vector and  $n_3 \neq 0$ . Solving for  $z$ , we obtain

$$z = \frac{c - n_1x - n_2y}{n_3}$$

Then the surface area element is

$$dS = \frac{1}{|n_3|} dx dy$$

*Proof.* Let

$$f(x) = \frac{c - n_1x - n_2y}{n_3}$$

Then  $f_x = -\frac{n_1}{n_3}$  and  $f_y = -\frac{n_2}{n_3}$ . Using the graph formula,

$$dS = \sqrt{1 + \frac{n_1^2}{n_3^2} + \frac{n_2^2}{n_3^2}} dx dy$$

Since  $\|\vec{n}\|^2 = n_1^2 + n_2^2 + n_3^2 = 1$ , we have

$$1 + \frac{n_1^2 + n_2^2}{n_3^2} = \frac{n_1^2 + n_2^2 + n_3^2}{n_3^2} = \frac{1}{n_3^2}$$

Thus,

$$dS = \frac{1}{|n_3|} dxdy$$

□

**Remark 3.2.17.** The formula

$$dS = \frac{1}{|n_3|} dxdy$$

says that the area of a piece of plane is the area of its projection onto the  $xy$ -plane, scaled by the factor  $\frac{1}{|n_3|}$ . If the plane is nearly horizontal, then  $|n_3|$  is close to 1, so the scaling factor is close to 1. If the plane is nearly vertical, then  $|n_3|$  is close to zero, so the projection onto the  $xy$ -plane becomes very small compared with the actual surface area.

A surface can have many different parametrizations. The area formula should not depend on which parametrization is used.

**Proposition 3.2.18.** *Let  $\vec{r}: D \rightarrow S$  be a parametrization of a surface, and let  $\vec{\phi}: \tilde{D} \rightarrow D$  be a smooth one-to-one change of parameters with nonzero Jacobian determinant. Then  $\tilde{\vec{r}} = \vec{r} \circ \vec{\phi}$  is another parametrization of  $S$ , and*

$$\iint_{\tilde{D}} \|\tilde{\vec{r}}_\alpha \times \tilde{\vec{r}}_\beta\| d\alpha d\beta = \iint_D \|\vec{r}_u \times \vec{r}_v\| dudv$$

*Proof Sketch.* By the chain rule, the tangent vectors of the new parametrization are linear combinations of the old tangent vectors

$$\tilde{\vec{r}}_\alpha = \vec{r}_u u_\alpha + \vec{r}_v v_\alpha, \quad \tilde{\vec{r}}_\beta = \vec{r}_u u_\beta + \vec{r}_v v_\beta$$

Using bilinearity and alternating properties of the cross product, one obtains

$$\tilde{\vec{r}}_\alpha \times \tilde{\vec{r}}_\beta = \det(D\vec{\phi})(\vec{r}_u \times \vec{r}_v)$$

Therefore,

$$\|\tilde{\vec{r}}_\alpha \times \tilde{\vec{r}}_\beta\| = |\det(D\vec{\phi})| \|\vec{r}_u \times \vec{r}_v\|$$

The ordinary Change of Variables Theorem then gives the desired equality. □

**Remark 3.2.19.** The absolute value  $|\det(D\vec{\phi})|$  is important because surface area is nonoriented. If the reparametrization reverses orientation, then  $\det(D\vec{\phi}) < 0$ , but the surface area remains positive.

**Remark 3.2.20 (The Schwarz Lantern Phenomenon).** It might seem natural to define surface area by approximating a surface with many small rectangles and taking the limit of the sum of their areas. For curves, approximating length by a polygonal paths works well. For surfaces, however, this idea is more delicate. Certain triangulations of a smooth surface can have total area tending to infinity, even when the surface has finite area.

This phenomenon shows that surface area must be defined carefully. The parametrization formula

$$\text{Area}(S) = \iint_D \|\vec{r}_u \times \vec{r}_v\| du dv$$

is a reliable way to compute surface area for smooth parametrized surfaces.



## Chapter 4

# Stokes's Theorem

In Chapters 2 and 3, we studied integration over curves and surfaces. Green's Theorem related a line integral around the boundary of a planar region to a double itnegral over the region itself

$$\int_{\partial S} \omega = \int_S d\omega$$

Stokes's Theorem extends this idea to higher-dimensional objects. Instead of working with planar regions, we will integrate differential forms over compact, oriented  $k$ -dimensional manifolds. The boundary of such a manifold is a  $(k - 1)$ -dimensional object, and Stokes's Theorem relates the integral of a form over that boundary to the integral of its exterior derivative over the manifold

$$\int_{\partial M} \omega = \int_M d\omega$$

This single formula contains several familiar theorems as special cases. When  $M$  is an interval, it becomes the ordinary Fundamental Theorem of Calculus. When  $M$  is a planar region, it becomes Green's Theorem. When  $M$  is a surface in  $\mathbb{R}^3$ , it becomes the classical Stokes's Theorem from vector calculus.

The main work in this chapter is to carefully define what it means to integrate a differential form over a general oriented manifold. Once this is done, Stokes's Theorem reveals the central message of differential forms: the integral of a derivative over a region is determined by the behaviour of the original form on the boundary.

## 4.1 Integrating over a General Compact, Oriented $k$ -Dimensional Manifold

So far, we have studied forms over objects that were described by a single parametrization. For example, a curve could be parametrized by  $\vec{r}: [a, b] \rightarrow \mathbb{R}^n$ , and a surface could be parametrized by  $\vec{r}: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ . In both cases, the integral was defined by pulling the form back to the parameter domain.

For a general manifold, however, one parametrization may not be enough. A curve, surface, or higher-dimensional object may need to be covered by many parametrized pieces. If these pieces overlap, we cannot simply add the integrals over all pieces, because the overlap would be counted more than once.

The purpose of this section is to explain how to define

$$\int_M \omega$$

when  $M$  is a compact, oriented  $k$ -dimensional manifold and  $\omega$  is a  $k$ -form. The key tool is a *partition of unity*, which allows us to split a global form into smaller pieces supported inside coordinate charts.

**Definition 4.1.1.** We define the closed upper half-space in  $\mathbb{R}^k$  by

$$\mathbb{H}^k = \{(u_1, u_2, \dots, u_k) \in \mathbb{R}^k : u_k \geq 0\}$$

Its boundary is

$$\partial\mathbb{H}^k = \{(u_1, u_2, \dots, u_k) \in \mathbb{R}^k : u_k = 0\}$$

Thus,  $\partial\mathbb{H}^k \simeq \mathbb{R}^{k-1}$ .

**Definition 4.1.2.** A subset  $M \subseteq \mathbb{R}^n$  is called a  *$k$ -dimensional manifold with boundary* if for every point  $\vec{p} \in M$ , there exists an open set  $W \subseteq \mathbb{R}^n$  containing  $\vec{p}$  and a smooth parametrization  $\vec{g}: U \rightarrow W \cap M$  where  $U$  is an open subset of either  $\mathbb{R}^k$  or of  $\mathbb{H}^k$  such that

1.  $\vec{g}$  is one-to-one
2.  $\vec{g}(U) = W \cap M$
3.  $D\vec{g}(\vec{u})$  has rank  $k$  for every  $\vec{u} \in U$ .
4. The inverse map  $\vec{g}^{-1}: W \cap M \rightarrow U$  is continuous.

The map  $\vec{g}: U \rightarrow W \cap M$  is called a *coordinate chart* or *local parametrization* for  $M$ .



**Remark 4.1.3.** A manifold is an object that locally looks like Euclidean space. If  $U \subseteq \mathbb{R}^k$  then the chart describes an interior piece of the manifold. If  $U \subseteq \mathbb{H}^k$ , then the chart may describe a piece near the boundary. The condition  $\text{rank}(D\vec{g}(\vec{u})) = k$  ensures that the tangent vectors  $\vec{g}_{u_1}, \vec{g}_{u_2}, \dots, \vec{g}_{u_k}$  are linearly independent and span a  $k$ -dimensional tangent space.

**Definition 4.1.4.** Let  $M \subseteq \mathbb{R}^n$  be a  $k$ -dimensional manifold with boundary. A point  $\vec{p} \in M$  is called a *boundary point* of  $M$  if, in some boundary chart  $\vec{g} : U \subseteq \mathbb{H}^k \rightarrow M$ , we have  $\vec{p} = \vec{g}(\vec{u})$  for some  $\vec{u} \in \partial\mathbb{H}^k$ . The set of all boundary points is denoted by  $\partial M$ . Points of  $M \setminus \partial M$  are called *interior points* of  $M$ .

**Example 4.1.5.** The closed interval  $M = [a, b] \subseteq \mathbb{R}$  is a compact 1-dimensional manifold with boundary. Its boundary is  $\partial M = \{a, b\}$  and the interior is  $(a, b)$ .

**Example 4.1.6.** The closed disk

$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$$

is a compact 2-dimensional manifold with boundary. Its boundary is the unit circle

$$\partial D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

**Example 4.1.7.** The sphere

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$$

is a compact 2-dimensional manifold without boundary. Thus,

$$\partial S^2 = \emptyset$$

**Example 4.1.8.** The closed solid ball

$$B^3 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$$

is a compact 3-dimensional manifold with boundary. Its boundary is the sphere

$$\partial B^3 = S^2$$

**Definition 4.1.9.** Let  $M \subseteq \mathbb{R}^n$  be a  $k$ -dimensional manifold, and let  $\vec{p} \in M$ . Suppose  $\vec{p} = \vec{g}(\vec{u})$  for a coordinate chart  $\vec{g} : U \subseteq \mathbb{R}^k \rightarrow M$ . The *tangent space* to  $M$  at  $\vec{p}$  is the  $k$ -dimensional vector space

$$T_{\vec{p}}(M) = \text{span}\{\vec{g}_{u_1}(\vec{u}), \vec{g}_{u_2}(\vec{u}), \dots, \vec{g}_{u_k}(\vec{u})\}$$

**Remark 4.1.10.** The tangent space does not depend on the choice of coordinate chart. Different charts give different bases, but they span the same tangent space. Thus, the tangent space is an intrinsic object attached to the manifold at the point  $\vec{p}$ .

**Example 4.1.11.** Let  $S = \vec{r}(D)$  be a surface in  $\mathbb{R}^3$ , where  $\vec{r}: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ . Then

$$T_{\vec{r}(u,v)}(S) = \text{span}\{\vec{r}_u(u, v), \vec{r}_v(u, v)\}$$

This agrees with our earlier description of tangent planes to parametrized surfaces.

**Definition 4.1.12.** Let  $M \subseteq \mathbb{R}^n$  be a  $k$ -dimensional manifold. A differential  $j$ -form  $\omega$  on  $M$  assigns to each point  $\vec{p} \in M$  an alternating  $j$ -covector on the tangent space  $T_{\vec{p}}(M)$ . Thus,  $\omega(\vec{p})$  acts on  $j$  tangent vectors  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_j \in T_{\vec{p}}(M)$ .

In these notes, we usually restrict forms on  $M$  by restricting differential forms defined on an open subset  $U \subseteq \mathbb{R}^n$  containing  $M$ .

**Remark 4.1.13.** If  $\omega$  is a  $j$ -form on an open set  $U \subseteq \mathbb{R}^n$  and  $M \subseteq U$  is a manifold, then  $\omega$  can be evaluated on tangent vectors to  $M$ . Thus, we may integrate a  $k$ -form over a  $k$ -dimensional manifold

$$\int_M \omega$$

However, we cannot integrate a  $j$ -form over a  $k$ -dimensional manifold when  $j \neq k$  in the same way. The degree of the form must match the degree of the manifold.

**Definition 4.1.14.** Let  $M \subseteq \mathbb{R}^n$  be a  $k$ -dimensional manifold. An *orientation* on  $M$  is a continuous choice of which ordered bases of each tangent space  $T_{\vec{p}}(M)$  are considered positively oriented. Equivalently,  $M$  is oriented if it can be covered by coordinate charts  $\vec{g}: U \rightarrow M$  such that the ordered basis  $(\vec{g}_{u_1}, \vec{g}_{u_2}, \dots, \vec{g}_{u_k})$  is positively oriented at every point of the chart. A manifold that admits an orientation is called *orientable*.

**Remark 4.1.15.** For curves, an orientation is a choice of direction. For surfaces in  $\mathbb{R}^3$ , an orientation is a choice of continuous normal unit vector field. For a general  $k$ -dimensional manifold, an orientation is a consistent choice of positive ordered bases for its tangent spaces.

**Definition 4.1.16.** Let  $M$  be an oriented  $k$ -dimensional manifold. A chart  $\vec{g} : U \rightarrow M$  is called *orientation-preserving* if  $(\vec{g}_{u_1}, \vec{g}_{u_2}, \dots, \vec{g}_{u_k})$  is a positively oriented basis of  $T_{\vec{g}(\vec{u})}(M)$  for every  $\vec{u} \in U$ . If the ordered basis has the opposite direction, then the chart is called *orientation-reversing*.

**Remark 4.1.17.** When we integrate over an oriented manifold, we use orientation-preserving charts. If a chart reverses orientation, the integral sign changes. This is why orientation must be handled carefully.

**Definition 4.1.18.** Let  $M \subseteq \mathbb{R}^n$  be an oriented  $k$ -dimensional manifold. The *oriented volume form* on  $M$  is the  $k$ -form  $\sigma$  with the property that, at each point  $\vec{p} \in M$ , the value  $\sigma(\vec{p})(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k)$  is the signed  $k$ -dimensional volume of the parallelepiped spanned by  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in T_{\vec{p}}(M)$ .

It is positive when  $(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k)$  is a positively oriented basis and negative when the orientation is reversed.

**Remark 4.1.19.** The volume form generalizes several familiar objects. For an oriented curve, the volume form is the signed arclength form. For an oriented surface in  $\mathbb{R}^3$ , the volume form is the oriented area form

$$\sigma = n_1 dy \wedge dz + n_2 dz \wedge dx + n_3 dx \wedge dy$$

For an oriented region in  $\mathbb{R}^n$ , the standard volume form is  $dx_1 \wedge dx_2 \wedge \dots \wedge dx_n$ .

Suppose  $M$  is covered by the coordinate charts  $\vec{g}_1, \vec{g}_2, \dots, \vec{g}_N$  where for each  $j$ ,  $\vec{g}_j : U_j \rightarrow M$ . If these charts overlap, then a naive definition such as

$$\sum_{i=1}^N \int_{U_i} \vec{g}_i^* \omega$$

will generally count the same part of  $M$  more than once. To avoid this, we use smooth scalar functions  $\rho_1, \rho_2, \dots, \rho_N$  that add up to 1 on  $M$ , where each function is supported inside one coordinate chart. Then the form  $\omega$  is split into pieces

$$\omega = \rho_1 \omega + \rho_2 \omega + \dots + \rho_N \omega$$

Each piece  $\rho_i \omega$  lives inside one chart, so it can be integrated using pullback.

**Definition 4.1.20.** Let  $\rho : M \rightarrow \mathbb{R}$  be a smooth scalar-valued function. The *support* of  $\rho$  is the closure of the set where  $\rho$  is nonzero

$$\text{supp}(\rho) = \overline{\{\vec{p} \in M : \rho(\vec{p}) \neq 0\}}$$

Informally, the support of  $\rho$  is the region where  $\rho$  actually contributes.

**Definition 4.1.21.** Let  $M$  be a compact manifold covered by finitely many coordinate charts. A *partition of unity subordinate to this cover* is a finite collection of smooth scalar-valued functions  $\rho_1, \rho_2, \dots, \rho_N : M \rightarrow \mathbb{R}$  such that

1. For every  $i$ ,  $0 \leq \rho_i \leq 1$ .
2. Each  $\rho_i$  is zero outside some coordinate chart.
3. The functions add up to 1 on  $M$ :  $\rho_1 + \rho_2 + \dots + \rho_N = 1$ .

**Theorem 4.1.22.** Let  $M \subseteq \mathbb{R}^n$  be a compact  $k$ -dimensional manifold with boundary. Then there exist smooth scalar-valued functions  $\rho_1, \rho_2, \dots, \rho_N : M \rightarrow \mathbb{R}$  such that

1. For each  $i$ ,  $0 \leq \rho_i \leq 1$ .
2. Each  $\rho_i$  is zero outside a coordinate ball.
3. The functions add up to 1

The collection  $\{\rho_i\}_{i=1}^N$  is called a *partition of unity on  $M$* .

*Proof Sketch.* The proof uses smooth bump functions. A bump function is a smooth scalar-valued function that is equal to 1 on a smaller region and is equal to 0 outside a slightly larger region. The idea is

1. Cover  $M$  by coordinate balls.
2. Since  $M$  is compact, choose finitely many coordinate balls that still cover  $M$ .
3. On each coordinate ball, build a bump function.
4. Normalize the bump functions so that their sum is exactly 1.

This produces smooth functions  $\rho_1, \rho_2, \dots, \rho_N$  that allow us to split the form  $\omega$  into local pieces.  $\square$

**Remark 4.1.23.** Partitions of unity are one of the main tools that make global integration possible. They allow us to reduce questions on a complicated manifold to questions on coordinate charts, where ordinary calculus applies.

**Definition 4.1.24.** Let  $M \subseteq \mathbb{R}^n$  be a compact, oriented  $k$ -dimensional manifold with piecewise smooth boundary. Let  $\omega$  be a smooth  $k$ -form on  $M$ . Choose a partition of unity  $\rho_1, \rho_2, \dots, \rho_N$  on  $M$  such that each  $\rho_i$  is supported inside an orientation-preserving coordinate chart  $\vec{g}_i : U_i \rightarrow M$ . We define

$$\int_M \omega = \sum_{i=1}^N \int_{U_i} \vec{g}_i^*(\rho_i \omega)$$

Equivalently,

$$\int_M \omega = \sum_{i=1}^N \int_{\vec{g}_i(U_i)} \rho_i \omega$$

Each integral is well-defined because  $\rho_i \omega$  is supported inside the image of a single coordinate chart.

**Remark 4.1.25.** The formula

$$\int_M \omega = \sum_{i=1}^N \int_{U_i} \vec{g}_i^*(\rho_i \omega)$$

should be read as follows: the functions  $\rho_i$  split  $\omega$  into local pieces  $\omega = \rho_1 \omega + \rho_2 \omega + \cdots + \rho_N \omega$ . Each local piece is supported inside one coordinate chart. The chart pulls that local piece back to a region in  $\mathbb{R}^K$ . Then we integrate using ordinary multivariable integration.

**Remark 4.1.26.** The definition looks dependent on several choices

- the coordinate charts
- the partition of unity
- the parametrizations used in the charts.

The fundamental fact is that the final value of

$$\int_M \omega$$

does not depend on these choices. This independence follows from the change of variables theorem and the compatibility of orientation-preserving coordinate changes.

**Proposition 4.1.27.** *Let  $\vec{g} : U \subseteq \mathbb{R}^k \rightarrow M$  and  $\vec{h} : V \subseteq \mathbb{R}^K \rightarrow M$  be two orientation-preserving coordinate charts whose images overlap. On the overlap, the transition map is  $\vec{\phi} : \vec{h}^{-1} \circ \vec{g}$ . If both charts preserve orientation, then  $\det(D\vec{\phi}) > 0$ . Therefore, for any compactly supported  $k$ -form  $\omega$  supported in the overlap,*

$$\int_U \vec{g}^* \omega = \int_V \vec{h}^* \omega$$

*Proof.* Since  $\vec{g} = \vec{h} \circ \vec{\phi}$ , we have

$$\vec{g}^* \omega = (\vec{h} \circ \vec{\phi})^* \omega$$

Using the composition rule for pullbacks,

$$(\vec{h} \circ \vec{\phi})^* \omega = \vec{\phi}^* (\vec{h}^* \omega)$$

The ordinary change of variables theorem applies because  $\det(D\vec{\phi}) > 0$ . Therefore, the integral computed in the  $\vec{g}$ -coordinates agrees with the integral computed in the  $\vec{h}$ -coordinates.  $\square$

**Remark 4.1.28.** This proposition is the reason orientation matters. If a transition map reverses orientation, then  $\det(D\vec{\phi}) < 0$  and the integral changes sign. Thus, a consistent orientation is necessary in order for the integral of a  $k$ -form over  $M$  to be well-defined.

**Example 4.1.29.** Let  $M = [a, b]$  with the standard orientation from left to right. A 1-form on  $M$  has the form  $\omega = f(x)dx$ . Then

$$\int_M \omega = \int_a^b f(x)dx$$

if the orientation is reversed, then

$$\int_{-M} \omega = - \int_M \omega$$

**Example 4.1.30.** Let  $M = S \subseteq \mathbb{R}^2$  be a compact region oriented by  $dx \wedge dy$ . If  $\omega = f(x, y)dx \wedge dy$ , then

$$\int_S \omega = \iint_S f(x, y)dA$$

This is the ordinary double integral written in differential-form language.

**Example 4.1.31.** Let  $S \subseteq \mathbb{R}^3$  be an oriented surface with unit normal vector field  $\vec{n} = (n_1, n_2, n_3)$ . The oriented area form is

$$\sigma = n_1 dy \wedge dz + n_2 dz \wedge dx + n_3 dx \wedge dy$$

Then

$$\int_S \sigma$$

is the oriented surface area of  $S$  with the chosen orientation. If  $\vec{F} = (P, Q, R)$  is a vector field, then

$$\omega_{\vec{F}} = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$$

then

$$\int_S \omega_{\vec{F}} = \int_S \vec{F} \cdot \vec{n} dS$$

Thus, flux is an example of integrating a 2-form over an oriented 2-dimensional manifold.

**Example 4.1.32.** Let  $S^2$  be the unit sphere oriented outward. This surface cannot be covered by one smooth coordinate chart without singularities. For example, spherical coordinates have singular behaviour at the poles.

Therefore, to integrate a 2-form over the entire space, we cover the sphere by several coordinate charts and use a partition of unity.

The final value of the integral is independent of the chosen charts and partition of unity.

**Proposition 4.1.33.** *Let  $M$  be an oriented compact  $k$ -dimensional manifold, and let  $-M$  denote the same manifold with the opposite direction. If  $\omega$  is a smooth  $k$ -form on  $M$ , then*

$$\int_{-M} \omega = - \int_M \omega$$

*Proof.* Reversing the orientation means that every orientation-preserving chart for  $M$  becomes orientation-reversing for  $-M$ . Equivalently, one may switch two coordinate variables in each chart. This changes the sign of the pullback of a  $k$ -form because wedge products are alternating.

Thus, every local integral changes sign, and so the global integral changes sign

$$\int_{-M} \omega = - \int_M \omega$$

□

To state Stokes's Theorem, we need to orient the boundary  $\partial M$  of an oriented manifold  $M$ . The rule is called the *outward-normal-first convention*.

For now, the idea is an ordered basis  $(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_{k-1})$  for  $T_{\vec{p}}(\partial M)$  is positively oriented if  $(\vec{n}_{\text{out}}, \vec{v}_1, \vec{v}_2, \dots, \vec{v}_{k-1})$  is a positively oriented basis for  $T_{\vec{p}}(M)$  where  $\vec{n}_{\text{out}}$  is an outward-pointing tangent vector to  $M$  normal to the boundary.

**Remark 4.1.34.** For a planar region in  $\mathbb{R}^2$  with the standard orientation  $dx \wedge dy$ , this convention gives the usual positive orientation of the boundary. The region stays on the left as one travels around the boundary.

For a surface in  $\mathbb{R}^3$ , the boundary orientation agrees with the right-hand rule.

We will use this convention in the next section when stating Stokes's Theorem.

## 4.2 Stokes's Theorem

In Section 4.1, we defined how to integrate a  $k$ -form over a compact, oriented  $k$ -dimensional manifold. We now state and study the central theorem of the course.

Stokes's Theorem says that integration and exterior differentiation are related by boundary

$$\int_{\partial M} \omega = \int_M d\omega$$

This formula generalizes several familiar theorems

- The ordinary Fundamental Theorem of Calculus.
- The Fundamental Theorem for Line Integrals.
- Green's Theorem
- The Classical Stokes's Theorem from vector calculus, and
- The Divergence Theorem.

The power of differential forms is that all of these results become special cases of one statement.

We now give a more complete proof of Stokes's Theorem. The proof is best understood in stages. The central idea is

1. Prove the theorem in a coordinate domain in  $\mathbb{R}^k$  or in the half-space  $\mathbb{H}^k$ .
2. Use partitions of unity to reduce the general case to this local case.
3. Check that the boundary orientations give the correct signs.

**Theorem 4.2.1 (Stokes's Theorem).** *Let  $M$  be a compact, oriented  $k$ -dimensional manifold with boundary  $\partial M$ . Let  $\omega$  be a smooth  $(k-1)$ -form on  $M$ . Then*

$$\int_{\partial M} \omega = \int_M d\omega$$

**Remark 4.2.2.** The degree of the form must match the dimension of the space over which it is integrated. Since  $M$  is  $k$ -dimensional, we integrate a



$k$ -form over  $M$ . Since  $\partial M$  is  $(k-1)$ -dimensional, we integrate a  $(k-1)$ -form over  $\partial M$ .

Thus, if  $\omega$  is a  $(k-1)$ -form, then  $d\omega$  is a  $k$ -form, and both sides of Stokes's Theorem make sense.

**Remark 4.2.3.** Stokes's Theorem says that the integral of a derivative over a region is determined by the behaviour of the original object on the boundary.

This is the same idea as the ordinary Fundamental Theorem of Calculus

$$\int_a^b f'(x)dx = f(b) - f(a)$$

In differential form notation, this becomes

$$\int_{[a,b]} df = \int_{\partial[a,b]} f$$

We first prove Stokes's Theorem for compactly supported forms on open subsets of Euclidean space.

**Lemma 4.2.4.** *Let  $U \subseteq \mathbb{R}^k$  be open, and let  $\eta$  be a smooth compactly supported  $(k-1)$ -form on  $U$ . Then*

$$\int_U d\eta = 0$$

*Proof.* Since  $\eta$  is a  $(k-1)$ -form on  $U \subseteq \mathbb{R}^k$  it can be written as

$$\eta = \sum_{i=1}^k f_i du_1 \wedge du_2 \wedge \cdots \wedge \widehat{du_i} \wedge \cdots \wedge du_k$$

where the hat means that the factor  $du_i$  is omitted.

For each term, we compute

$$d(f_i du_1 \wedge du_2 \wedge \cdots \wedge \widehat{du_i} \wedge \cdots \wedge du_k) = df_i \wedge du_1 \wedge du_2 \wedge \cdots \wedge \widehat{du_i} \wedge \cdots \wedge du_k$$

Since

$$df_i = \sum_{j=1}^k (f_i)_{u_j} du_j$$

all terms vanish except the term with  $j = i$ . Indeed, if  $j \neq i$ , then  $du_j$  already appears in

$$du_1 \wedge du_2 \wedge \cdots \wedge \widehat{du_i} \wedge \cdots \wedge du_k$$

so the wedge product contains a repeated factor, and is zero. Thus,

$$d\eta = \sum_{i=1}^k (f_i)_{u_j} du_i \wedge du_1 \wedge du_2 \wedge \cdots \wedge \widehat{du_i} \wedge \cdots \wedge du_k$$

To move  $du_i$  into the  $i$ th position, it must pass over  $i-1$  forms. Therefore

$$du_i \wedge du_1 \wedge du_2 \wedge \cdots \wedge \widehat{du_i} \wedge \cdots \wedge du_k = (-1)^{i-1} du_1 \wedge du_2 \wedge \cdots \wedge du_k$$

Hence,

$$d\eta = \sum_{i=1}^k (-1)^{i-1} (f_i)_{u_j} du_1 \wedge du_2 \wedge \cdots \wedge du_k$$

Therefore,

$$\int_U d\eta = \sum_{i=1}^k (-1)^{i-1} \int_U (f_i)_{u_j} du_1 du_2 \cdots du_k$$

Because  $\eta$  has compact support on  $U$ , each coefficient function  $f_i$  has compact support in  $U$ . Choose a rectangle  $R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_k, b_k]$  large enough to contain the support of every  $f_i$ . Extend each  $f_i$  by zero outside  $U$ . Then

$$\int_U (f_i)_{u_j} du_1 du_2 \cdots du_k = \int_R (f_i)_{u_j} du_1 du_2 \cdots du_k$$

By Fubini's Theorem and the ordinary Fundamental Theorem of Calculus,

$$\int_R (f_i)_{u_j} du_1 du_2 \cdots du_k = 0$$

because  $f_i$  is zero on the two faces  $u_i = a_i$  and  $u_i = b_i$ . Thus, each term vanishes, so

$$\int_U d\eta = 0$$

as desired. □

**Remark 4.2.5.** Lemma 4.2.4 corresponds to an interior coordinate chart. If a form is supported entirely inside the interior of a manifold, then it does not contribute anything to the boundary integral.

So locally, away from the boundary, Stokes's Theorem says

$$\int_U d\eta = 0$$

We now prove the case where the coordinate chart meets the boundary.

**Lemma 4.2.6.** *Give  $\mathbb{H}^k$  the standard orientation  $du_1 \wedge du_2 \wedge \cdots \wedge du_k$ . Then the boundary orientation on  $\partial\mathbb{H}^k$  is*

$$(-1)^k du_1 \wedge du_2 \wedge \cdots \wedge du_{k-1}$$

*Equivalently, the standard basis  $(\frac{\partial}{\partial u_1}, \frac{\partial}{\partial u_2}, \dots, \frac{\partial}{\partial u_{k-1}})$  has the boundary orientation when  $k$  is even, and opposite orientation when  $k$  is odd.*

*Proof.* At a boundary point of  $\mathbb{H}^k$  the outward-pointing normal vector is  $-\frac{\partial}{\partial u_k}$  because the half-space lies in the direction  $u_k \geq 0$ .

By the outward-normal-first convention, an ordered basis  $(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_{k-1})$  is positively oriented in  $\mathbb{H}^k$ .

Now take  $\vec{v}_1 = \frac{\partial}{\partial u_1}, \vec{v}_2 = \frac{\partial}{\partial u_2}, \dots, \vec{v}_{k-1} = \frac{\partial}{\partial u_{k-1}}$ . Then

$$\left( -\frac{\partial}{\partial u_k}, \frac{\partial}{\partial u_1}, \frac{\partial}{\partial u_2}, \dots, \frac{\partial}{\partial u_{k-1}} \right)$$

differs from the standard ordered basis

$$\left( \frac{\partial}{\partial u_1}, \frac{\partial}{\partial u_2}, \dots, \frac{\partial}{\partial u_k} \right)$$

by moving  $\frac{\partial}{\partial u_k}$  from the last position to the first position and adding a minus sign.

Moving  $\frac{\partial}{\partial u_k}$  past  $k-1$  vectors contributes a factor  $(-1)^{k-1}$ . the extra minus sign contributes another factor  $-1$ . Thus, the total sign is  $(-1)^k$ . Therefore, the boundary orientation is

$$(-1)^k du_1 \wedge du_2 \wedge \cdots \wedge du_{k-1}$$

□

**Lemma 4.2.7.** *Let  $U \subseteq \mathbb{H}^k$  be open in the relative topology of  $\mathbb{H}^k$ . Let  $\eta$  be a smooth compactly-supported  $(k-1)$ -form on  $U$ . Then*

$$\int_U d\eta = \int_{\partial U} \eta$$

Here,  $\partial U$  means the part of the boundary lying in  $\partial\mathbb{H}^k$  with the boundary orientation.

*Proof.* As before, write

$$\eta = \sum_{i=1}^k f_i du_1 \wedge du_2 \wedge \cdots \wedge \widehat{du_i} \wedge \cdots \wedge du_k$$

From the computation in Lemma 4.2.4,

$$d\eta = \sum_{i=1}^k (-1)^{i-1} (f_i)_{u_j} du_1 \wedge du_2 \wedge \cdots \wedge du_k$$

Choose a large rectangle

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [0, b_k]$$

inside the half-space large enough to contain the support of the functions  $f_i$ . Extend the functions by zero outside  $U$ . For  $i < k$ , the integral

$$\int_R (f_i)_{u_i} du_1 du_2 \cdots du_k$$

vanishes by the ordinary Fundamental Theorem of Calculus, because  $f_i$  is zero on the faces  $u_i = a_i$  and  $u_i = b_i$ . Thus, only the term  $i = k$  can contribute. Thus, we have

$$(-1)^{k-1} \int_R (f_k)_{u_k} du_1 du_2 \cdots du_k$$

Using the ordinary Fundamental Theorem of Calculus in the  $u_k$  direction,

$$\int_0^{b_k} (f_k)_{u_k} du_k = f_k(u_1, u_2, \dots, u_{k-1}, b_k) - f(u_1, u_2, \dots, u_{k-1}, 0)$$

Since  $f_k$  is zero near the top face  $u_k = b_k$ , this becomes  $-f_k(u_1, u_2, \dots, u_{k-1}, 0)$ .

Therefore,

$$\int_U d\eta = (-1)^{k-1} \left( - \int_{\partial U} f_k(u_1, u_2, \dots, u_{k-1}, 0) du_1 du_2 \cdots du_{k-1} \right)$$

Hence,

$$\int_U d\eta = (-1)^k \int_{\partial U} f_k(u_1, u_2, \dots, u_{k-1}, 0) du_1 du_2 \cdots du_{k-1}$$

Now restrict  $\eta$  to the boundary  $u_k = 0$ . Every term in  $\eta$  containing  $du_k$  vanishes when restricted to the boundary. The only term that survives is the term omitting  $du_k$ . Thus,

$$\eta|_{\partial U} = f_k(u_1, u_2, \dots, u_{k-1}, 0) du_1 \wedge du_2 \wedge \cdots \wedge du_{k-1}$$

But by Lemma 4.2.6, the boundary orientation is

$$(-1)^k du_1 \wedge du_2 \wedge \cdots \wedge du_{k-1}$$

Therefore,

$$\int_{\partial U} \eta = (-1)^k \int_{\partial U} f_k(u_1, u_2, \dots, u_{k-1}, 0) du_1 du_2 \cdots du_{k-1}$$

This agrees with the value of

$$\int_U d\eta$$

Hence,

$$\int_U d\eta = \int_{\partial U} \eta$$

as desired.  $\square$

**Remark 4.2.8.** The half-space calculation is the only place where the boundary sign matters. The term  $(-1)^k$  comes from the outward-normal-front convention. Without that convention, the signs in Stokes's Theorem would not match the signs in the Fundamental Theorem of Calculus.

We now transfer the local result from a coordinate domain to a manifold.

**Proposition 4.2.9.** *Let  $M$  be an oriented  $k$ -dimensional manifold with boundary. Suppose  $\omega$  is a smooth  $(k-1)$ -form supported inside the image of a single orientation-preserving coordinate chart  $\vec{g}: U \rightarrow M$ , where  $U$  is open either in  $\mathbb{R}^k$  or in  $\mathbb{H}^k$ . Then*

$$\int_M d\omega = \int_{\partial M} \omega$$

*Proof.* Since  $\omega$  is supported inside the image of the chart, we may compute the integral using this chart.

By the definition of integration over a manifold,

$$\int_M d\omega = \int_U \vec{g}^*(d\omega)$$

Pullback commutes with exterior differentiation, so

$$\bar{g}^*(d\omega) = d(\bar{g}^*\omega)$$

Thus,

$$\int_M d\omega = \int_U d(\bar{g}^*\omega)$$

Now we consider the following cases.

Case 1 (Interior Chart): If  $U \subseteq \mathbb{R}^k$ , then the support of  $\omega$  does not meet the boundary of  $M$ . Hence,

$$\int_{\partial M} \omega = 0$$

By Lemma 4.2.4,

$$\int_U d(\bar{g}^*\omega) = 0$$

Therefore,

$$\int_M d\omega = \int_{\partial M} \omega$$

Case 2 (Boundary Chart): If  $U \subseteq \mathbb{H}^k$ , then by Lemma 4.2.7,

$$\int_U d(\bar{g}^*\omega) = \int_{\partial U} \bar{g}^*\omega$$

Since  $\bar{g}$  is orientation preserving and the boundary orientation is defined consistently with the half-space model, we have

$$\int_{\partial U} \bar{g}^*\omega = \int_{\partial M} \omega$$

for the portion of the boundary covered by the chart. Thus,

$$\int_M d\omega = \int_{\partial M} \omega$$

This proves the proposition. □

We now prove the full theorem.

**Proposition 4.2.10.** *Let  $M$  be a compact, oriented  $k$ -dimensional manifold with boundary. Let  $\omega$  be a smooth  $(k-1)$ -form on  $M$ . Choose a partition of unity  $\rho_1, \rho_2, \dots, \rho_N$  such that each  $\rho_i$  is supported inside a coordinate chart. Then  $\omega = \sum_{i=1}^N \rho_i \omega$ . Moreover,*

$$d\omega = \sum_{i=1}^N d(\rho_i \omega)$$

*Proof.* Since  $\rho_1 + \rho_2 + \cdots + \rho_N = 1$ , we have

$$\omega = (\rho_1 + \rho_2 + \cdots + \rho_N)\omega = \sum_{i=1}^N \rho_i \omega$$

Now apply  $d$

$$d\omega = d\left(\sum_{i=1}^N \rho_i \omega\right) = \sum_{i=1}^N d(\rho_i \omega)$$

by linearity of the exterior derivative.

Equivalently, using the product rule,

$$d(\rho_i \omega) = d\rho_i \wedge \omega + \rho_i d\omega$$

Summing over  $i$  gives

$$\sum_{i=1}^N d(\rho_i \omega) = \left(\sum_{i=1}^N d\rho_i\right) \wedge \omega + \left(\sum_{i=1}^N \rho_i\right) d\omega$$

Since

$$\sum_{i=1}^N \rho_i = 1$$

we have

$$\sum_{i=1}^N d\rho_i = 0$$

Therefore,

$$\sum_{i=1}^N d(\rho_i \omega) = d\omega$$

as desired.  $\square$

Using all of the results above, we now prove Stokes's Theorem.

*Proof of Stokes's Theorem (Theorem 4.2.1).* Let  $M$  be a compact, oriented  $k$ -dimensional manifold with boundary, and let  $\omega$  be a smooth  $(k-1)$ -form on  $M$ . Choose a partition of unity  $\rho_1, \rho_2, \dots, \rho_N$  on  $M$  such that each  $\rho_i$  is supported inside a single orientation-preserving coordinate chart.

Then

$$\omega = \sum_{i=1}^N \rho_i \omega$$

By Proposition 4.2.10, we have

$$d\omega = \sum_{i=1}^N d(\rho_i\omega)$$

Therefore,

$$\int_M d\omega = \int_M \sum_{i=1}^N d(\rho_i\omega) = \sum_{i=1}^N \int_M d(\rho_i\omega)$$

Each form  $\rho_i\omega$  is supported inside a single coordinate chart. Therefore, by Proposition 4.2.9,

$$\int_M d(\rho_i\omega) = \int_{\partial M} \rho_i\omega$$

Hence,

$$\int_M d\omega = \sum_{i=1}^N \int_{\partial M} \rho_i\omega = \int_{\partial M} \sum_{i=1}^N \rho_i\omega = \int_{\partial M} \omega$$

Therefore,

$$\int_M d\omega = \int_{\partial M} \omega$$

□

**Remark 4.2.11.** The proof shows why Stokes's Theorem is fundamentally a repeated version of the ordinary Fundamental Theorem of Calculus.

The exterior derivative creates a partial derivative terms, and integration converts those derivative terms into boundary contributions.

The role of orientation is to make all the signs agree.

**Theorem 4.2.12 (The Fundamental Theorem of Calculus).** *Let  $M = [a, b] \subseteq \mathbb{R}$  with the standard orientation. Let  $f : M \rightarrow \mathbb{R}$  be smooth. Then*

$$\int_a^b df = f(b) - f(a)$$

*Proof.* Here,  $M$  is 1-dimensional, so  $\omega = f$  is a 0-form. Then

$$d\omega = df = f'(x)dx$$

By Stokes's Theorem (Theorem 4.2.1),

$$\int_M df = \int_{\partial M} f$$



Since  $\partial M = \{b\} \setminus \{a\}$ , we get

$$\int_{\partial M} f = f(b) - f(a)$$

Therefore,

$$\int_a^b f'(x)dx = f(b) - f(a)$$

□

**Theorem 4.2.13 (The Fundamental Theorem for Line Integrals).** *Let  $C$  be an oriented curve in  $\mathbb{R}^n$  starting at  $A$  and ending at  $B$ . Let  $f$  be a smooth scalar-valued function on an open set containing  $C$ . Then*

$$\int_C df = f(B) - f(A)$$

*Proof.* The curve  $C$  is a compact oriented 1-dimensional manifold with boundary  $\partial C = \{B\} \setminus \{A\}$ . Apply Stokes's Theorem (Theorem 4.2.1) to the 0-form  $f$ . Then

$$\int_C df = \int_{\partial C} f = f(B) - f(A)$$

□

**Remark 4.2.14.** This is exactly the Fundamental Theorem for Line Integrals (Theorem 2.1.18). If  $df = \omega$ , then

$$\int_C \omega$$

depends only on the endpoints of  $C$ . Thus, exact 1-forms are path independent.

**Theorem 4.2.15 (Green's Theorem).** *Let  $S \subseteq \mathbb{R}^2$  be a compact, oriented planar region with positively oriented boundary  $\partial S$ . Let  $\omega = Pdx + Qdy$  be a smooth 1-form on an open set containing  $S$ . Then*

$$\int_{\partial S} Pdx + Qdy = \iint_S (Q_x - P_y)dA$$

*Proof.* By Stokes's Theorem (Theorem 4.2.1)

$$\int_{\partial S} \omega = \int_S d\omega$$

Compute

$$d\omega = dP \wedge dx + dQ \wedge dy$$

Since

$$dP = P_x dx + P_y dy, \quad dQ = Q_x dx + Q_y dy$$

we get

$$d\omega = (Q_x - P_y) dx \wedge dy$$

Therefore,

$$\int_{\partial S} P dx + Q dy = \int_S (Q_x - P_y) dx \wedge dy$$

With the standard orientation,  $dx \wedge dy$  corresponds to  $dA$ . Therefore,

$$\int_{\partial S} P dx + Q dy = \iint_S (Q_x - P_y) dA$$

□

**Remark 4.2.16.** Green's Theorem is Stokes's Theorem in dimension 2. The boundary is a curve, and the region is a surface in  $\mathbb{R}^2$ . The exterior derivative  $d(Pdx + Qdy)$  measures infinitesimal circulation, and Green's Theorem says that total boundary circulation equals total interior circulation.

**Theorem 4.2.17 (Classical Stokes's Theorem).** *Let  $S \subseteq \mathbb{R}^3$  be a compact, oriented surface with positively oriented boundary  $\partial S$ . Let  $\vec{F} = (P, Q, R)$  be a smooth vector field on an open set containing  $S$ . Then*

$$\int_{\partial S} \vec{F} \cdot \vec{T} ds = \int_S (\nabla \times \vec{F}) \cdot \vec{n} dS$$

*Proof.* Associate  $\vec{F} = (P, Q, R)$  the work 1-form

$$\omega_{\vec{F}} = P dx + Q dy + R dz$$

Then

$$\int_{\partial S} \omega_{\vec{F}} = \int_{\partial S} \vec{F} \cdot \vec{T} ds$$

Now compute

$$d\omega_{\vec{F}} = dP \wedge dx + dQ \wedge dy + dR \wedge dz$$

Using the coordinate formula from Chapter 1, we have

$$d\omega_{\vec{F}} = (R_y - Q_z) dy \wedge dz + (P_z - R_x) dz \wedge dx + (Q_x - P_y) dx \wedge dy$$

But

$$\nabla \times \vec{F} = (R_y - Q_z, P_z - R_x, Q_x - P_y)$$

Therefore,  $d\omega_{\vec{F}}$  is the flux 2-form associated with  $\nabla \times \vec{F}$ . Hence,

$$\int_S d\omega_{\vec{F}} = \int_S (\nabla \times \vec{F}) \cdot \vec{n} dS$$

By Stokes's Theorem (Theorem 4.2.1),

$$\int_{\partial S} \omega_{\vec{F}} = \int_S d\omega_{\vec{F}}$$

Therefore,

$$\int_{\partial S} \vec{F} \cdot \vec{T} ds = \int_S (\nabla \times \vec{F}) \cdot \vec{n} dS$$

□

**Remark 4.2.18.** The classical Stokes's Theorem says that the circulation of a vector field around the boundary of a surface equals the flux of its curl through the surface.

In differential-form language, this is simply

$$\int_{\partial S} \omega_{\vec{F}} = \int_S d\omega_{\vec{F}}$$

The curl appears because the exterior derivative of a work 1-form is the flux 2-form of the curl.

**Example 4.2.19.** Let  $\vec{F}(x, y, z) = (-y, x, z)$ . Let  $S$  be the upper hemisphere  $x^2 + y^2 + z^2 = 1$  with  $z \geq 0$ , oriented upward, meaning that the normal has positive  $z$ -component. The boundary  $\partial S$  is the unit circle  $x^2 + y^2 = 1$  with  $z = 0$  oriented counterclockwise when viewed from above.

Compute the curl

$$\nabla \times \vec{F} = \nabla \times (-y, x, z) = (0, 0, 2)$$

Thus,

$$(\nabla \times \vec{F}) \cdot \vec{n} = (0, 0, 2) \cdot \vec{n} = 2n_3$$

From the upward-oriented hemisphere,  $n_3 dS$  is the projection area element onto the  $xy$ -plane. Therefore,

$$\int_S 2n_3 dS = 2 \iint_D dA$$

where  $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ . Hence,

$$\int_{\partial S} \vec{F} \cdot \vec{T} ds = 2 \text{Area}(D) = 2\pi$$

**Remark 4.2.20.** The answer could also be found by integrating directly around the unit circle. On  $\partial S$ , we have

$$\vec{r}(t) = (\cos(t), \sin(t), 0)$$

for  $0 \leq t \leq 2\pi$ . Then

$$\vec{F}(\vec{r}(t)) = (-\sin(t), \cos(t), 0), \quad \vec{r}'(t) = (-\sin(t), \cos(t), 0)$$

Thus,

$$\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = 1$$

so

$$\int_{\partial S} \vec{F} \cdot \vec{T} ds = \int_0^{2\pi} dt = 2\pi$$

Stokes's Theorem replaces this boundary integral with a surface integral. The value is the same for any oriented surface with the same boundary, provided the orientations are compatible.

**Proposition 4.2.21.** *Let  $M_1$  and  $M_2$  be compact, oriented  $k$ -dimensional manifolds with the same oriented boundary  $\partial M_1 = \partial M_2$ . Let  $\omega$  be a smooth  $(k-1)$ -form on an open set containing both manifolds. Then*

$$\int_{M_1} d\omega = \int_{M_2} d\omega$$

*Proof.* By Stokes's Theorem (Theorem 4.2.1)

$$\int_{M_1} d\omega = \int_{\partial M_1} \omega = \int_{\partial M_2} \omega = \int_{M_2} d\omega$$

□

**Remark 4.2.22.** This is why, in classical Stokes's Theorem, we may often replace a complicated surface with a simpler surface having the same boundary. For example, a hemisphere can often be replaced by a disk, as long as the boundary orientation is preserved.

Although the Divergence Theorem will be especially important in the physics applications, it is already a direct consequence of Stokes's Theorem.

**Theorem 4.2.23 (The Divergence Theorem).** *Let  $M \subseteq \mathbb{R}^3$  be a compact solid region with outward-oriented boundary  $\partial M$ . Let  $\vec{F} = (P, Q, R)$  be a smooth vector field on an open set containing  $M$ . Then*

$$\int_{\partial M} \vec{F} \cdot \vec{n} dS = \iiint_M \nabla \cdot \vec{F} dV$$

*Proof.* Associate to  $\vec{F} = (P, Q, R)$  the flux 2-form

$$\omega_{\vec{F}} = P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy$$

Then

$$\int_{\partial M} \omega_{\vec{F}} = \int_{\partial M} \vec{F} \cdot \vec{n} dS$$

Now compute

$$\begin{aligned} d\omega_{\vec{F}} &= dP \wedge dy \wedge dz + dQ \wedge dz \wedge dx + dR \wedge dx \wedge dy \\ &= (P_x + Q_y + R_z) dx \wedge dy \wedge dz \end{aligned}$$

But  $P_x + Q_y + R_z = \nabla \cdot \vec{F}$ . Therefore,

$$d\omega_{\vec{F}} = (\nabla \cdot \vec{F}) dx \wedge dy \wedge dz$$

By Stokes's Theorem (Theorem 4.2.1),

$$\int_{\partial M} \omega_{\vec{F}} = \int_M d\omega_{\vec{F}}$$

Hence,

$$\int_{\partial M} \vec{F} \cdot \vec{n} dS = \iiint_M \nabla \cdot \vec{F} dV$$

□

**Remark 4.2.24.** The Divergence Theorem says that the total outward flux through the boundary of a solid region equals the total divergence inside the region. In differential form language, it is just Stokes's Theorem applied to a flux 2-form.

**Definition 4.2.25.** A compact manifold  $M$  is called *closed* if it has no boundary  $\partial M = \emptyset$ .

For example,  $S^1$  and  $S^2$  are closed manifolds.

**Corollary 4.2.26.** *Let  $M$  be a compact, oriented  $k$ -dimensional manifold without boundary. Let  $\eta$  be a smooth  $(k-1)$ -form on  $M$ . Then*

$$\int_M d\eta = 0$$

*Equivalently, if  $\omega$  is an exact  $k$ -form, then*

$$\int_M \omega = 0$$

**Remark 4.2.27.** This result is useful for detecting when a form is not exact. If

$$\int_M \omega \neq 0$$

for some closed oriented manifold  $M$ , then  $\omega$  cannot be exact. This is one of the first connections between differential forms and topology.

**Corollary 4.2.28.** *For an oriented manifold  $M$ ,  $\partial(\partial M) = \emptyset$ .*

Stokes's Theorem gives

$$\int_{\partial M} d\omega = \int_M d(d\omega)$$

But  $d(d\omega) = 0$ . Thus, the above integral is 0.

## Chapter 5

# Applications to Physics

In the previous chapters, we developed the language of differential forms, pullbacks, surface integrals, flux, and Stokes's Theorem. We now apply this language to physical problems.

The central idea is that many objects from vector calculus have natural interpretations as differential forms. For example, a vector field  $\vec{F} = (P, Q, R)$  can be interpreted in two different ways.

As a work field, it corresponds to the 1-form

$$\omega_{\vec{F}} = Pdx + Qdy + Rdz$$

As a flux field, it corresponds to the 2-form

$$\eta_{\vec{F}} = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$$

The exterior derivative then encodes the familiar vector calculus operators. For a scalar-valued function  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ , the differential  $df$  corresponds to the gradient  $\nabla f$ .

For a work 1-form  $\omega_{\vec{F}}$  the exterior derivative  $d\omega_{\vec{F}}$  corresponds to the curl of  $\vec{F}$ . For a flux 2-form  $\eta_{\vec{F}}$ , the exterior derivative  $d\eta_{\vec{F}}$  corresponds to the divergence of  $\vec{F}$ . Thus the sequence from 0-forms to 1-forms to 2-forms to 3-forms matches the vector calculus sequence of scalar fields to vector fields to vector fields to scalar fields, where the operators are  $\nabla$ ,  $\nabla \times$ , and  $\nabla \cdot$ .

The identity  $d^2 = 0$  also explains two familiar vector calculus identities

$$\nabla \times (\nabla f) = \vec{0}_3, \quad \nabla \cdot (\nabla \times \vec{F}) = 0$$

In this chapter, we use this dictionary to study physical laws. First, we describe the correspondence between differential forms and vector calculus in

$\mathbb{R}^3$ . Then we apply Stokes's Theorem to flux integrals and derive Gauss-type laws. Finally, we discuss Maxwell's equations and see how the language of differential forms organizes the electric field, magnetic field, charge density, and current density into a compact geometric framework.

## 5.1 The Dictionary in $\mathbb{R}^3$

In  $\mathbb{R}^3$ , differential forms give a unified language for scalar fields, vector fields, work, flux, curl, and divergence.

The main idea is that a vector field  $\vec{F} = (P, Q, R)$  can be represented in two ways:

- as a 1-form, when we want to integrate along curves,
- as a 2-form, when we want to integrate through surfaces.

The exterior derivative  $d$  then becomes the familiar vector-calculus operators  $\nabla$ ,  $\nabla \times$ , and  $\nabla \cdot$ . Thus, the language of differential forms gives a single framework for the gradient, curl, divergence, circulation, and flux.

**Definition 5.1.1.** A smooth scalar-valued function  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$  is a differential 0-form. Its exterior derivative is

$$df = f_x dx + f_y dy + f_z dz$$

This 1-form corresponds to the gradient vector field  $\nabla f = (f_x, f_y, f_z)$ .

**Remark 5.1.2.** The differential  $df$  acts on a tangent vector  $\vec{v} = (v_1, v_2, v_3)$  by

$$df(\vec{v}) = f_x v_1 + f_y v_2 + f_z v_3$$

This is the dot product  $df(\vec{v}) = \nabla f \cdot \vec{v}$ . So the 1-form  $df$  measures directional change of  $f$ , while the vector field  $\nabla f$  points in the direction of steepest increase.

**Example 5.1.3.** Let  $f(x, y, z) = x^2 y + e^z$ . Then

$$df = 2xy dx + x^2 dy + e^z dz$$

The corresponding gradient vector field is

$$\nabla f = (2xy, x^2, e^z)$$

Thus, the 1-form  $df$  and the vector field  $\nabla f$  carry the same information.



The 1-form  $\omega_{\vec{F}} = Pdx + Qdy + Rdz$  is used for line integrals. If  $C$  is an oriented curve with parametrization

$$\vec{r}(t) = (x(t), y(t), z(t))$$

for  $a \leq t \leq b$ , then

$$\int_C \omega_{\vec{F}} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt$$

Therefore,

$$\int_C \omega_{\vec{F}} = \int_C \vec{F} \cdot \vec{T} ds$$

This is the work or circulation integral of the vector field along the curve.

**Example 5.1.4.** Let  $\vec{F}(x, y, z) = (-y, x, z)$ . the associated work 1-form is

$$\omega_{\vec{F}} = -ydx + xdy + zdz$$

If  $C$  is a closed curve, then

$$\int_C \omega_{\vec{F}}$$

measures the circulation of  $\vec{F}$  around  $C$ .

**Theorem 5.1.5.** Let  $\vec{F} = (P, Q, R)$  be a smooth vector field on an open set  $U \subseteq \mathbb{R}^3$ . Let  $\omega_{\vec{F}} = Pdx + Qdy + Rdz$ . Then

$$d\omega_{\vec{F}} = (R_y - Q_z)dy \wedge dz + (P_z - R_x)dz \wedge dx + (Q_x - P_y)dx \wedge dy$$

The coefficient vector is exactly the curl

$$\nabla \times \vec{F} = (R_y - Q_z, P_z - R_x, Q_x - P_y)$$

Thus,  $d\omega_{\vec{F}} = \eta_{\nabla \times \vec{F}}$ , where  $\eta_{\nabla \times \vec{F}}$  is the flux 2-form associated with  $\nabla \times \vec{F}$ .

**Remark 5.1.6.** The curl appears naturally because the exterior of a 1-form is a 2-form. A work form  $\omega_{\vec{F}}$  measures motion along curves. Its derivative  $d\omega_{\vec{F}}$  measures infinitesimal circulation through surfaces. This is why Stokes's Theorem becomes

$$\int_{\partial S} \omega_{\vec{F}} = \int_S d\omega_{\vec{F}}$$

or, in vector calculus language

$$\int_{\partial S} \vec{F} \cdot \vec{T} ds = \int_S (\nabla \times \vec{F}) \cdot \vec{n} dS$$

**Corollary 5.1.7.** *For every smooth scalar-valued function  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ ,  $\nabla \times (\nabla f) = \vec{0}_3$ . Therefore, the 3-form corresponding to  $\nabla \cdot (\nabla \times \vec{F})$  is*

$$d(d\omega_{\vec{F}}) = d^2\omega_{\vec{F}}$$

But  $d^2 = 0$ . Hence,

$$\nabla \cdot (\nabla \times \vec{F}) = 0$$

*Proof.* The 1-form corresponding to  $\nabla f$  is  $df$ . Therefore, the 2-form corresponding to  $\nabla \times (\nabla f)$  is  $d(df) = d^2f$ . Since  $d^2 = 0$ , we have

$$\nabla \times (\nabla f) = \vec{0}_3$$

as desired. □

The flux 2-form  $\eta_{\vec{F}}$  is used for surface integrals. If  $S$  is an oriented surface with unit normal vector field  $\vec{n}$ , then

$$\int_S \eta_{\vec{F}} = \int_S \vec{F} \cdot \vec{n} dS$$

Thus, the same vector field can appear in two differential forms, for work integrals and for flux integrals.

**Example 5.1.8.** Let  $\vec{F}(x, y, z) = (x, y, z)$ . The associated flux 2-form is

$$\eta_{\vec{F}} = xdy \wedge dz + ydz \wedge dx + zdx \wedge dy$$

If  $S$  is the unit sphere oriented outward, then

$$\int_S \eta_{\vec{F}} = \int_S \vec{F} \cdot \vec{n} dS$$

On the unit sphere,  $\vec{n} = (x, y, z)$ , so

$$\vec{F} \cdot \vec{n} = x^2 + y^2 + z^2 = 1$$

Therefore,

$$\int_S \eta_{\vec{F}} = \int_S dS = 4\pi$$

**Theorem 5.1.9.** *Let  $\vec{F} = (P, Q, R)$  be a smooth vector field on an open set  $U \subseteq \mathbb{R}^3$ . Let*

$$\eta_{\vec{F}} = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$$

*Then*

$$d\eta_{\vec{F}} = (P_x + Q_y + R_z)dx \wedge dy \wedge dz$$

*The coefficient  $P_x + Q_y + R_z$  is the divergence of  $\vec{F}$ .*

**Remark 5.1.10.** Divergence appears naturally because the exterior derivative of a 2-form is a 3-form. A flux form  $\eta_{\vec{F}}$  measures flow through surfaces. Its derivative  $d\eta_{\vec{F}}$  measures source density or sink density inside volumes.

This is why Stokes's Theorem becomes

$$\int_{\partial M} \eta_{\vec{F}} = \int_M d\eta_{\vec{F}}$$

or in vector calculus language,

$$\int_{\partial M} \vec{F} \cdot \vec{n} dS = \iiint_M \nabla \cdot \vec{F} dV$$

**Corollary 5.1.11.** For every smooth vector field  $\vec{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ , we have

$$\nabla \cdot (\nabla \times \vec{F}) = 0$$

*Proof.* The work 1-form corresponding to  $\vec{F}$  is  $\omega_{\vec{F}}$  and the flux 2-form corresponding to  $\nabla \times \vec{F}$  is  $d\omega_{\vec{F}}$ . Therefore, the 3-form corresponding to  $\nabla \cdot (\nabla \times \vec{F})$  is  $d(d\omega_{\vec{F}}) = 0$ .  $\square$

**Definition 5.1.12.** Let  $\rho : \mathbb{R}^3 \rightarrow \mathbb{R}$  be a smooth scalar-valued function. The associated 3-form is

$$\rho dx \wedge dy \wedge dz$$

This form is integrated over regions in  $\mathbb{R}^3$ .

If  $M \subseteq \mathbb{R}^3$  is a compact region, then

$$\int_M \rho dx \wedge dy \wedge dz = \iiint_M \rho dV$$

**Remark 5.1.13.** A 3-form represents a density with respect to volume. For example:

- mass density gives total mass by integration over volume,
- charge density gives total charge by integration over volume,
- divergence gives net source density by integration over volume.

Since there are no nonzero 4-forms in  $\mathbb{R}^3$ , the exterior derivative of every 3-form is zero

$$d(\rho dx \wedge dy \wedge dz) = 0$$

The correspondence between work 1-forms and flux 2-forms can be described using the Hodge star operator.

**Definition 5.1.14.** In the standard orientation and Euclidean metric on  $\mathbb{R}^3$ , the *Hodge star operator* satisfies

$$\star dx = dy \wedge dz, \quad \star dy = dz \wedge dx, \quad \star dz = dx \wedge dy$$

Therefore, if

$$\omega_{\vec{F}} = Pdx + Qdy + Rdz$$

then

$$\star \omega_{\vec{F}} = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$$

Thus,

$$\star \omega_{\vec{F}} = \eta_{\vec{F}}$$

**Remark 5.1.15.** The Hodge star converts the work form of a vector field into its flux form. In other words,  $\vec{F} \longleftrightarrow \omega_{\vec{F}}$  for line integrals, while  $\vec{F} = \eta_{\vec{F}} = \star \omega_{\vec{F}}$  for surface integrals. This is one reason the same vector field can naturally appear in two different differential-form degrees.

## 5.2 Gauss's Law

In Section 5.1, we saw that a vector field  $\vec{F} = (P, Q, R)$  can be represented by the flux 2-form

$$\eta_{\vec{F}} = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$$

The exterior derivative for this flux form is

$$d\eta_{\vec{F}} = (\nabla \cdot \vec{F})dx \wedge dy \wedge dz$$

Therefore, Stokes's Theorem (Theorem 4.2.1) gives the Divergence Theorem (Theorem 4.2.23)

$$\int_{\partial V} \eta_{\vec{F}} = \int_V d\eta_{\vec{F}}$$

or equivalently,

$$\int_{\partial V} \vec{F} \cdot \vec{n} dS = \iiint_V \nabla \cdot \vec{F} dV$$

Gauss's Law is a physical law of exactly this form. It says that the total flux of a field through the boundary of a region is determined by the total amount of source inside the region.

For electric fields, the source is charge. For gravitational fields, the source is mass.

**Definition 5.2.1.** A surface  $S \subseteq \mathbb{R}^3$  is called *closed* if it is the boundary of a solid region  $V \subseteq \mathbb{R}^3$ . Thus,  $S = \partial V$ . We orient  $S$  using the outward unit normal vector field  $\vec{n}$ .

**Definition 5.2.2.** Let  $\vec{F}$  be a smooth vector field on an open set containing  $S$ . The outward flux of  $\vec{F}$  through  $S$  is

$$\int_S \vec{F} \cdot \vec{n} dS$$

Equivalently, if  $\eta_{\vec{F}}$  is the flux 2-form associated with  $\vec{F}$ , then

$$\int_S \eta_{\vec{F}} = \int_S \vec{F} \cdot \vec{n} dS$$

**Remark 5.2.3.** Flux measures how much of a vector field passes through a surface.

- If  $\vec{F} \cdot \vec{n} > 0$ , then the field points outward through the surface.
- If  $\vec{F} \cdot \vec{n} < 0$ , then the field points inward through the surface.

Thus, flux is signed.

**Remark 5.2.4.** The Divergence Theorem (Theorem 4.2.23) says that the total outward flux through the boundary equals total divergence inside the region.

Thus, divergence measures source density

- If  $\nabla \cdot \vec{F} > 0$ , then the field has a source.
- If  $\nabla \cdot \vec{F} < 0$ , then the field has a sink.
- If  $\nabla \cdot \vec{F} = 0$ , then the field has no net local source or sink.

**Definition 5.2.5.** Let  $\rho : \mathbb{R}^3 \rightarrow \mathbb{R}$  be a smooth scalar-valued function. We interpret  $\rho(\vec{x})$  as the electric charge density at the point  $\vec{x} \in \mathbb{R}^3$ . The total charge inside a solid region  $V$  is

$$Q_{\text{enc}} = \iiint_V \rho dV$$

In differential form language, the charge density is represented by the 3-form  $\rho dx \wedge dy \wedge dz$ .

**Theorem 5.2.6 (Gauss's Law).** Let  $\vec{E}$  be an electric field in  $\mathbb{R}^3$ . Let  $V \subseteq \mathbb{R}^3$  be a compact solid region with outward-oriented boundary  $\partial V$ . Then Gauss's Law says

$$\int_{\partial V} \vec{E} \cdot \vec{n} dS = \frac{1}{\varepsilon_0} \iiint_V \rho dV$$

Equivalently,

$$\int_{\partial V} \vec{E} \cdot \vec{n} dS = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

Here,  $\varepsilon_0$  is the permittivity constant.

**Remark 5.2.7.** Gauss's law says that the outward electric flux through a closed surface depends only on the total charge enclosed by that surface.

Charges outside the surface may affect the electric field at points on the surface, but their net contribution to the total flux through the closed surface is zero.

**Definition 5.2.8.** Let  $\vec{E} = (E_1, E_2, E_3)$  be an electric field. The associated flux 2-form is

$$n_{\vec{E}} = E_1 dy \wedge dz + E_2 dz \wedge dx + E_3 dx \wedge dy$$

Then

$$\int_{\partial V} n_{\vec{E}} = \int_{\partial V} \vec{E} \cdot \vec{n} dS$$

**Theorem 5.2.9 (Gauss's Law).** *Electric Gauss's Law can be written as*

$$d\eta_{\vec{E}} = \frac{\rho}{\varepsilon_0} dx \wedge dy \wedge dz$$

Equivalently,

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

*Proof.* From Theorem 5.1.9,

$$d\eta_{\vec{E}} = (\nabla \cdot \vec{E}) dx \wedge dy \wedge dz$$

If

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

then

$$d\eta_{\vec{E}} = \frac{\rho}{\varepsilon_0} dx \wedge dy \wedge dz$$

Conversely, if

$$d\eta_{\vec{E}} = \frac{\rho}{\varepsilon_0} dx \wedge dy \wedge dz$$

then comparing coefficients of  $dx \wedge dy \wedge dz$  gives

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

□

**Proposition 5.2.10.** *Assume  $\vec{E}$  and  $\rho$  are smooth. Then the following are equivalent.*

1. For every compact solid region  $V \subseteq \mathbb{R}^3$ ,

$$\int_{\partial V} \vec{E} \cdot \vec{n} dS = \frac{1}{\varepsilon_0} \iiint_V \rho dV$$

2. At every point,

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

*Proof.* Assume first that

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

By the Divergence Theorem (Theorem 4.2.23),

$$\int_{\partial V} \vec{E} \cdot \vec{n} dS = \iiint_V \nabla \cdot \vec{E} dV$$

Therefore, the differential form implies the differential form.

Conversely, suppose the integral form holds for every compact solid region  $V \subseteq \mathbb{R}^n$ . Then

$$\int_{\partial V} \vec{E} \cdot \vec{n} dS = \frac{1}{\varepsilon_0} \iiint_V \rho dV$$

By the Divergence Theorem (Theorem 4.2.23), the left-hand side is

$$\iiint_V \nabla \cdot \vec{E} dV$$

Hence,

$$\iiint_V \left( \nabla \cdot \vec{E} - \frac{\rho}{\varepsilon_0} \right) dV = 0$$

for every compact solid region  $V$ .

Since the integrand is continuous and its integral over every region is zero, the integrand must be zero everywhere. Therefore,

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

□

**Example 5.2.11.** Let a point charge  $q$  be located at the origin. For  $\vec{x} \neq \vec{0}_3$ , the electric field is

$$\vec{E}(\vec{x}) = \frac{q}{4\pi\epsilon_0} \frac{\vec{x}}{\|\vec{x}\|^3}$$

This field is defined on  $\mathbb{R}^3 \setminus \{\vec{0}_3\}$ . Away from the origin,  $\nabla \cdot \vec{E} = 0$ . Thus, the field has no source at any point where it is smooth. The source is concentrated at the origin.

**Proposition 5.2.12.** Let  $S_a = \{\vec{x} \in \mathbb{R}^3 : \|\vec{x}\| = a\}$  be a sphere of radius  $a > 0$  centered at the origin, oriented outward. For the point-charge field

$$\vec{E}(\vec{x}) = \frac{q}{4\pi\epsilon_0} \frac{\vec{x}}{\|\vec{x}\|^3}$$

we have

$$\int_{S_a} \vec{E} \cdot \vec{n} dS = \frac{q}{\epsilon_0}$$

*Proof.* On the sphere  $S_a$ , the outward unit normal is

$$\vec{n}(\vec{x}) = \frac{\vec{x}}{\|\vec{x}\|} = \frac{\vec{x}}{a}$$

and also,  $\|\vec{x}\| = a$ . Therefore,

$$\vec{E}(\vec{x}) = \frac{q}{4\pi\epsilon_0} \frac{\vec{x}}{a^3}$$

Then

$$\vec{E}(\vec{x}) \cdot \vec{n}(\vec{x}) = \frac{q}{4\pi\epsilon_0} \frac{\vec{x}}{a^3} \cdot \frac{\vec{x}}{a} = \frac{q}{4\pi\epsilon_0} \frac{a^2}{a^4} = \frac{q}{4\pi\epsilon_0 a^2}$$

Thus,

$$\int_{S_a} \vec{E} \cdot \vec{n} dS = \int_{S_a} \frac{q}{4\pi\epsilon_0 a^2} dS = \frac{4}{4\pi\epsilon_0 a^2} (4\pi a^2) = \frac{q}{\epsilon_0}$$

□

**Remark 5.2.13.** The flux through the sphere does not depend on the radius  $a$ . This is a signature of an inverse-square field. The field strength decreases like  $\frac{1}{a^2}$ , but the surface area of the sphere increases like  $a^2$ . These two effects cancel.



**Proposition 5.2.14.** *Let  $S = \partial V$  be a closed surface oriented outward. Suppose a point charge  $q$  is located at a point  $\vec{a} \notin V$ . Then the total electric flux through  $S$  due to this point charge is*

$$\int_S \vec{E} \cdot \vec{n} dS = 0$$

*Proof.* The electric field of the point charge at  $\vec{a}$  is smooth on an open set containing  $V$ , because the singularity lies outside  $V$ . Moreover,  $\nabla \cdot \vec{E} = 0$  on that open set.

By the Divergence Theorem (Theorem 4.2.23)

$$\int_S \vec{E} \cdot \vec{n} dS = \iiint_V \nabla \cdot \vec{E} dV = 0$$

□

**Remark 5.2.15.** A charge outside a closed surface may produce field lines passing through the surface, but every field line that enters must also leave. Therefore, its net outward flux through the closed surface is zero. Only enclosed charge contributes to the total flux.

**Theorem 5.2.16.** *Suppose point charges  $q_1, q_2, \dots, q_N$  are located at points  $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_N$ . Let  $S = \partial V$  be a closed surface oriented outward, and suppose no charge lies directly on  $S$ . Then*

$$\int_S \vec{E} \cdot \vec{n} dS = \frac{1}{\epsilon_0} \sum_{\vec{a}_i \in V} q_i$$

*Proof.* By the principle of superposition, the total electric field is

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_N$$

where  $\vec{E}_i$  is the field produced by the charge  $q_i$ . Flux is linear, so

$$\int_S \vec{E} \cdot \vec{n} dS = \sum_{i=1}^N \int_S \vec{E}_i \cdot \vec{n} dS$$

If  $\vec{a}_i \notin V$ , then Proposition 5.2.14 gives

$$\int_S \vec{E}_i \cdot \vec{n} dS = 0$$

If  $\vec{a}_i \in V$ , then the flux contribution is  $\frac{q_i}{\varepsilon_0}$ . Therefore,

$$\int_S \vec{E} \cdot \vec{n} dS = \frac{1}{\varepsilon_0} \sum_{\vec{a}_i \in V} q_i$$

□

**Remark 5.2.17.** This theorem explains why Gauss's Law is especially useful for closed surfaces. The field itself may be complicated, but the total flux only detects the net source enclosed by the surface.

**Definition 5.2.18.** Let  $\rho : \mathbb{R}^3 \rightarrow \mathbb{R}$  be a charge density. The charge enclosed by a solid region  $V$  is

$$Q_{\text{enc}} = \iiint_V \rho dV$$

Then Gauss's Law becomes

$$\int_{\partial V} \vec{E} \cdot \vec{n} dS = \frac{1}{\varepsilon_0} \iiint_V \rho dV$$

**Remark 5.2.19.** In differential form language, this says

$$\int_{\partial V} \eta_{\vec{E}} = \int_V \frac{\rho}{\varepsilon_0} dx \wedge dy \wedge dz$$

By Stokes's Theorem (Theorem 4.2.1)

$$\int_{\partial V} \eta_{\vec{E}} = \int_V d\eta_{\vec{E}}$$

Therefore, Gauss's Law says that  $d\eta_{\vec{E}}$  is the charge-density 3-form scaled by  $\frac{1}{\varepsilon_0}$ . That is,

$$d\eta_{\vec{E}} = \frac{\rho}{\varepsilon_0} dx \wedge dy \wedge dz$$

The same idea applies to Newtonian graviy, but the sign is different. Gravity is attractive, so gravitational field lines point inward toward mass.

**Definition 5.2.20.** Let a point mass  $M$  be located at the origin. For  $\vec{x} \neq \vec{0}_3$ , the gravitational field is

$$\vec{F}(\vec{x}) = -GM \frac{\vec{x}}{\|\vec{x}\|^3}$$

Here,  $G$  is the gravitational constant. The negative sign indicates that the field points toward the origin.

**Proposition 5.2.21.** *Let  $S_a = \{\vec{x} \in \mathbb{R}^3 : \|\vec{x}\| = a\}$  be the sphere of radius  $a > 0$  centered at the origin, oriented outward. Then*

$$\int_{S_a} \vec{F} \cdot \vec{n} dS = -4\pi GM$$

*Proof.* On  $S_a$ , the outward unit normal is

$$\vec{n}(\vec{x}) = \frac{\vec{x}}{a}$$

The gravitational field is

$$\vec{F}(\vec{x}) = -GM \frac{\vec{x}}{a^3}$$

Thus,

$$\vec{F}(\vec{x}) \cdot \vec{n}(\vec{x}) = -GM \frac{\vec{x}}{a^3} \cdot \frac{\vec{x}}{a} = -\frac{GM}{a^2}$$

Therefore,

$$\int_{S_a} \vec{F} \cdot \vec{n} dS = \int_{S_a} -\frac{GM}{a^2} dS = -\frac{GM}{a^2} (4\pi a^2) = -4\pi GM$$

□

**Remark 5.2.22.** The negative sign appears because the gravitational field points inward, while the normal vector on a closed surface is chosen outward. Thus,  $\vec{F} \cdot \vec{n} < 0$  for a sphere enclosing a positive mass.

**Theorem 5.2.23.** *Let  $\mu : \mathbb{R}^3 \rightarrow \mathbb{R}$  be a mass density. Let  $\vec{F}$  be the gravitational field produced by this mass distribution. Then*

$$\int_{\partial V} \vec{F} \cdot \vec{n} dS = -4\pi G \iiint_V \mu dV$$

*Equivalently,*

$$\nabla \cdot \vec{F} = -4\pi G\mu$$

In differential form, if  $\eta_{\vec{F}}$  is the flux 2-form associated with  $\vec{F}$ , then

$$d\eta_{\vec{F}} = -4\pi G\mu dx \wedge dy \wedge dz$$

**Example 5.2.24.** Suppose the earth is modeled as a ball of radius  $R$  centered at the origin, with total mass  $M$  uniformly distributed. The mass density is

$$\mu = \frac{M}{\frac{4}{3}\pi R^3} = \frac{3M}{4\pi R^3}$$

Let  $\vec{x} \in \mathbb{R}^3$  with  $\|\vec{x}\| = r < R$ . Let  $S_r = \{\vec{y} \in \mathbb{R}^3 : \|\vec{y}\| = r\}$ . By symmetry, the gravitational field points radially inward and has constant magnitude on  $S_r$ . Write

$$\vec{F}(\vec{x}) = -A(r) \frac{\vec{x}}{r}$$

where  $A(r) \geq 0$  is the magnitude of the field at radius  $r$ . The mass enclosed by  $S_r$  is

$$M_{\text{enc}} = \mu \left( \frac{4}{3} \pi r^3 \right) = \frac{3M}{4\pi R^3} \cdot \frac{4}{3} \pi r^3 = M \frac{r^3}{R^3}$$

By gravitational Gauss's Law,

$$\int_{S_r} \vec{F} \cdot \vec{n} dS = -4\pi G M_{\text{enc}}$$

Since  $\vec{F} \cdot \vec{n} = -A(r)$  on  $S_r$ , we have

$$\int_{S_r} \vec{F} \cdot \vec{n} dS = -A(r)(4\pi r^2) = -4\pi G M \frac{r^3}{R^3}$$

Cancelling  $-4\pi r^2$  gives

$$A(r) = \frac{GM}{R^3} r$$

Therefore,

$$\vec{F}(\vec{x}) = -\frac{GM}{R^3} \vec{x}$$

for  $\|\vec{x}\| < R$ . Inside a uniformly dense sphere, the gravitational field is proportional to the distance from the center.

**Remark 5.2.25.** This is surprising at first because outside a spherical mass, gravity behaves like an inverse-square field:

$$\|\vec{F}(\vec{x})\| = \frac{GM}{\|\vec{x}\|^2}$$

But inside the sphere, only the mass enclosed within radius  $r$  contributes to the net flux through  $S_r$ . Since the enclosed mass grows like  $r^3$  and the surface grows like  $r^2$ , the field strength grows like  $r$ .

**Proposition 5.2.26.** Let  $\vec{F}$  be a smooth vector field near  $\vec{a} \in \mathbb{R}^3$ . Then

$$\nabla \cdot \vec{F}(\vec{a}) = \lim_{r \rightarrow 0^+} \frac{1}{\text{Vol}(B_r(\vec{a}))} \int_{\partial B_r(\vec{a})} \vec{F} \cdot \vec{n} dS$$

Thus, divergence is the outward flux per unit volume through very small spheres.

**Remark 5.2.27.** Gauss's Law says that source density is divergence. For electric fields,

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

For gravitational fields,

$$\nabla \cdot \vec{F} = -4\pi G\mu$$

The difference in sign reflects the difference between electric and gravitational sources: Electric fields point outward from positive charge, while gravitational fields point inward toward positive mass.

### 5.3 Maxwell's Equations

In Sections 5.1 and 5.2, we built the dictionary between vector calculus and differential forms in  $\mathbb{R}^3$ . A vector field  $\vec{F} = (P, Q, R)$  can be represented as a work 1-form

$$\omega_{\vec{F}} = Pdx + Qdy + Rdz$$

or as a flux 2-form

$$\eta_{\vec{F}} = Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$$

The exterior derivative then encodes curl and divergence

$$d\omega_{\vec{F}} \longleftrightarrow \nabla \times \vec{F}, \quad d\eta_{\vec{F}} \longleftrightarrow \nabla \cdot \vec{F}$$

Maxwell's equations describe how electric and magnetic fields interact with charge and current. In differential forms, these equations become especially natural because they are built from circulation, flux, exterior derivatives, and Stokes's Theorem.

Throughout this section, we work in vacuum using SI units.

**Definition 5.3.1.** The *electric field* is a time-dependent vector field

$$\vec{E}(x, y, z, t) = (E_1(x, y, z, t), E_2(x, y, z, t), E_3(x, y, z, t))$$

The *magnetic field* is a time-dependent vector field

$$\vec{B}(x, y, z, t) = (B_1(x, y, z, t), B_2(x, y, z, t), B_3(x, y, z, t))$$

For each fixed time  $t$ , we regard  $\vec{E}(\cdot, t)$  and  $\vec{B}(\cdot, t)$  as vector fields on the space  $\mathbb{R}^3$ .

**Remark 5.3.2.** The same physical vector field can appear as either a 1-form or a 2-form depending on what kind of integral we want. For circulation around a curve, we use the work 1-form. For flux through a surface, we use the flux 2-form. This distinction is essential in Maxwell's equations.

**Definition 5.3.3.** The *charge density* is a scalar-valued function  $\rho(x, y, z, t)$ . The total charge contained in a solid region  $V \subseteq \mathbb{R}^3$  at time  $t$  is

$$Q(V, t) = \iiint_V \rho(x, y, z, t) dV$$

In differential form language, charge density is represented by the 3-form

$$\rho dx \wedge dy \wedge dz$$

**Definition 5.3.4.** The *current density* is a time-dependent vector field

$$\vec{J}(x, y, z, t) = (J_1(x, y, z, t), J_2(x, y, z, t), J_3(x, y, z, t))$$

It measures the flow of charge through surfaces. The associated current flux 2-form is

$$\eta_{\vec{J}} = J_1 dy \wedge dz + J_2 dz \wedge dx + J_3 dx \wedge dy$$

If  $S$  is an oriented surface, then

$$\int_S \eta_{\vec{J}} = \int_S \vec{J} \cdot \vec{n} dS$$

is the total current passing through  $S$ .

**Theorem 5.3.5 (Maxwell's Equations).** *In vacuum, Maxwell's equations are*

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \cdot \vec{B} = 0, \quad \nabla \times \vec{E} = -\vec{B}_t, \quad \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \vec{E}_t$$

Here,  $\varepsilon_0$  is the electric permittivity of free space, and  $\mu_0$  is the magnetic permeability of free space.

**Remark 5.3.6.** Each Maxwell equation has an interpretation:

- The equation  $\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$  says that electric charge is a source of electric field.
- The equation  $\nabla \cdot \vec{B} = 0$  says that there are no magnetic monopoles.
- The equation  $\nabla \times \vec{E} = -\vec{B}_t$  says that a changing magnetic field creates circulating electric field.

- The equation  $\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \vec{E}_t$  says that magnetic circulation is produced by electric current and by changing electric fields.

We now translate Maxwell's equations using the dictionary from Section 5.1. For this section, the exterior derivative  $d$  means the spatial exterior derivative. That is, it differentiates with respect to  $x, y, z$ , but not with respect to  $t$ . Time is treated as a parameter.

**Theorem 5.3.7.** *In vacuum, Maxwell's equations can be written as*

$$d\eta_{\vec{E}} = \frac{\rho}{\varepsilon_0} dx \wedge dy \wedge dz, \quad d\eta_{\vec{B}} = 0, \quad d\omega_{\vec{E}} = -(\eta_{\vec{B}})_t, \quad d\omega_{\vec{B}} = \mu_0 \eta_{\vec{J}} + \mu_0 \varepsilon_0 (\eta_{\vec{E}})_t$$

*Proof.* These immediately follow from the definitions of work and flux, as well as using Maxwell's Equations (Theorem 5.3.5).  $\square$

Using Stokes's Theorem, each differential equation can be converted into an integral equation.

**Theorem 5.3.8 (Gauss's Law for Electric Fields).** *Let  $V \subseteq \mathbb{R}^3$  be a compact solid region with oriented-outward boundary  $\partial V$ . Then*

$$\int_{\partial V} \vec{E} \cdot \vec{n} dS = \frac{1}{\varepsilon_0} \iiint_V \rho dV$$

*Proof.* Starting from  $d\eta_{\vec{E}} = \frac{\rho}{\varepsilon_0} dx \wedge dy \wedge dz$ , integrating over  $V$  and using Stokes's Theorem (Theorem 4.2.1), we have

$$\int_V d\eta_{\vec{E}} = \frac{1}{\varepsilon_0} \int_V \rho dx \wedge dy \wedge dz = \int_{\partial V} \eta_{\vec{E}}$$

and thus,

$$\int_{\partial V} \vec{E} \cdot \vec{n} dS = \frac{1}{\varepsilon_0} \iiint_V \rho dV$$

$\square$

**Theorem 5.3.9 (Gauss's Law for Magnetic Fields).** *Let  $V \subseteq \mathbb{R}^3$  be a compact solid region with oriented-outward boundary  $\partial V$ . Then*

$$\int_{\partial V} \vec{B} \cdot \vec{n} dS = 0$$

*Proof.* Starting from  $d\eta_{\vec{B}} = 0$ , integrating over  $V$  and using Stokes's Theorem,

$$0 = \int_V d\eta_{\vec{B}} = \int_{\partial V} \eta_{\vec{B}} = \int_{\partial V} \vec{B} \cdot \vec{n} dS = 0$$

$\square$

**Remark 5.3.10.** The equation

$$\int_{\partial V} \vec{B} \cdot \vec{n} dS = 0$$

says that the net magnetic flux through every closed surface is zero. In physical terms, magnetic field lines do not begin or end. They form closed loops or extend infinitely. This is the statement that magnetic monopoles do not appear in classical Maxwell's theory.

**Theorem 5.3.11 (Faraday's Law).** *Let  $S$  be an oriented surface with positively oriented boundary  $\partial S$ . Then*

$$\int_{\partial S} \vec{E} \cdot \vec{T} ds = - \left( \int_S \vec{B} \cdot \vec{n} dS \right)'$$

*Proof.* Starting from  $d\omega_{\vec{E}} = -(\eta_{\vec{B}})_t$ , integrate over  $S$  and using Stokes's Theorem (Theorem 4.2.1),

$$\int_{\partial S} \omega_{\vec{E}} = \int_S d\omega_{\vec{E}} = - \int_S (\eta_{\vec{B}})_t$$

If  $S$  is fixed in time, then

$$\int_S (\eta_{\vec{B}})_t = \left( \int_S \eta_{\vec{B}} \right)'$$

Therefore,

$$\int_{\partial S} \vec{E} \cdot \vec{T} dS = \int_{\partial S} \omega_{\vec{E}} = - \left( \int_S \eta_{\vec{B}} \right)' = - \left( \int_S \vec{B} \cdot \vec{n} dS \right)'$$

□

**Remark 5.3.12.** Faraday's Law says that changing magnetic flux induces electric circulation. The negative sign is important. It expresses Lenz's law, which states that the induced electric circulation opposes the change in magnetic flux. In differential-form language, Faraday's law is the statement

$$d\omega_{\vec{E}} = -(\eta_{\vec{B}})_t$$

Thus, time variation in magnetic flux creates spatial circulation of the electric field.



**Theorem 5.3.13 (Ampere's Law).** *Let  $S$  be an oriented surface with positively oriented boundary  $\partial S$ . Then*

$$\int_{\partial S} \vec{B} \cdot \vec{T} ds = \mu_0 \int_S \vec{J} \cdot \vec{n} dS + \mu_0 \varepsilon_0 \left( \int_S \vec{E} \cdot \vec{n} dS \right)'$$

*Proof.* Starting from  $d\omega_{\vec{B}} = \mu_0 \eta_{\vec{J}} + \mu_0 \varepsilon_0 (\eta_{\vec{E}})_t$ , integrating over  $S$  and using Stokes's Theorem (Theorem 4.2.1)

$$\int_S d\omega_{\vec{B}} = \mu_0 \int_S \eta_{\vec{J}} + \mu_0 \varepsilon_0 \int_S (\eta_{\vec{E}})_t = \int_{\partial S} \omega_{\vec{B}}$$

If  $S$  is fixed in time, then

$$\int_S (\eta_{\vec{E}})_t = \left( \int_S \eta_{\vec{E}} \right)'$$

Therefore,

$$\int_{\partial S} \omega_{\vec{B}} = \mu_0 \int_S \eta_{\vec{J}} + \mu_0 \varepsilon_0 \left( \int_S \eta_{\vec{E}} \right)'$$

In vector calculus notation, this becomes

$$\int_{\partial S} \vec{B} \cdot \vec{T} ds = \mu_0 \int_S \vec{J} \cdot \vec{n} dS + \mu_0 \varepsilon_0 \left( \int_S \vec{E} \cdot \vec{n} dS \right)'$$

□

**Remark 5.3.14.** The term

$$\mu_0 \int_S \vec{J} \cdot \vec{n} dS$$

comes from the actual electric current passing through the surface. The term

$$\mu_0 \varepsilon_0 \left( \int_S \vec{E} \cdot \vec{n} dS \right)'$$

is called the *displacement current term*. It says that changing the electric flux also produces magnetic circulation. This term is essential for charge conservation and for electromagnetic waves.

Maxwell's equations imply the conservation of electric charge.

**Theorem 5.3.15.** *The charge density  $\rho$  and current density  $\vec{J}$  satisfy*

$$\rho_t + \nabla \cdot \vec{J} = 0$$

*In differential-form language,*

$$(\rho dx \wedge dy \wedge dz)_t + d\eta_{\vec{J}} = 0$$

*Proof.* Start from Amere's Law (Theorem 5.3.13) in differential form

$$d\omega_{\vec{B}} = \mu_0 \eta_{\vec{J}} + \mu_0 \varepsilon_0 (\eta_{\vec{E}})_t$$

Apply  $d$  to both sides, and using  $d^2 = 0$  to get

$$0 = d(d\omega_{\vec{B}}) = \mu_0 d\eta_{\vec{J}} + \mu_0 \varepsilon_0 d((\eta_{\vec{E}})_t) = \mu_0 d\eta_{\vec{J}} + \mu_0 \varepsilon_0 (d\eta_{\vec{E}})_t$$

Now using Gauss's Law (Theorem 5.3.8),

$$0 = \mu_0 d\eta_{\vec{J}} + \mu_0 \varepsilon_0 \left( \frac{\rho}{\varepsilon_0} dx \wedge dy \wedge dz \right)_t$$

Cancelling  $\mu_0$  gives

$$0 = d\eta_{\vec{J}} + (\rho dx \wedge dy \wedge dz)_t$$

and so

$$d\eta_{\vec{J}} = (\nabla \cdot \vec{J}) dx \wedge dy \wedge dz$$

In particular,

$$(\rho_t + \nabla \cdot \vec{J}) dx \wedge dy \wedge dz = 0$$

Therefore,

$$\rho_t + \nabla \cdot \vec{J} = 0$$

□

**Remark 5.3.16.** The continuity equation says that the charge cannot be created or destroyed.

- If charge decreases inside a region, then it must be because charge flows out through the boundary.
- If charge inside inside a region, then it must be because charge flows in through the boundary.

In integral form, the continuity equation says

$$\left( \iiint_V \rho dV \right)' = - \int_{\partial V} \vec{J} \cdot \vec{n} dS$$

One of Maxwell's great predictions is that electromagnetic waves can travel through vacuum. Assume there is no charge and no current, so  $\rho = 0$  and  $\vec{J} = \vec{0}_3$ . Then Maxwell's equation become

$$\nabla \cdot \vec{E} = 0, \quad \nabla \cdot \vec{B} = 0, \quad \nabla \times \vec{E} = -\vec{B}_t, \quad \nabla \times \vec{B} = \mu_0 \varepsilon_0 \vec{E}_t$$

**Proposition 5.3.17.** *In vacuum,  $\vec{E}$  satisfies the wave equation*

$$\vec{E}_{tt} = \frac{1}{\mu_0 \varepsilon_0} \Delta \vec{E}$$

*Equivalently,*

$$\Delta \vec{E} - \mu_0 \varepsilon_0 \vec{E}_{tt} = 0$$

*Proof.* Start with Faraday's law (Theorem 5.3.11) and take the curl on both sides

$$\nabla \times (\nabla \times \vec{E}) = -(\nabla \times \vec{B})_t$$

Using the vacuum Ampere's Law (Theorem 5.3.13) and taking the curl,

$$\nabla \times (\nabla \times \vec{E}) = -\mu_0 \varepsilon_0 \vec{E}_{tt}$$

Now use the vector identity

$$\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \Delta \vec{E}$$

In vacuum,  $\nabla \cdot \vec{E} = 0$ . Therefore,

$$\nabla \times (\nabla \times \vec{E}) = -\Delta \vec{E}$$

Hence, cancelling the negative sign

$$\Delta \vec{E} = \mu_0 \varepsilon_0 \vec{E}_{tt}$$

Therefore,

$$\vec{E}_{tt} = \frac{1}{\mu_0 \varepsilon_0} \Delta \vec{E}$$

□

**Proposition 5.3.18.** *In vacuum,  $\vec{B}$  satisfies the wave equation*

$$\vec{B}_{tt} = \frac{1}{\mu_0 \varepsilon_0} \Delta \vec{B}$$

*Equivalently,*

$$\Delta \vec{B} - \mu_0 \varepsilon_0 \vec{B}_{tt} = \vec{0}_3$$

**Remark 5.3.19.** The wave speed is

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

Thus, Maxwell's equations predict electromagnetic waves travelling at speed  $c$ . This speed is the speed of light in vacuum.

Maxwell's equations also explain why electric and magnetic fields can be expressed in terms of potentials.

**Proposition 5.3.20.** *Suppose  $d\eta_{\vec{B}} = 0$  on a star-shaped region  $U \subseteq \mathbb{R}^3$ . Then there exists a 1-form  $\omega_{\vec{A}} = A_1 dx + A_2 dy + A_3 dz$  such that  $d\omega_{\vec{A}} = \eta_{\vec{B}}$ . Equivalently, there exists a vector field  $\vec{A} = (A_1, A_2, A_3)$  such that  $\nabla \times \vec{A} = \vec{B}$ . The vector field  $\vec{A}$  is called a magnetic vector potential.*

**Remark 5.3.21.** The existence of  $\vec{A}$  comes from the fact that closed forms are exact on star-shaped regions. Since  $d\eta_{\vec{B}} = 0$ , the magnetic flux 2-form is closed. On a star-shaped region, it is exact, so  $\eta_{\vec{B}} = d\omega_{\vec{A}}$ . In vector-calculus language, this means

$$\vec{B} = \nabla \times \vec{A}$$

**Proposition 5.3.22.** *Suppose  $\eta_{\vec{B}} = d\omega_{\vec{A}}$  on a star-shaped region. Then Faraday's law implies that there exists a scalar-valued function  $\phi$  such that*

$$\omega_{\vec{E}} = -d\phi - (\omega_{\vec{A}})_t$$

*Equivalently,*

$$\vec{E} = -\nabla\phi - \vec{A}_t$$

*Proof.* Using Faraday's law (Theorem 5.3.11) and  $\eta_{\vec{B}} = d\omega_{\vec{A}}$ , we have

$$d\omega_{\vec{E}} = -(d\omega_{\vec{A}})_t = -d((\omega_{\vec{A}})_t)$$

Therefore,

$$d(\omega_{\vec{E}} + (\omega_{\vec{A}})_t) = 0$$

On a star-shaped region, every closed 1-form is exact, so there exists a scalar-valued function  $\phi$  such that

$$\omega_{\vec{E}} + (\omega_{\vec{A}})_t = -d\phi$$

Therefore,

$$\omega_{\vec{E}} = -d\phi - (\omega_{\vec{A}})_t$$

or

$$\vec{E} = -\nabla\phi - \vec{A}_t$$

□

The equations  $d\eta_{\vec{B}} = 0$  and  $d\omega_{\vec{E}} = -(\eta_{\vec{B}})_t$  can be combined into one equation on spacetime.

**Definition 5.3.23.** On spacetime coordinates  $(x, y, z, t)$ , define the electromagnetic 2-form

$$\mathcal{F} = \eta_{\vec{B}} + \omega_{\vec{E}} \wedge dt$$

Explicitly,

$$\mathcal{F} = B_1 dy \wedge dz + B_2 dz \wedge dx + B_3 dx \wedge dy + E_1 dx \wedge dt + E_2 dy \wedge dt + E_3 dz \wedge dt$$

**Proposition 5.3.24.** The two equations  $d\eta_{\vec{B}} = 0$  and  $d\omega_{\vec{E}} = -(\eta_{\vec{B}})_t$  are equivalent to the single spacetime equation  $d\mathcal{F} = 0$ .

*Proof.* Using the spacetime exterior derivative, we compute

$$d\mathcal{F} = d(\eta_{\vec{B}}) + d(\omega_{\vec{E}} \wedge dt)$$

Since  $\eta_{\vec{B}}$  depends on space and time,

$$d(\eta_{\vec{B}}) = d_{\text{space}}\eta_{\vec{B}} + (\eta_{\vec{B}})_t \wedge dt$$

Also,

$$d(\omega_{\vec{E}} \wedge dt) = d_{\text{space}}\omega_{\vec{E}} \wedge dt$$

since  $dt \wedge dt = 0$ . Therefore,

$$d\mathcal{F} = d_{\text{space}}\eta_{\vec{B}} + ((\eta_{\vec{B}})_t + d_{\text{space}}\omega_{\vec{E}}) \wedge dt$$

Thus,  $d\mathcal{F} = 0$  if and only if  $d_{\text{space}}\eta_{\vec{B}} = 0$  and  $(\eta_{\vec{B}})_t + d_{\text{space}}\omega_{\vec{E}} = 0$ .  $\square$

**Remark 5.3.25.** The equation  $d\mathcal{F} = 0$  contains the two Maxwell equations that do not involve charge or current directly.



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